

Metabolic Stability and Targeted Delivery of Oligonucleotides: Advancing RNA Therapeutics Beyond The Liver

Puneet Anand,* Yu Zhang, Spoorthi Patil, and Keerat Kaur



Cite This: *J. Med. Chem.* 2025, 68, 6870–6896



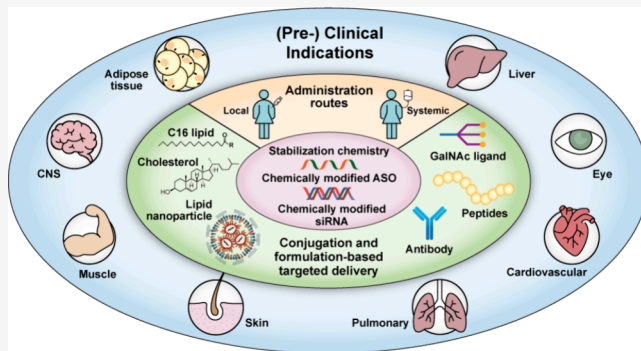
Read Online

ACCESS |

Metrics & More

Article Recommendations

ABSTRACT: Oligonucleotides have emerged as a formidable new class of nucleic acid therapeutics. Fully modified oligonucleotides exhibit enhanced metabolic stability and display successful clinical applicability for targets formerly considered “undruggable”. Accumulating studies show that conjugation to targeting modalities of stabilized oligonucleotides, especially small interfering RNAs (siRNAs), has enabled robust delivery to intended cells/tissues. However, the major challenge in the field has been the stability and targeted delivery of oligonucleotides (siRNAs and antisense oligonucleotides (ASOs)) to extrahepatic tissues. In this Perspective, we review chemistry innovations and emerging delivery approaches that have revolutionized oligonucleotide drug discovery and development. We explore findings from both academia and industry that highlight the potential of oligonucleotides for indications involving different extrahepatic organs—including skeletal muscles, brain, lungs, skin, heart, adipose tissue, and eyes. In all, continued advances in chemistry coupled with conjugation-based approaches or novel administration routes will further advance the delivery of oligonucleotides to extrahepatic tissues.



SIGNIFICANCE

siRNAs and ASOs have become powerful therapeutic tools, evidenced by a surge in approvals over the past decade. The significance of oligonucleotides for the treatment of rare and common diseases cannot be overstated, considering their precise silencing ability and therapeutic potential. This Perspective presents strategies and examples for applying these modalities to extrahepatic tissues, underscoring the importance of innovative design and delivery in achieving effective oligonucleotide therapeutics.

INTRODUCTION

The nucleic acid therapeutic (NAT) field has flourished in recent years with the FDA approval of 24 drugs, including small interfering RNAs (siRNAs), ASOs, aptamers, and mRNAs (Figure 1).¹ Of note, 20 of these 24 NATs were approved during the past decade. In particular, the discovery of two classes of oligonucleotides, namely siRNAs and ASOs, has brought to bear the true therapeutic potential of genetic medicines by providing precision therapy (via selective gene silencing) for many diseases.² In a landmark Nobel Prize-winning discovery, Fire et al. demonstrated double-stranded siRNA-mediated gene knockdown in *Caenorhabditis elegans*; they also found that this process was catalytic, thereby requiring only a few molecules of siRNA per cell.³ Separately, Elbashir et al. showed that a 21-mer

siRNA duplex caused sequence-dependent knockdown of mRNA in cultured mammalian cells.⁴ The initial example of successful gene silencing in vivo by systemic delivery was demonstrated almost two decades ago.⁵ In this seminal study the authors administered cholesterol-conjugated siRNA (for *ApoB*) iv in mice, and found that cholesterol enabled binding of conjugated siRNA to LDL-C, which subsequently led to ApoB silencing and thereby cholesterol reduction in the liver and jejunum.⁵ Further, Davis et al. provided the earliest evidence of gene silencing in humans mediated by systemic administration of nanoparticles encapsulating an siRNA.⁶ The primary subcellular location where an siRNA performs its action is the cytoplasm;² however, emerging data indicate that it may also operate in the nucleus.⁷ The first siRNA therapeutic, Onpattro, was approved by the FDA in 2018;⁸ remarkably, it was followed by the approval of five additional siRNA drugs in a span of 5 years.

Received: October 17, 2024

Revised: December 11, 2024

Accepted: December 26, 2024

Published: January 8, 2025



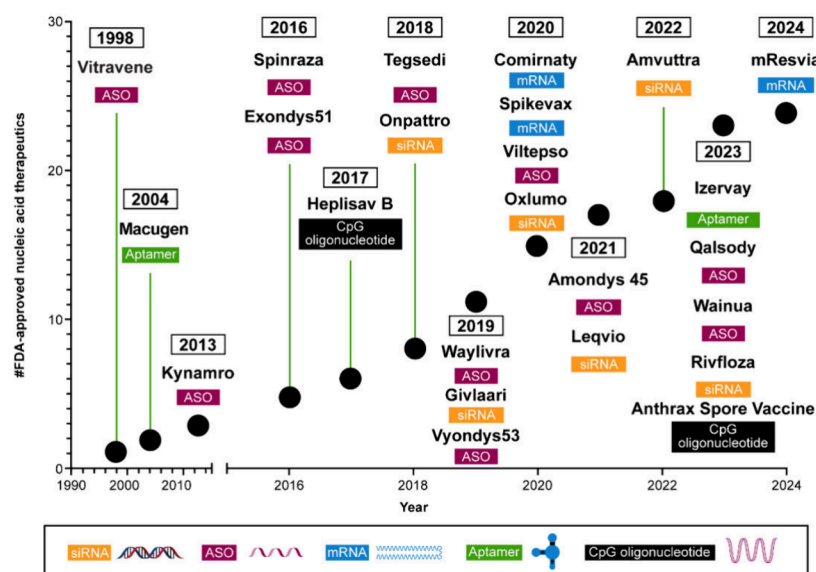


Figure 1. FDA-approved NATs. There are currently 24 approved NATs, which include siRNAs (6), ASOs (11), CpG oligonucleotides (2), mRNAs (3), and aptamers (2). Interestingly, 20 out of the 24 NATs were approved in the past decade, signifying a resurgence for this class of therapeutics. NAT: nucleic acid therapeutic; siRNA: small interfering RNA.

The discovery of gene silencing by an ASO was first reported in 1978. In this study, the authors, including Paul Zamecnik (regarded as the father of ASO therapy), demonstrated that adding a synthetic 13-mer oligodeoxynucleotide—complementary to 13 nucleotides at the 3' and 5' reiterated terminal sequence of Rous sarcoma virus 35S RNA—to chick embryo fibroblast tissue cultures infected with the Rous sarcoma virus led to the inhibition of virus production.⁹ Further, they suggested that this 13-mer ASO was likely working as an antiviral agent by specifically blocking Rous sarcoma virus viral protein translation.¹⁰ ASOs operate in the nucleus and cytoplasm to target mRNA.¹¹ Foundational work carried out by Izant and Weinrub in 1985 revealed that ASO delivery in *Xenopus laevis* oocytes inhibited actin gene activity, thereby resulting in a decreased growth rate.¹² The first ASO drug, vitravene, was approved in 1998.¹³ It is worth pointing out that both siRNA and ASO drugs downregulate protein production, whereas mRNA-based therapeutics (e.g., COVID-19 vaccines) upregulate the production of specific proteins.

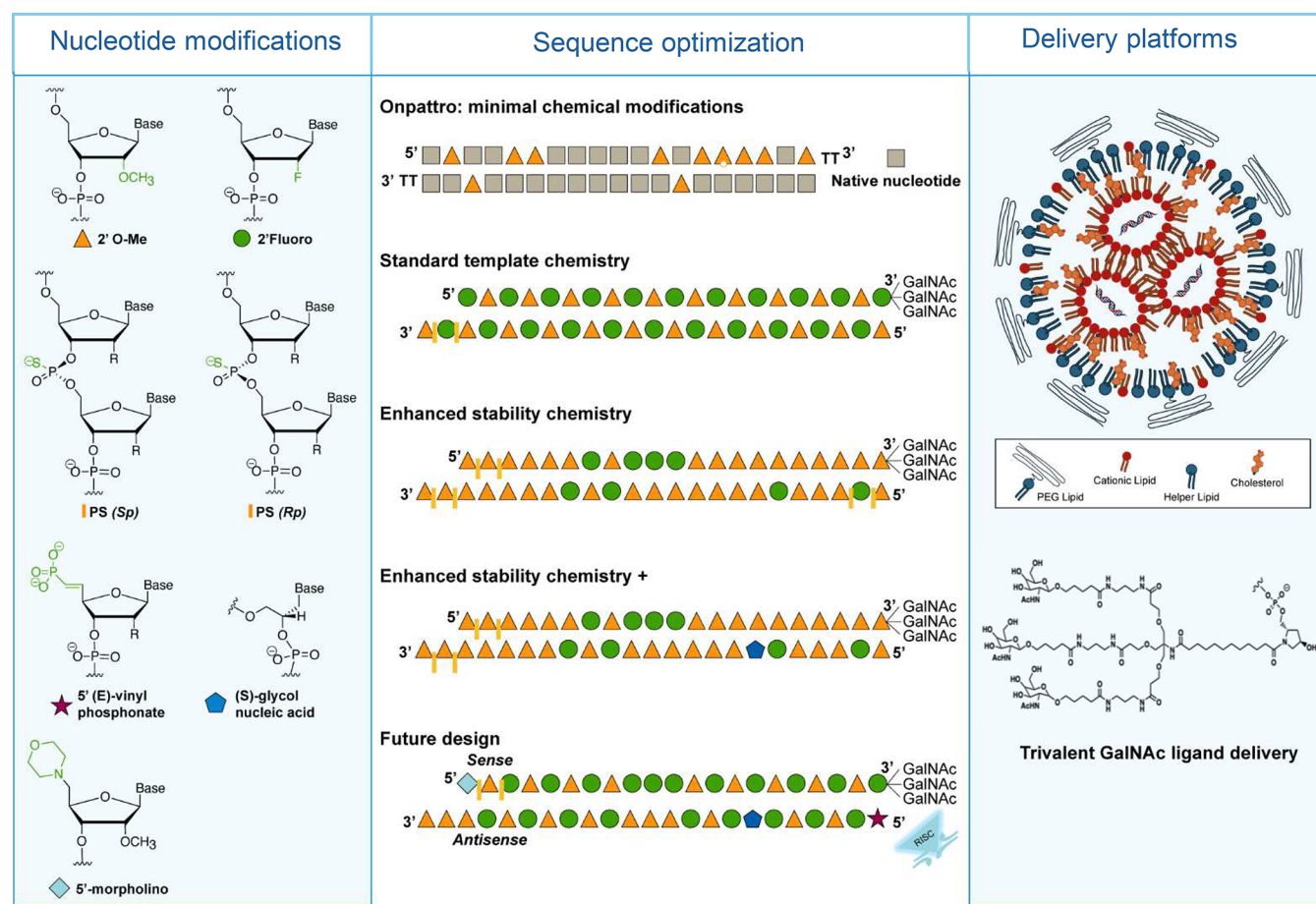
Unmodified siRNAs and ASOs are labile in the cellular milieu and can be degraded by nucleases, thereby necessitating the need to develop therapeutic oligonucleotides with enhanced stability. Novel chemical designs have been introduced and implemented, and these have greatly facilitated the advancement of more stable oligonucleotides.¹⁴ Remarkably, chemical innovations in the field have propelled oligonucleotide therapeutics toward clinical applications for multiple therapeutic indications. Interestingly, it took precisely two decades for the first ASO siRNA therapeutics to receive approval following their initial discoveries in 1978 and 1998.

Delivery of oligonucleotides to the target organ is another major challenge, which requires uptake (entry), endolysosomal escape, and in vivo efficacy.^{5,15} The oligonucleotides must extravasate into the tissue of interest, exhibit cellular uptake, display internalization into endosomes, and then be released from the endosomes to carry out its function.^{16–19} Recent findings reveal that the delivery of siRNA to the liver is achieved either via encapsulation into lipid nanoparticles or conjugation to trivalent N-acetylgalactosamine (GalNAc).^{20,21} Naked ASOs

have been successfully administered locally for CNS and eye indications but systemically for other indications. Notably, 21 oligonucleotides have been FDA approved, and there has been a remarkable surge in the number of preclinical and clinical studies demonstrating the safety, efficacy, and durability of responses to chemically stabilized and/or conjugated oligonucleotides for various therapeutic indications.²² While therapeutic oligonucleotides were initially used for rare diseases, a growing body of clinical studies suggests that these modalities can be applied to both rare and common diseases. Designing novel targeting ligands, expanding the repertoire of chemical designs, enhancing metabolic stability, and applying innovative administration routes are some areas of intense ongoing investigations to enable the delivery of oligonucleotides to extrahepatic tissues. In this review, we detail the progress made to the design and delivery of oligonucleotides, provide examples of preclinical and clinical advancements, and put forth some of the remaining challenges and our perspectives on the oligonucleotide therapeutic landscape.

CHEMISTRY AND DESIGN OF OLIGONUCLEOTIDES

siRNAs. Mechanism of Action. An siRNA is a short stretch of 19–23 mer double-stranded oligonucleotides (~15 kDa), and its mechanism of action initially involves interaction and binding to the RNA-induced silencing complex (RISC) in the cytoplasm.²³ Subsequently, the sense strand is removed leaving functional RISC with antisense loaded.²⁴ The antisense strand directs the RISC to its target mRNA, and the argonaute 2 protein within the RISC acts as a slicer for the degradation of mRNA.²⁵ Argonaute 2 has a duck-shaped structure consisting of a mid-domain, a PAZ domain, and a catalytic PIWI domain.²⁶ The 5' phosphate end of the antisense strand binds to the mid-domain, the 3' antisense end binds to the PAZ domain, and the PIWI domain serves as a slicer for the target mRNA.²⁶ siRNA therapeutics are potent (picomolar range) inhibitors in vitro and have an EC₅₀ of 100 ng/g of tissue or 1 ng of drug/g of tissue in the RISC.²⁷ In addition, siRNAs are catalytic, exhibit specificity for their target mRNAs, and have a durable mechanism of action.¹ Notably, it has been demonstrated that small amounts



Regeneron - Internal

Figure 2. Chemical designs and FDA-approved delivery platforms for siRNAs. (Left) Commonly employed modifications include 2' ribose modifications (2'-O-Me, 2'F) and backbone modification (PS, 5'VP). Glycol nucleic acid modification at position 7 on the antisense strand prevents seed region-mediated off-target toxicity. 5' Morpholino modification at the sense strand mitigates loading of the sense strand into the RISC. (Middle) Evolution of chemical designs for siRNAs. Onpattro (patisiran) was the first approved siRNA therapeutic with minimal chemical modifications, whereas all other approved members in this class contain extensive chemical modifications. As displayed using future design chemistry, the mechanism of action of siRNA involves binding of the antisense strand to the RISC machinery to generate an antisense-RISC, which carries out catalytic degradation of multiple copies of the targeted mRNA. (Right) LNPs and trivalent GalNAc are the two approved platforms for delivery of siRNA to the liver. 2'F: 2'-fluoro; 2'O-Me: 2' O-methyl; 5'VP: 5'-vinylphosphonate; GalNAc: N-acetylgalactosamine; PS: phosphorothioate; RISC: RNA-induced silencing complex; siRNA: small interfering RNA.

Table 1. Six Approved siRNA Therapeutics^a

siRNA	Company	Target	Formulation of siRNA	Method of delivery	Indications	Year approved	Dose administered
Onpattro (patisiran)	Alnylam	TTR	Lipid nanoparticles	iv	ATTR amyloidosis	2018	0.3 mg/kg once every 3 weeks
Givlaari (givosiran)	Alnylam	Aminolevulinic acid synthase 1	Trivalent GalNAc	sc	Acute hepatic porphyria	2019	2.5 mg/kg administered once daily
Oxlumo (lumasiran)	Alnylam	Glycolate oxidase	Trivalent GalNAc	sc	Primary hyperoxaluria type 1	2020	Based on body weight, 3 mg/kg or 6 mg/kg. Dosage given once every month or every 3 months
Leqvio (inclisiran)	Alnylam/Novartis	PCSK9	Trivalent GalNAc	sc	Hypercholesterolemia	2021	284 mg administered as a single injection initially, again at 3 months, and then every 6 months
Amvuttra (vutrisiran)	Alnylam	TTR	Trivalent GalNAc	sc	TTR polyneuropathy	2022	25 mg administered once every 3 months
Rivfloza (nedosiran)	Novo Nordisk	LDHA	Trivalent GalNAc	sc	Primary hyperoxaluria type 1	2023	Based on body weight, 128 mg or 160 mg administered once monthly

^aATTR: transthyretin-mediated; GalNAc: N-acetylgalactosamine; LDHA: lactate dehydrogenase A; PCSK9: proprotein convertase subtilisin/kexin type 9; siRNA: small interfering RNA; TTR: transthyretin.

of siRNA molecules (~ 1000) per cell are required to achieve a maximum siRNA response.²⁸

Naked siRNA Molecules Have Poor Drug-Like Properties. While siRNA-based drugs have potential broad therapeutic applications across multiple indications, these molecules have poor drug-like properties—primarily due to their charge and size.²⁹ The highly negative charge on siRNAs renders them incapable of passively diffusing through the lipid bilayer, and therefore they need a targeting moiety for cellular uptake and endocytosis.²⁹ Once inside the endosomes, the siRNA molecules form a depot and are slowly released into the cytoplasm. In fact, less than 1% of siRNA molecules escape the endosomes and more than 99% of siRNA molecules are trapped within the endosomes.³⁰ Therefore, the endosomal escape problem is considered to be a major rate-limiting step in advancing siRNA therapeutic drug discovery, particularly for extrahepatic tissues.²⁹ Further, due to its unfavorable physicochemical properties, siRNA is rapidly cleared via first-pass metabolism through the kidneys.³¹ In addition, siRNAs can be degraded by nucleases (e.g., ribonuclease [RNase]) that recognizes 2'-OH of the ribose to cleave the phosphodiester linkage.²⁹ Moreover, siRNA may also activate an innate immune response through its interaction with the TLR3 receptor, which recognizes the sugar and phosphate backbone of siRNA.^{32,33}

Chemical Modifications of siRNA. Despite some of its inherent limitations and initial hiccups, the siRNA field has evolved since its discovery almost two decades ago. The therapeutic potential of siRNA was brought to realization by shifting the focus from minimal to extensive modifications of siRNA and conjugation of siRNA to targeted ligands for delivery to specific tissues¹⁴ (Figure 2). Notably, all six FDA-approved siRNA therapeutics currently target the liver (Table 1);^{1,34} however, there has been substantial preclinical and clinical data that support the targeted delivery of siRNA to extrahepatic tissues. Chemical stabilization of siRNA, coupled with either formulation into lipid nanoparticles (LNPs) or conjugation to trivalent GalNAc, has had transformative implications in the field of siRNA therapeutics. GalNAc can be attached to either the 3' (preferred) or 5' end of the sense strand, as the sense strand does not enter the RISC.³⁵ The two principal chemical modifications that are employed to stabilize siRNA include modifications to either the phosphate backbone (phosphorothioate [PS]) or to the 2' position (2' O-methyl [2'OMe] or 2' deoxy-2'-fluoro [2'F]) of ribose (Figure 2).³⁶ 2'OMe and 2'-deoxy-2'F modifications on the ribose sugar display an attenuated immune response and increased potency of siRNA for its target mRNA.^{37,38} These modifications do not interfere with RISC loading, and result in enhanced metabolic stability.³⁶ The 2'F modification locks the 3'-endo conformation, thereby increasing the affinity for the target mRNA. However, the 2'F modification provides decreased resistance against nuclease degradation compared with the 2'OMe modification. The PS modification was discovered by Eckstein in 1966,³⁹ and it was successfully applied to oligonucleotide synthesis by Stec et al. in 1984.⁴⁰ PS modifications at the ends of duplex siRNA provide protection against exonucleases.⁴¹

Chirality of PS Modification Within The siRNA Duplex. The PS linkage is chiral, and it configures as right-handed (Rp) or left-handed (Sp) isomers (Figure 2).⁴² Manoharan et al. recently published a study demonstrating that the PS chirality of the antisense strand is an important parameter that may govern Ago2 loading, RNA interference activity, and metabolic stability.⁴² Using siRNAs against two different targets (trans-

thyretin [TTR] and complement factor 5), the authors reported that the Rp isomer exhibited greater stability for 5' exonuclease, whereas the Sp isomer displayed greater stability for 3' exonuclease, in a mouse model.⁴² Specifically, the authors found that Rp diastereomers at the 5' end and Sp diastereomers at the 3' end of the antisense siRNA strand improved the PK and PD properties of siRNAs in vivo.⁴² Mechanistically, in silico modeling suggested that the Rp isomer at the 5' end of antisense strand bound with greater affinity to the mid-domain of the AGO2 protein, and the Sp isomer at the 3' end of the antisense strand bound with increased affinity to the PAZ domain.⁴² These data suggest that chirally pure PS may be an important consideration in the design of oligonucleotides. While PS modifications generally enhance siRNA stability, it is established that PS linkage results in lower affinity for the target mRNA than natural phosphodiester (PO) linkage. Also, it has previously been reported in mice that higher PS content (13 vs 8) was associated with decreased activity of blunt siRNA.⁴³

Introduction of Vinyl Phosphate Modification for Improved RISC Loading. More recently, another phosphate modification, (5'-(E)-vinyl phosphonate (5'-(E)-VP)) was applied into siRNA design by Alnylam Pharmaceuticals at the 5' end of the antisense strand, and it was confirmed that the presence of 5'-(E)-VP increased siRNA stability and potency⁴⁴ (Figure 2). Phosphorylation of the 5' end of the antisense strand is potentially not a rate-limiting step for the liver, but it may be for extrahepatic tissues. Therefore, the delivery of oligonucleotides to these tissues may require the addition of 5'-(E)-VP.

Collectively, the overarching goal of these siRNA modifications is to increase metabolic stability and reduce both the immune response and off-target effects.

Extensive Modification of siRNA Molecules Affects Metabolic Stability In Vivo. Previous studies have shown that fully stabilized siRNA conjugated to cholesterol leads to increased accumulation of siRNA in tissues such as the liver, kidney, and spleen following systemic administration.⁴⁵ In contrast, systemic delivery of a partially modified siRNA ($\sim 60\%$ chemical modification) conjugated to cholesterol resulted in significantly lower tissue accumulation.⁴⁵ Interestingly, these findings were recapitulated with various other conjugated ligands (cholesterol, docosahexaenoic acid [DHA], and GalNAc) or with different systemic administration routes (iv vs sc).⁴⁵ Several studies have reported that extensive chemical stabilization is critical for developing therapeutic siRNA.³⁶ A case in point is the FDA approval of fully stabilized vutrisiran vs the discontinuation (due to mortality imbalance) of partially stabilized revusiran.^{38,46–48} Revusiran, targeting TTR (encoding TTR protein) for TTR amyloidosis, is a first-generation GalNAc conjugate that is based on a standard chemistry platform and has an alternating pattern of 2'F and 2'OMe modifications at all ribose positions.^{38,48} In contrast, vutrisiran, which also targets TTR, is based on an enhanced stability chemistry approach in which the siRNA contains a greater percentage of 2'OMe vs 2'F; it also incorporates two additional PS linkages at the 5' end of the antisense and sense strands.³⁸ Owing to metabolic stability limitations, revusiran was administered weekly at relatively high dosages, whereas vutrisiran is administered once every 3 months at lower dosages (Table 1). Both revusiran and vutrisiran have the same sequence but different chemistry, and these results show the translation of siRNA chemical modifications to clinical success.

Impact of the Position of Chemical Modifications on siRNA Activity. Alnylam showed that the position of chemical

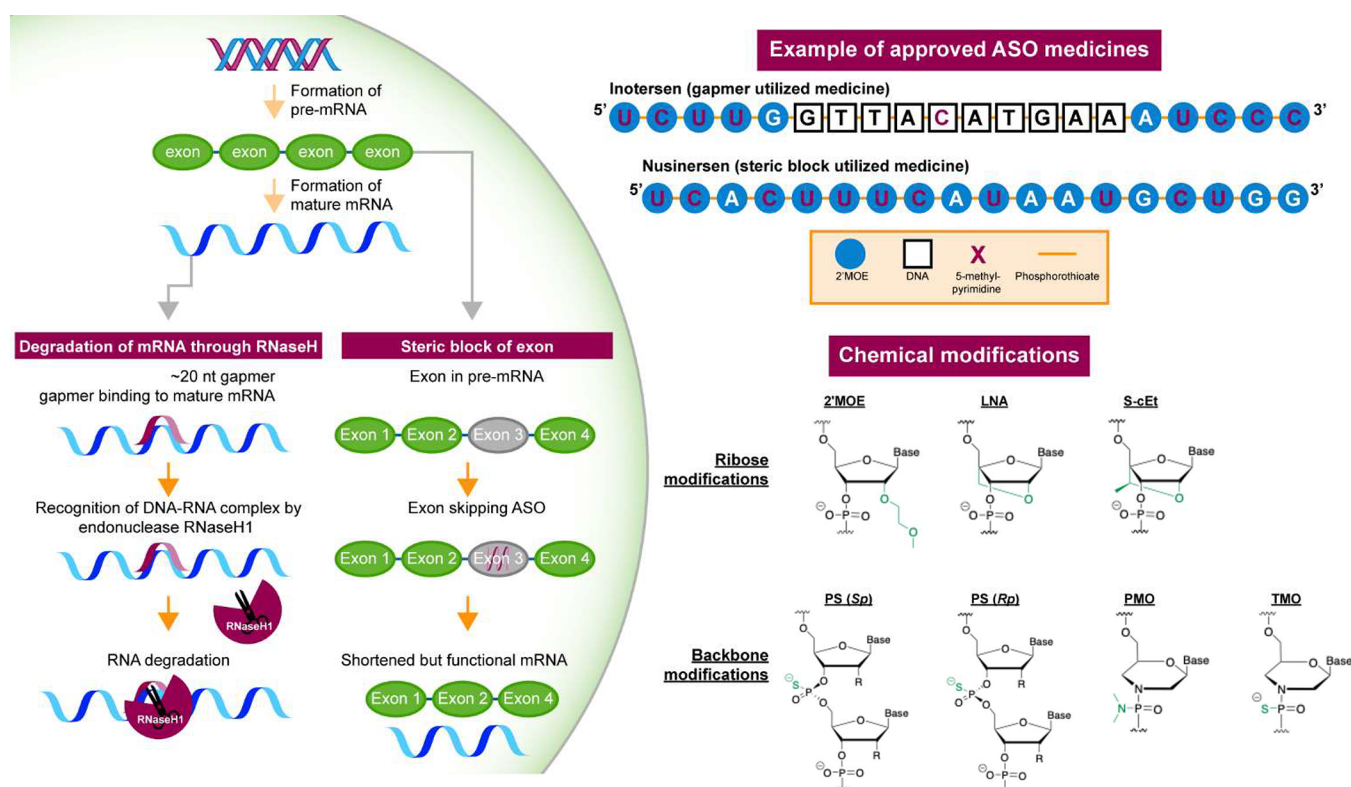


Figure 3. Mechanisms of action and chemical modifications of ASOs. (Left) ASO operates via two main mechanisms: mRNA degradation through endonuclease RNase H1, and mRNA modulation by steric blocking of the faulty exon. The RNase H1 mechanism is activated by the binding of a gapmer ASO to mature mRNA, ultimately resulting in the degradation of target mRNA. Steric-blocking ASOs bind to pre-mRNA containing an exon to form a heteroduplex. The ASO masks a portion of the exon, allowing for the formation of a shortened but functional mRNA. (Right) Depiction of the structures of approved ASO medicines Inotersen and Nusinersen. Inotersen is a gapmer ASO while Nusinersen is predominately made up of fully modified RNA. Typical chemical modifications to ASOs include backbone modifications (PS, TMO, or PMO), 2' ribose modifications (2'-O-MOE, LNA, 2'-cEt) or base modification (5-methyl-pyrimidine). 2'-cEt: 2'-O-(2-methoxyethyl) ethyl; 2'MOE: 2'-O-methoxyethyl; gapmer (ASO with a central DNA gap flanked by modified RNA); LNA: locked nucleic acid; PMO: phosphorodiamidate morpholino oligonucleotide; PS: phosphorothioate; RNase H1: ribonuclease H1; TMO: thiomorpholino oligonucleotide.

modifications within the siRNA duplex impacts activity. Specifically, their team demonstrated that the replacement of 2'-deoxy-2-fluoro with 2'-O-Me at positions 2, 6, or 14 of the antisense strand or at position 11 of the sense strand negatively impacted siRNA activity, thereby providing evidence that steric hindrance subserved by 2'OMe must be considered in the design of siRNAs.²⁷ Further, it was demonstrated that increasing the ratio of 2'-O-Me to 2'-deoxy 2-F (80/20) in siRNA chemistry improved the potency and durability of the response in mice and nonhuman primates (NHPs).²⁷

The Structure of siRNA may Influence its Activity. Besides chemical modifications and conjugation to targeting ligands, the siRNA structure is also an important consideration for siRNA activity, and various companies use different types of siRNA structures. Overhangs of 2–3 bases at the 3' end of the antisense strand have been associated with increased RISC stability.⁴⁹ Most companies, including Alnylam, employ a conventional format (two nucleotide overhang at the 3' end of the antisense strand), whereas other companies such as Silence Therapeutics have a blunt siRNA, and Dicerna (now Novo Nordisk) uses a tetraloop in their siRNA design.³⁶ Recent studies have demonstrated that the systemic delivery of lipid-conjugated, asymmetric siRNA structures leads to more effective mRNA silencing in mouse extrahepatic tissues compared to lipid-conjugated blunt siRNA structures.⁴³

siRNA-Mediated Off-Target Effects and Potential Solutions. As with other classes of therapeutics, siRNA may mediate off-target effects by binding to unrelated mRNA.⁵⁰ To counter the problem of seed region-mediated off-target effects, the addition of glycol nucleic acid (GNA) at position 7 of the antisense strand has been successfully employed (Figure 2),⁵¹ which promotes binding of the antisense strand to the kink region of the argonaute 2 protein, and prevents binding to nontargeted mRNA through thermodynamic destabilization.⁵² The purpose of GNA modification at the seventh nucleotide is to weaken the binding between the antisense strand and any unrelated mRNA that depends on base pairing in the seed region. However, GNA modification does not disrupt on-target binding since it is controlled by the entire sequence. Further, morpholino modification at the 5' position of the sense strand has been shown to prevent phosphorylation of the sense strand, thereby preventing the loading of the sense strand onto the RISC⁵³ (Figure 2). Additionally, RNA sequencing (RNA-seq) is employed to identify potential off-target effects during the screening stage. Application of these approaches will further advance the discovery and development of safer siRNA therapeutics.

ASOs. Mechanism of Action. An ASO consists of 16–20 nucleotides that bind to complementary RNA and which may be delivered locally or systemically.⁵⁴ Upon administration, ASOs (similar to siRNAs) are internalized into endosomes and are

Table 2. Eleven approved ASOs^a

ASO	Company	Target	Class of ASO	Mechanism of action	Method of delivery	Indication	Year approved	Dose administered
Vitravene (fomivirsen)	Ionis/Novartis	CMC IE-2	21-mer deoxyoligonucleotide with PS linkages	RNase H-mediated	Intravitreal	CMV retinitis	1998	300 µg every 4 weeks
Kynamro (mipomersen)	Ionis/Sanofi	APOB-100	5-10-5 MOE gapmer	RNase H-mediated	sc	Homozygous familial hypercholesterolemia	2013	200 mg once weekly
Spinraza (nusinersen)	Ionis	SMN2	RNA oligonucleotides modified with 2'MOE	Steric block	Intrathecal	SMA	2016	Four doses of 12 mg at 0, 4, 8, and 63 days; 12 mg maintenance dose after every 4 months
Exondys 51 (eteplirsen)	Sarepta	Exon 51 skipping	PMO	Steric block	iv	DMD	2016	30 mg/kg every week
Tegsedi (inotersen)	Ionis/Ackea	TTR	5-10-5 gapmer	RNase H-mediated	sc	Hereditary ATTR	2018	300 mg once weekly
Vyondys 53 (golodirsen)	Sarepta	Exon 53 skipping	PMO	Steric block	iv	DMD	2019	30 mg/kg every week
Waylivra (volanesorsen)	Ionis/Ackea	Apolipoprotein CIII	5-10-5 MOE gapmer	RNase H-mediated	sc	Familial chylomicronemia syndrome	2019	285 mg once every week
Vilepso (vitolarsen)	NS Pharma	Exon 53	PMO	Steric block	iv	DMD	2021	80 mg/kg every week
Amondys 45 (casimersen)	Sarepta	Exon 45 skipping	PMO	Steric block	iv	DMD	2021	30 mg/kg every week
Qalsody (tofersen)	Ionis/Biogen	SOD1	5-10-5 MOE gapmer	RNase H-mediated	Intrathecal	ALS	2023	100 mg every 14 days for three doses; 100 mg maintenance dose every 28 days
Wainua (eplontersen)	AstraZeneca/Ionis	TTR protein	GalNAc ₃ -conjugated 5-10-5 MOE gapmer	RNase H-mediated	sc	Hereditary TTR-mediated amyloidosis	2023	45 mg every 4 weeks

^aNote that six ASOs perform their action via an RNase H-mediated mechanism while five are steric blockers. ATTR: transthyretin-mediated; DMD: Duchenne muscular dystrophy; GalNAc: N-acetylgalactosamine; MOE: methoxyethylribose; PMO: phosphorodiamidate morpholino oligonucleotide; PS: phosphorothioate; RNase: ribonuclease; SMA: spinal muscular dystrophy; TTR: transthyretin.

slowly released from the endosomes to the nucleus to affect target mRNA silencing. ASOs can mediate their action via several different mechanisms, and two commonly used modes of operation include ribonuclease H1 (RNase H1)-mediated (1) target mRNA degradation (by DNA gapmer ASO) or (2) target mRNA modulation (by splice switching/steric blocking) (Figure 3).⁵⁵ Ionis Pharmaceuticals and Sarepta Therapeutics have been pioneers in the discovery and development of therapeutic ASOs. Currently, there are 11 approved ASOs for multiple diseases (Table 2). Early work from Ionis demonstrated that binding of the DNA gapmer ASO to RNA (to form a DNA:RNA heteroduplex) triggers the recruitment of RNase H1, therefore leading to degradation of a specific mRNA.^{56,57} It was shown that a gapmer must contain 8–10 deoxynucleotides for optimal RNase H1 activity.⁵⁴ It has been illustrated that there are “hotspots” within a target mRNA molecule that are more susceptible to an ASO-based RNase H1-mediated mechanism of action.⁵⁸

ASO Molecules Have Poor Drug-Like Properties. Like the physio-chemical nature of siRNA-based drugs, naturally occurring ASOs are labile and have unfavorable drug-like properties; these therefore require chemical stabilization.⁵⁴ The commonly used chemical modifications in ASO design include backbone PS linkage and 2′-O-methoxyethyl (2′MOE) on the 2′ position of the ribose (Figure 3).

First-Generation ASO Design. The first generation ASO design by Ionis Pharmaceuticals involved chemical modification of ASO drugs by using PS backbone modification (Figure 3).^{58,59} PS is more lipophilic than PO, which facilitates the binding of PS-containing ASOs to serum proteins and also prevents rapid clearance through the kidneys.^{60,61} At least 10–12 PS are required to obtain optimum binding to serum proteins.⁶² Systematic delivery of PS ASOs largely resulted in delivery to the liver, kidney, lymph nodes, spleen, bone marrow, and fat cells.⁶³ Further, PS ASOs have a tissue elimination half-life of up to 48 h.⁶²

Second-Generation ASO Design. The second-generation ASO design included the 2′MOE modification (Figure 3) to ribose in addition to the PS backbone, and it was shown that the presence of 2′MOE increased the tissue half-life of PS ASOs to 2–4 weeks.^{64,65} The 2′MOE modification rigidifies the overall structure, increases affinity for the target mRNA, hinders nuclease attacks, and decreases nonspecific interactions. Compared with first-generation ASOs, those from the second generation displayed 10-fold greater potency. Furthermore, systemic delivery of second-generation ASOs was shown to preferentially biodistribute to the kidneys, liver, and white adipose tissues, whereas local administration was found to be active in difficult-to-reach organs including the CNS. Interestingly, a second generation ASO mipomersen (targeting APOB for homozygous familial hypercholesterolemia) showed potential for oral delivery with resultant pharmacological responses in several species, including humans;⁶⁶ however, mipomersen was discontinued due to its high dose requirement, which led to severe treatment-associated adverse effects. While second-generation ASOs have an overall favorable safety profile, thrombocytopenia was observed in patients who were taking either inotersen or volanesorsen (300 mg of these drugs was administered sc weekly).^{67,68}

Generation 2.5 ASO Design. Ionis also introduced a 2.5-generation design, which contained a constrained ethyl-modified oligonucleotide (2′-cEt) modification^{54,69} (Figure 3); it was found that 2′-cEt ASOs exhibited comparable potency

to locked nucleic acid (LNA) ASOs. Further, they showed that 2′-cEt ASOs displayed favorable safety profiles in clinical trials.^{70–72} Remarkably, due to potency gain (compared with generation 2.0 ASO chemistry), generation 2.5 chemistry enabled efficacy of ASOs in tier 2 tissues, including lung, heart, fat, bone, and bone marrow. 2′-cEt ASO modification was introduced following the discovery of LNA ASOs, which were the first examples of constrained sugars within nucleic acids (Figure 3); these ASOs were shown to display 5–10 times greater potency for the target mRNA than 2′MOE gapmers. Unfortunately, several clinical trials found that PS LNA ASOs were associated with hepatic, renal, and platelet toxicities;^{73,74} however, emerging studies suggest that certain chemical modifications applied to the ASO sequence may potentially mitigate these associated toxicities (discussed below).

Ligand-Conjugated Antisense Design. Ionis has also developed a ligand-conjugated antisense platform for targeted delivery of ASOs to specific tissues with the aim of increasing the therapeutic index of ASOs. Their team conjugated ASO to GalNAc and showed that the resultant ASO–GalNAc conjugate bound strongly to asialoglycoprotein receptors (ASGPRs) expressed on hepatocytes, thereby resulting in increased potency for liver targets by 30-fold in humans.⁷⁵ In fact, Ionis and AstraZeneca recently reported the FDA approval of their GalNAc-conjugated ASO eplontersen (targeting TTR), for the treatment of the polyneuropathy of hereditary TTR-mediated amyloidosis in adults.⁷⁶

Toxicity Associated with Gapmer ASOs, and Potential Solutions. It is well-established that toxic PS ASOs in all chemical classes may exhibit sequence-dependent toxicity.⁵⁴ It was reported that toxic PS ASOs bound to paraspeckle proteins with greater affinity than safer PS ASOs.^{77,78} This resulted in the formation of an RNA-PS protein complex, and it was observed that RNase H1 interaction (via its spacer domain) with the paraspeckle protein within this complex ultimately led to mislocalization of these complexes to the nucleolus.^{77,78} Together, these events resulted in apoptosis and associated toxicity.^{77,78} Remarkably, modification of toxic PS-ASOs at position 2 of the DNA gapmer with 2′MOE markedly diminished hepatotoxicity in mice, thereby significantly enhancing the therapeutic index of these ASOs; this may in turn allow for higher systemic doses of these ASOs, with the aim of delivering pharmacological levels of the drug to extrahepatic tissues.⁷⁸ The PS backbone in ASOs can enhance toxicity, and Ionis recently showed that replacement of the PS linkage at position 2 or 3 in the gap with a mesyl-phosphoramidate (MsPA) linkage increased ASO stability and reduced its nonspecific protein binding while maintaining RNase H1 activity.⁷⁹ In addition, they reported that two MsPA modifications, either in the 5′ region of the gapmer or in the 3′ wing, enhanced activities of 2′OMe-modified gapmer ASOs, which may likely be due to enhanced RNase H1 cleavage activity.⁸⁰ These studies therefore provide evidence that elegant medicinal chemistry approaches can address the problem of sequence-mediated toxicity of gapmer ASOs.

Phosphorodiamidate Morpholino Oligonucleotides. Phosphorodiamidate morpholino oligonucleotides (PMOs) are gaining traction as efficient steric block ASOs (Figure 3). PMOs are short single-strand DNA analogs in which the ribofuranose ring has been replaced by morpholino rings, and the backbone is connected by phosphorodiamidates.⁸¹ Compared with classical ASOs, PMOs are uncharged (Figure 3). PMOs function as steric-block ASOs to modulate splicing,

thereby resulting in exclusion or inclusion of a specific exon. Sarepta Therapeutics has pioneered the use of PMOs for the treatment of Duchenne muscular dystrophy (DMD). PMOs induce exon skipping in DMD,⁸² which is caused by a frameshift mutation that results in the translation of a defective dystrophin protein.⁸³ PMOs bind to their target sites (containing the mutation) on the mRNA and block its interaction with spliceosomes, which results in the removal of the mutated exon and subsequent restoration of the truncated (but functional) dystrophin protein. While four PMO drugs are currently FDA approved (Table 2), the major drawback with PMOs is their rapid clearance through the kidneys (they have a neutral charge, and the molecular mass of PMOs is below the cutoff of renal clearance) after systemic administration, which means they require frequent and high dosage (Table 2). Notably, the tissue half-lives of PMOs are 7–14 days and these molecules are excreted in intact form via urine.⁸⁴

Thiomorpholino Oligonucleotides. More recently another class of ASO, termed thiomorpholino oligonucleotides (TMOs) (Figure 3), was proposed for the treatment of DMD.⁸⁵ TMOs are similar in structure to PMOs but contain a negatively charged sulfur atom directly connected to a phosphorus atom.⁸⁵ In a recently published study, the authors used myotubes from mdx mice that are amenable to exon 23 skipping, and showed that self-delivery of nanomolar concentrations of 20-mer TMO (but not 20-mer PMO) led to efficient deletion of exon 23.⁸⁵ In addition, it was demonstrated that TMOs, 2'MOE PS, and 2'OMe PS were all efficient in exon 23 deletion in myotubes; however, TMOs displayed better effects at low nanomolar concentrations.⁸⁵ Further, TMO chimeras containing either DNA gapmer or DNA TMO mixmers displayed efficacy in exon 23 splice switching.⁸⁵ Given that DNA gapmer ASOs are known to induce an RNase H1-mediated response, it was surprising to see that these can also operate via steric blocking (and not activating RNase H) to induce exon 23 skipping. Stability studies with Tm assays revealed that TMOs exhibit greater destabilizing effects compared to either 2'OMe PS or PMOs, necessitating further investigation to elucidate the underlying mechanism.⁸⁵ Notably, systemic delivery (intravenous) of 50 mg/kg/week of TMOs for 4 weeks in mdx mice led to an increased density of dystrophin protein in muscle fibers, as demonstrated by immunohistochemistry (IHC). These results differed significantly from the treatment of mdx mice with either PMO or 2'OMe PS, where the dystrophin protein density was considerably lower.⁸⁵ These promising preclinical data support the continued exploration of TMOs as a potential therapeutic modality for DMD.

Chirality of ASOs. Like siRNAs, the PS group in ASOs introduces chirality, resulting in a mixture of Rp and Sp diastereomers. The specific configuration of these diastereomers can affect the ASO's binding affinity to target RNA and its interaction with proteins involved in RNA processing.⁸⁶ However, the concrete benefits of generating chirally pure stereoisomers in the context of oligonucleotides are still being evaluated. Wave Life Sciences has also explored the use of phosphoramidate (PN) linkages in their oligonucleotides. The PN backbone chemistry introduces additional chiral centers, allowing for further optimization of ASO properties.⁸⁷ For example, the PN-1 chemistry (1,3 dimethylimidazolin-2-ylidene) PN developed by Wave Life Sciences has shown to increase potency and durability of silencing in preclinical studies.⁸⁷

Off-Target Effects of ASOs. ASOs may also exhibit off-target effects that are hybridization-dependent or hybridization-independent. Hybridization-dependent effects may arise from Watson–Crick interactions of DNA gapmer ASOs with nontargeted mRNA, which can induce RNase H1-mediated degradation of the corresponding mRNA.⁸⁸ In addition, ASOs may cause hybridization-independent steric blocking at a nontarget site, thereby preventing downstream protein synthesis. One of the commonly employed strategies to mitigate hybridization-dependent off-target effects is to introduce targeted mismatches.⁸⁹ In addition, ASOs may mediate off-target effects that are secondary and are mediated through interactions with paraspeckle proteins.^{78,90} Therefore, harmful off-target effects of ASOs need to be considered during the *in silico* design stage.

DELIVERY OF OLIGONUCLEOTIDES

Oligonucleotides are conjugated to various ligands or encapsulated in retargeted nanocarriers for targeted tissue delivery. The ligands may include sugars, lipids, aptamers, or antibodies. The two principal approaches employed for oligonucleotide conjugate administration are local and systemic routes. Commonly used local delivery methods involve IT and icv administration for CNS indications, intravitreal (ivt) administration for ocular indications, and inhalation for pulmonary indications (discussed below). In contrast, systemic delivery mostly encompasses iv or sc injections. While systemic delivery of siRNA is clearly more convenient for patients, this administration route requires crossing the BBB (for CNS indications), and (possibly) higher doses that may lead to nonspecific biodistribution in peripheral organs. These considerations necessitate the need to carefully examine the delivery route of conjugated siRNA to achieve a favorable therapeutic index. Unlike ASOs, siRNAs have been shown to work only when conjugated to a targeted ligand or encapsulated into LNPs. Below is a description of the targeting ligands or formulation systems currently being used for delivery of RNA therapeutics in clinical and preclinical studies.

Targeting Ligands and Delivery Platforms. Lipid Nanoparticles. Nanoparticles (NPs) with different physicochemical properties and from various sources have been successfully employed for oligonucleotide delivery. Most NPs are relatively large, and the encapsulation of siRNA into NPs (>6–8 nm) precludes clearance of siRNA by the kidneys. Synthetic sources for NPs include inorganic NPs,⁹¹ metal organic frameworks,⁹² and lipid and/or polymer-based assemblies^{93–95}/nanogels⁹⁶/dendrimers.⁹⁷ Natural sources include protein- or DNA-based nanovectors,⁹⁸ viral⁹⁹/virus-like particles,⁹⁹ exosomes,¹⁰⁰ and cell membrane-derived NPs.¹⁰¹ Among nanotechnologies for RNA delivery, LNPs are currently the frontrunner delivery platform for clinical translation, with four FDA approved drugs including one loading siRNA and three loading mRNAs.⁹⁵

LNPs consist of a multicomponent lipid formulation of a nucleic acid.¹⁰² The first FDA-approved siRNA therapeutic, Patisiran (Onpattro), was encompassed into LNPs. INN Patisiran is approved for polyneuropathy associated with hereditary transthyretin-mediated amyloidosis (hATTR) and is used for the treatment of transthyretin-mediated amyloidosis (ATTR).¹⁰³ Patisiran consists of a minimally modified siRNA (for TTR mRNA) and is encapsulated into four lipid constituents: MC3 (ionizable cationic lipid), PEG-DMG (PEG-lipid), cholesterol (sterol), and distearoylphosphatidyl-

Table 3. Selected Examples of Antibody-Mediated Retargeting Lipid-Based NPs for Extrahepatic Oligonucleotides Delivery^a

Antibodies	Cargo	Lipid composition	Modification method	In vitro/in vivo	Targeted cells or tissues	Results	References
Anti-β7 integrin (FIBS04)	siRNA	PC/DPPE/cholesterol	Amine-NHS ester reaction	In vivo	Leukocyte subsets	The targeted gene is silenced	120
Anti-LEA-1	siRNA	Soybean PC/cholesterol/DPPE	Amine coupling	In vitro	Leukocytes	Selective uptake of siRNA by T cells and macrophages	121
Anti-CD45RO	siRNA	PC/DSPE-PEG2K-biotin	Biotin-streptavidin interaction	In vivo	Effector T cells	Selective attachment and endocytosis are achieved	122
Anti-CD20	Antisense therapeutics	PEI/HEPC/DOPE/DC-cholesterol/DSPE-PEG2K/DSPE-PEG2K-maleimide	Thiol-maleimide reaction (amine on antibody thiolated by Traut's reagent)	In vivo	Leukemia cells	High transfection efficiency to targeted cells	123
Anti-DEC-205	siRNA	POPC/DDAB/DSPE-PEG2K/DSPE-PEG2K-maleimide	Thiol-maleimide reaction (thiolated antibody for reaction)	In vivo	Dendritic cells	Dendritic cell-specific siRNA delivery and gene silencing in vivo	124
Anti-CD38	siRNA	MC3/cholesterol/DSPC/DMG-PEG/DSPE-PEG-maleimide	Thiol-maleimide reaction (antibody reduced by DTT)	In vivo	B cell	LNPs are specifically taken up by human mantle cell lymphoma in bone marrow	125
Anti-CD4	siRNA	MC3/cholesterol/DSPC/DMG-PEG2K/DSPE-PEG2K-maleimide	Thiol-maleimide reaction	In vivo	T cells	CD45 silencing was restricted to the CD4+ T cells	126
Anti-DEC-205 (scFv)	siRNA	DSPC/cholesterol/DSPE-PEG2K-maleimide/DLindMA	Thiol-maleimide reaction (engineering cysteine reduced by TCEP)	In vivo	Dendritic cells	LNPs preferentially delivered to DEC205+ dendritic cells	127
Anti-Iggb7, CD3, CD4, CD25, CD29, CD44, CD34, Ly6C Abs	siRNA	MC3/DSPC/cholesterol/DMG-PEG2K/DSPE-PEG2K	Fc-binding lipoprotein post inserted* onto LNP to bind added antibody	Ex vivo	Treg, T helper, CTL, B cell, monocytes	Versatile and effective platform for targeted antibody-LNP delivery	128
Multiple antibodies	siRNA	MC3/DSPC/cholesterol/DMG-PEG2K/DSPE-PEG2K	Fc-binding lipoprotein post inserted* onto LNP to bind added antibody	In vivo	Leukocytes	Successful transfection ex vivo of the hard-to-transfect leukocytes	129
Antipodoplanin	siRNA	DMG-PEG2K/DSPE-PEG/SS-OP/cholesterol	DBCO-azide click reaction	In vivo	Lymphatic endothelial cells	Selectively bound by endothelium-free zones of arterial segments	130
Anti EGFR	siRNA	O,O'-dimyristyl-N-lysyl glutamate/cholesterol/DSPE-PEG2K/DSPE-PEG2K-maleimide	Thiol-maleimide reaction (amine on antibody thiolated by Traut's reagent)	In vivo	Lung cancer	Effectively targeted to cancer cells growing in mouse lungs	131
Anti-HB-EGF (Fabs)	siRNA	DOPE/cholesterol/DMPG/DSPE-PEG2K-maleimide	Thiol-maleimide reaction	In vitro	Triple-negative breast cancer	The targeted protein is suppressed in tumors	132
Anti-CD117	siRNA and mRNA	Various ionizable lipids/cholesterol/DSPC/ various PEG lipids	Thiol-maleimide reaction (antibodies' interchain disulfide reduced by TCEP)	In vivo	HSPCs in bone marrow	Almost all HSPCs in bone marrow were edited with anti-CD117 LNPs	133

^aIn this table cargo refers to the type of nucleic acid delivered by the lipid nanoparticles. Lipid Composition defines the specific lipids used to formulate the nanoparticles and modification method is the conjugation chemistry approach used to attach the antibody to lipid nanoparticles. 2K: 2000; CTL: cytotoxic T-cell; DBCO: dibenzocyclooctyn; DC-cholesterol: 3β-[N-(N',N'-dimethylaminoethane)-carbamoyl]cholesterol; DDAB: dimethyldioctadecylammonium bromide; DLindMA: 1,2-dilimolexyo-*n,n*-dimethyl-3-aminopropane; DMG-PEG: 1,2-dimyristoyl-sn-glycero-3-phosphoethanolamine-N-[methoxy(PEG)]; DMPG: 1, 2-dimyristoyl-sn-glycero-3-phospho-rac-(1-glycerol); DOPE: dioleoylphosphatidylethanolamine; DPPE: 1,2-dipalmitoyl-sn-glycero-3-phosphoethanolamine; DSPC: distearoylphosphatidylcholine; DSPE-PEG: 1,2-distearoyl-sn-glycero-3-phosphoethanolamine-N-[methoxy(PEG)]; HEPG: hydrogenated egg phosphatidylcholine; HSPC: ematopoietic stem and progenitor cell; LNP: lipid nanoparticle; MC3: (6Z,9Z,28Z,31Z)-heptatriaconta-6,9,28,31-tetraen-19-yl 4-(dimethylamino)butanoate; NHS: N-hydroxysuccinimide; PC: phosphatidylcholine; PEG: polyethylene glycol; PEI: polyethylenimine; POPC: 1-palmitoyl-2-oleoyl-sn-glycero-3-phosphocholine; SS-OP: stearyl-stearyl-oleoyl-phosphatidylcholine; TCEP: tris(2-carboxyethyl) phosphine. *Postinsertion of lipid conjugates refers to a method where lipid-conjugated molecules are inserted into preformed LNPs after their initial formation.

Table 4. Ongoing Clinical Trials for Peptide Conjugates and Antibody–Oligonucleotide Conjugates^a

Investigational drug	Company	Target	Class	Disease indication	Phase
SRP-5051 (Vesleteplirsen)	Sarepta Therapeutics	Exon 51	Peptide conjugate	DMD	2
CONNECT1-EDOS1	PepGen	Exon 51	Peptide conjugate	DMD	2
CONNECT2-EDOS1	PepGen	Exon 51	Peptide conjugate	DMD	2
DYNE-251	Dyne Therapeutics	Exon 51	AOC	DMD	1/2
Del-zota™ (AOC 1044)	Avidity Biosciences	Exon 44	AOC	DMD	1/2
FREEDOM-DM1	PepGen	DMPK	Peptide conjugate	DM1	1
FREEDOM2-DM1	PepGen	DMPK	Peptide conjugate	DM1	2
ARO-DM1	Arrowhead	DMPK	Peptide conjugate	DM1	1/2 A
Del-Desiran™ (AOC 1001)	Avidity Biosciences	DMPK	AOC	DM1	3
DYNE-101	Dyne Therapeutics	DMPK	AOC	DM1	1/2
ARO-DUX4	Arrowhead	DUX4	Peptide conjugate	Facioscapulohumeral muscular dystrophy	1
Del-brax™ (AOC 1020)	Avidity Biosciences	DUX4	AOC	Facioscapulohumeral muscular dystrophy	1/2
DNL310 (ETV:IDS)	Denali Therapeutics	Iduronate 2-sulfatase	AOC	MPS II (Hunter Syndrome)	2/3
INCLINE-101 (TAC-001)	Tallac	TLR-9	CpG	Cancer	1/2

^aAOC: antibody–oligonucleotide conjugate; DMD: Duchenne muscular dystrophy; DM1: type 1 myotonic dystrophy.

choline (DSPC; phospho¹⁰ helper lipid).¹ LNPs are formed by mixing the lipids in ethanol and the nucleic acids in an acidic aqueous solution. Subsequently, ionizable cationic lipid MC3 interacts with negatively charged nucleic acids to form the core. There is an enhancement of polarity during the fabrication process, and the more polar lipids (PEG-lipid and DSPC) coat the surface of the LNPs.¹⁰⁴ PEG-DMG is important for LNP size control and stability.¹⁰² Notably, PEG-DMG's sheddable nature allows LNPs to bind to ApoE in the blood, which facilitates the delivery of LNPs to hepatocytes through the interaction of ApoE with the low-density lipoprotein receptor (LDLR).

LNPs have many benefits, such as large RNA cargo capacity, high encapsulation efficiency, redosability (safety), scalability, and ease of modification and functionalization.^{94,105} However, traditional LNPs tend to go to the liver for the following reasons: 1) the liver harbors a substantially larger blood volume and exhibits a slower flow rate compared to other organs, plus the fact that liver sinusoidal endothelial cells have fenestrations that allow LNPs to pass through; 2) ApoE-mediated delivery is via interactions with the LDLR, which is highly expressed by hepatocytes; and 3) the liver contains a high density of scavenger receptors, particularly on hepatic reticuloendothelial system cells, which can recognize and bind to the LNPs leading to preferential uptake by the liver.¹⁰⁶ To broaden the application of LNP/RNA technologies, it is important to detarget LNPs from the liver and retarget them to the desired cells and tissues.^{107,108} Tuning the administration route is a straightforward and effective approach to improve the biodistribution and therapeutic results of LNPs.^{95,109} For systemic administration, it is important to design and develop targeted LNPs that could circumvent the biological barriers during blood circulation and extravasation to target nonliver cells.^{95,110}

For the detargeting of LNPs from the liver, an overarching goal is to ensure LNPs evade detection by the immune system and their binding to APOE and/or other serum proteins that mediate nonspecific cellular uptake during blood circulation. Strategies for generating NPs with such attributes include cell membrane coating, protein modifications, and a nonfouling biomaterials layer.¹¹¹ However, these strategies need to be optimized so that there is no deleterious effect on cellular uptake and endosomal escape.^{112,113} Another approach for the detargeting of LNPs was recently introduced by Saunders et al. where the authors developed a “nanoprimer” that serves as a

pretreatment approach to transiently occupy liver cells, thereby facilitating the delivery of LNPs to other tissues.¹¹⁴ In a related approach, Ouyang et al. discovered the threshold dose of nanoprimers that is required to overwhelm Kupffer cell uptake rates.¹¹⁵

For the retargeting of LNPs, researchers have found that particle size, surface charge, and lipid composition play an important role in the biodistribution of nanoparticles after iv injection.^{116,117} Recently, a technology dubbed selective organ targeting (SORT) has been reported that incorporates SORT molecules with different charges, therefore precisely altering the biodistribution of the LNPs and further mediating nonliver tissue-specific delivery.^{118,119} Incorporating targeting ligands into LNPs has been effective in enhancing the targeting precision and efficacy. Several ligands (e.g., aptamers, sugars, antibodies and antibody fragments, vitamins, and aptamers) have been coupled on the LNP surface through various approaches, including direct conjugation of targeting ligands onto preformed LNPs and postinsertion of lipid conjugated molecules onto preformed LNPs, thus enabling targeted delivery. Antibodies and their various formats have been conjugated to LNPs, and the resultant conjugates have shown successful retargeting of LNPs to multiple extrahepatic cells and tissues, including immune cells, lymph nodes, bone marrow, cancer cells, and endothelial cells (Table 3).

N-Acetyl Galactosamine. Conjugation of chemically stabilized siRNA to trivalent GalNAc has resulted in the FDA approval of five therapeutics that are administered sc for various liver indications. Trivalent GalNAc is conjugated to the 3' end of the sense strand using a pH labile hydroxy proline linker. GalNAc allows targeting to the liver where it binds to ASGPRs expressed on hepatocytes¹³⁴ (note that the ASGPR is conserved across species), and there are ~1 million ASGPRs on liver cells that recycle every 15 min.³¹ While the endosomal escape rate is very low, efficient recycling of the ASGPR enables pharmacological levels of siRNA to be delivered to the cytoplasm within 3–4 days. Importantly, chemically stabilized siRNA is very stable in the low acidic pH of endosomes and forms a long-term depot.³⁰ In addition, the liver-to-kidney distribution ratio is greater than 30 at pharmacological doses. Although these GalNAc conjugates have a short plasma life due to rapid liver uptake, residence time in the liver is long.⁴¹ Further, the liver PK properties of siRNA conjugates are inversely correlated to their

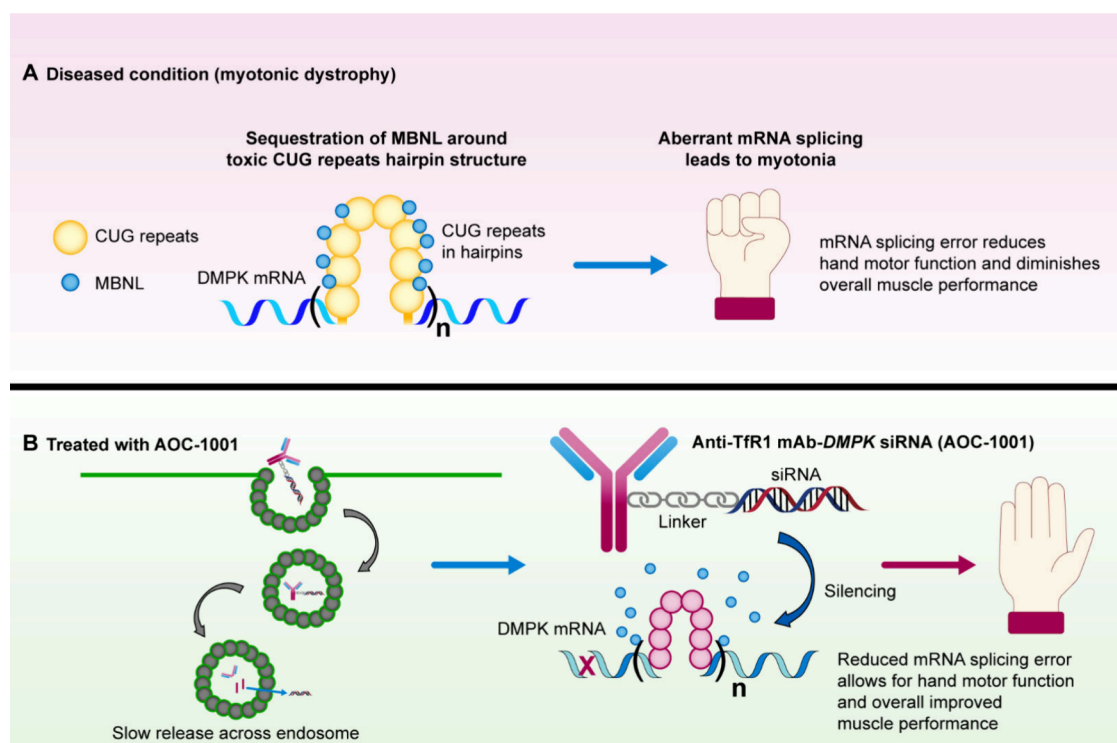


Figure 4. Proposed mechanism of action of Avidity Biosciences' investigational drug, AOC 1001, for myotonic dystrophy. (A) In patients with myotonic dystrophy, DMPK mRNA forms toxic (CUG) $_n$ repeats that result in the formation of a hairpin structure. Subsequently, MBLN—which is an essential splicing factor, is sequestered by these repeats, which leads to a splicing aberration. (B) AOC 1001 is generated by the conjugation of a TfR1 mAb to DMPK siRNA. Following systemic administration, AOC 1001 binds to TfR1 expressed on skeletal muscles. Further, AOC 1001 is internalized into endosomes, and the siRNA is released to cause DMPK silencing, which likely leads to the release of MBLN followed by the restoration of normal splicing machinery. In their phase 2 clinical trials, Avidity recently reported that this molecular correction (subserved by AOC 1001) is associated with an overall improved muscle performance. MBLN: muscle bind-like protein; TfR: transferrin receptor.

RISC loading.⁴¹ Therefore, the PD response of siRNA–GalNAc conjugates is dependent on their liver PK and RISC PK profiles, but not their plasma PK profiles.⁴¹ It is important to emphasize that five of the six approved siRNA therapeutics employ GalNAc as a targeting ligand for liver delivery. Given the evolution of enhanced metabolic stability of siRNAs, targeting of siRNAs to the liver via GalNAc is currently considered the preferred route compared with siRNA delivery with LNPs.

Antibodies. Antibodies convey a more targeted approach to the delivery of oligonucleotides to extrahepatic tissues. Oligonucleotides are conjugated to mAbs to generate antibody–oligonucleotide conjugates (AOCs) (Table 4). An AOC marries the specificity of the mAb with the precision therapy provided by oligonucleotides. Previous studies reported that antibody–siRNA conjugates can bind to their target antigens on the cell surface, become internalized, and mediate target mRNA silencing (Figure 4).^{135,136} Sugo et al. conducted the first study demonstrating antibody-mediated delivery of siRNA to muscular organs.¹³⁷ Specifically, they conjugated an anti-CD71 antibody fragment to siRNA targeting hypoxanthine-guanine phosphoribosyltransferase (HPRT) to generate anti-CD71 siHPRT.¹³⁷ Systemic administration of anti-CD71-siHPRT in mice led to robust and durable HPRT gene silencing in skeletal muscle and the heart.¹³⁷ Further, intramuscular injection of an anti-CD71 antibody fragment conjugated to siRNA targeting myostatin was administered in a mouse model of peripheral arterial disease, which led to a 72% reduction in MSTN mRNA levels and a subsequent increase of 17% in muscle mass.¹³⁷

AOCs may be generated via different conjugation chemistries including random or site-specific conjugation. Genentech has pioneered the thiomab platform (a mAb with an engineered cysteine), and their team has shown that conjugation of siRNA (via a maleimide handle) to thiomabs led to a homogeneous product.¹³⁸ Further, thiomab–siRNA conjugates were demonstrated to mediate modest target-dependent knockdown in vitro and within tumor cells in vivo.¹³⁹ Another conjugation platform that is widely used in the antibody–drug conjugate (ADC) field employs the native cysteine thiol maleimide Michael addition reaction, and 10 of the 13 FDA-approved ADCs are based on this chemistry.¹⁴⁰ Importantly, conjugation of an oligonucleotide with an antibody to generate an AOC has been found to improve the PK properties of the oligonucleotide.¹⁴¹

Many companies are using mAbs or their formats for the targeted delivery of oligonucleotides. Denali Therapeutics is developing an oligonucleotide transport vehicle (OTV) platform for antibody-mediated systemic delivery of ASOs to the CNS.¹⁴² Their team has engineered a human transferrin receptor 1 (TfR1) binding site into one Fc portion of the antibody, which contains nonbinding Fabs. The fused Fab portion has enhanced the PK properties of the entire molecule, which is termed an antibody transport vehicle. As an exemplar, they have demonstrated proof of concept using the MALAT1 ASO, which was conjugated to an antibody transport vehicle via noncleavable maleimide linker to obtain an OTV with an oligonucleotide-to-antibody ratio (OAR) of 1.¹⁴² The ASO is composed of a 3-10-3 locked nucleic acid DNA gapper with a fully modified phosphorothioate backbone.¹⁴² The OTV

displayed long-term stability (3 weeks) at 5 and 25 °C, and it was stable in plasma for 48 h at 37 °C.¹⁴² Systemically delivered OTV bound with a low monovalent affinity (100–300 nM) to the apical domain of TfR1 on brain endothelial cells. It was then transcytosed by TfR1, and the *MALAT1* ASO was eventually released into the brain parenchyma.¹⁴² Studies using SH-SY5Y cells revealed that both naked ASO and OTV traffics in a similar fashion (via the endolysosomal system).¹⁴² Additionally, the in vivo plasma PK properties of the OTV were significantly more favorable compared to the naked ASO. While the naked ASO exhibited a plasma half-life of only minutes, the OTV extended this duration to several hours.¹⁴² Remarkably, multiple systemic (iv) dosing of OTV (containing doses of 2.5 mg/kg of ASO; three doses were administered weekly for 3 weeks and the last one for 3 days) resulted in ASO accumulation and *MALAT1* knockdown (~50%) throughout the mouse CNS:¹⁴² the OTV conjugate was able to cross the BBB to knockdown *MALAT1* throughout the CNS.

Consistent with TfR1 expression in muscle, robust *MALAT1* knockdown was also observed in the quadriceps, heart, and diaphragm. Interestingly, the level of *MALAT1* knockdown observed in clearance organs (liver and kidney) was similar between OTV and naked ASO administration, supporting the notion that the OTV does not cause further knockdown in nontargeted tissues beyond what is achieved by administration of the naked ASO.¹⁴² Importantly, systemic administration (through multiple dosing) led to *MALAT1* knockdown across all the major CNS cell types and cortical layers of the brain.¹⁴² Furthermore, greater accumulation in brain ASO levels was observed in human TfR knock-in mice after multiple iv injections with the OTV vs a bivalent TfR:ASO (notably, the anti-TfR1 bivalent antibody had stronger receptor affinity).¹⁴² The bivalent TfR:ASO was localized to the vasculature whereas the OTV was also distributed to the brain parenchyma.¹⁴² Systemic delivery of the bivalent TfR:ASO in mice likely led to receptor dimerization, followed by internalization and degradation of the TfR protein in the vasculature.¹⁴² This contrasted with the systemic delivery of the OTV, which did not result in TfR degradation in the vasculature and parenchyma. In addition, IT administration of the naked ASO in NHPs enabled modest and uneven distribution of the ASO in the brain, whereas multiple iv treatment of NHPs with the OTV (each dose: 30 mg/kg) resulted in high and uniform distribution of the ASO across the CNS; this increased accumulation of the ASO in OTV-treated NHPs was accompanied by greater target gene silencing.¹⁴² Thus, the OTV platform provides a novel delivery method to cross the BBB and deliver ASOs to the brain.

Lipids. A growing body of studies suggests that lipids are emerging as robust ligands for enhanced biodistribution of oligonucleotides to tissues. In one study, it was reported that optimization of the hydrophobicity of targeting lipids led to siRNA being preferentially delivered to either the kidney or the liver.¹⁴³ The authors systemically injected mice with 20 mg/kg of different lipid-conjugated siRNAs and found that more hydrophobic lipids (e.g., cholesterol) distributed siRNA to the liver, while less hydrophobic lipids (e.g., DHA) had a greater propensity for delivering siRNA to the kidney.¹⁴³ The liver-to-kidney partitioning of siRNA by lipids may be explained by the fact that enhanced hydrophobicity of lipids leads to greater LDL-C binding, thereby resulting in greater uptake by the liver.¹⁴⁴

For local delivery to the CNS, lipids (as targeting ligands) have been reported to enhance siRNA retention and

accumulation in the brain and stabilized siRNA chemistry was shown to improve the durability of the response. For example, it was found that intrastriatal administration of a DHA–siRNA conjugate enabled broad tissue distribution and subsequent knockdown of *HTT* mRNA in the striatum of mice.¹⁴⁵ However, systemic administration (iv or sc) of a DHA–siRNA conjugate precluded successful delivery of siRNA to the brain, which was ascribed to unfavorable PK properties resulting from the first-pass metabolism via the liver and kidneys.^{145,146} In addition, transvascular delivery of DHA–siRNA conjugates revealed binding of DHA to endothelial cells of the brain, but this binding did not result in siRNA being transcytosed into the brain.¹⁴⁷ However, disruption by mannitol of the BBB led to transvascular delivery of the DHA–siRNA conjugate to the neurons and glia.¹⁴⁷ Further, mannitol-mediated transvascular delivery of the DHA–*HTT*–siRNA conjugate showed a strong relationship between PK and PD properties, as revealed by enhanced *HTT* mRNA knockdown in brain regions that had the highest siRNA accumulation.¹⁴⁷ In another study, introduction of a lipophilic moiety (C16) was shown to enhance siRNA uptake across the CNS following IT administration of a C16–siRNA conjugate in mice (discussed below). Prakash et al. demonstrated that attaching ASOs to fatty acids of varying lengths (16–22 carbons) enhanced their binding to plasma proteins, particularly albumin, which correlated with increased ASO activity in mouse muscle tissue.¹⁴⁸ Therefore, the addition of a lipophilic molecule to an siRNA or an ASO can significantly impact the PK/PD properties. Systemic delivery of siRNA to extrahepatic tissues has been achieved via conjugation of siRNA to docosanoic acid (DCA) or phosphatidyl choline (PC)-DCA.¹⁴³ A recent study reported that functionalizing DCA with a PC headgroup facilitated the delivery of chemically stabilized siRNA to many extrahepatic tissues, including the kidney, heart, lungs, muscle, pancreas, bladder, duodenum, and fat.¹⁴⁹ In addition, the authors showed that the biodistribution of siRNA to these tissues can be altered by either modifying the linker length between the phosphate and choline moiety of the PC group, or by increasing the PC headgroup valency.¹⁴⁹ This study highlights the impact of the PC headgroup in the delivery of siRNA to extrahepatic tissues. Taken together, these studies provide evidence that the conjugation of lipids to siRNA enhances the ability of siRNA to cross the lipid bilayer and enter cells. The mechanism likely involves binding of lipid-conjugated siRNA to serum proteins or cell surface proteins, which facilitates the cellular uptake of the conjugate via endocytosis.

Peptides. Peptides are emerging as robust delivery vehicles for the transport of various cargos, including oligonucleotides.^{150,151} (Table 4). For instance, cell-penetrating peptides (CPPs) facilitate the intracellular transport of cargo molecules that exhibit difficulty in crossing cellular membranes alone.¹⁵² CPPs are gaining popularity considering their potential for many therapeutic indications.¹⁵³ CPPs are generally polycationic peptides composed of positively charged amino acids, such as arginine, lysine, ornithine, and histidine. Upcoming tumor-targeting therapies find polycationic CPPs to be efficient since they hold a positive charge in physiological conditions, allowing them to bind to carbohydrate chains on the cell membrane and enter the cytoplasm of the cell.¹⁵⁴ Sarepta Therapeutics released data on their investigational drug, RC-1001, for the treatment of DMD, a fatal disease that progresses due to frameshift mutations in the *DMD* gene responsible for producing dystrophin protein.¹⁵⁵ RC-1001 is a CPP conjugated to the PMO backbone, with the hypothesis that the resultant conjugate promotes exon

23 skipping and enhanced production of functional but attenuated dystrophin protein. Following in vivo studies of RC-1001 in mdx mice, greater levels of exon skipping and enhanced functional dystrophin protein were observed in the CPP-conjugated PMOs compared with PMOs alone. Further, a greater durability effect was observed, with dystrophin production in muscle lasting for up to 60 days. In a recent study, Klein et al. conjugated a PMO (specific for *DMPK* mRNA) to an arginine rich CPP peptide, Pip6a, and found that systemic administration of the resultant conjugate significantly enhanced ASO delivery in striated muscles of DM1 mice compared with unconjugated PMO.¹⁵⁶ In another study, systemic delivery in mdx mice of a cyclic CPP–PMO conjugate, but not unconjugated PMO, was shown to exhibit increased skeletal muscle delivery coupled with enhanced exon skipping.¹⁵⁷ Although currently no FDA approved medicines are on the market due to insufficient in vivo data on the stability, toxicity, and endosomal escape capabilities of CPPs, recent clinical trials show promising results for future treatments.¹⁵⁸

Another class of peptide-based delivery systems, termed bicyclic peptides, has been introduced by Bicycle Therapeutics. Bicycle molecules are made up of short linear peptides with three points of reactivity via cysteine residues.¹⁵⁹ These peptides are constrained by a central chemical scaffold that forms a bicyclic structure containing 9–20 amino acids and two loops. The small size of bicycle molecules promotes rapid tissue penetration, and their constrained structure contributes to their high affinity and increased selectivity to the desired target. Since these peptides are synthetic, PK properties can be modulated by introducing various chemical modifications. An especially unique characteristic of bicycles is their ability to be conjugated together to form multimeric bicycles that can significantly improve the functionality and affinity for the target. Bicycle Therapeutics has explored versatile uses of these peptides and has most recently developed bicycle–toxin conjugates (BTCs) to allow for the delivery of toxin payloads to tumors. An important advantage of BTCs over conventional antibody conjugates is that bicycle molecules allow for precision-guided, rapid distribution and faster clearance, thereby reducing the exposure time of the payload in systemic circulation. Bicycle Therapeutics recently reported that their nectin-4-targeting BTC (BT8009) has entered phase 2/3 clinical trials for metastatic urothelial cancer.¹⁶⁰ While the payload used in this case was monomethyl auristatin E, oligonucleotides (siRNA or ASO) can also be potentially conjugated to bicycles via a linker to generate bicycle–oligonucleotide conjugates. In fact, Ionis Pharmaceuticals and Bicycle Therapeutics recently announced a partnership to explore ASO delivery using bicycles with a high affinity to the TfR1 receptor.

Arrowhead also uses peptides for the targeted delivery of siRNA to skeletal muscles. Their targeted RNA interference molecule (TRiM) platform for muscle delivery consists of a peptide-targeting ligand (linked to the 5' end of the sense strand) that is specific for α v β 6 integrin receptor, linker, plus a PK/PD modulator (linked to the 3' end of the sense strand).¹⁶¹ The PK/PD modulator consists of a lipid moiety that facilitates the delivery and cellular uptake. ARO–DM1 leverages this platform with an siRNA specific for *DMPK* mRNA.¹⁶¹ Systemic administration (iv) of a single dose of 15 mg/kg of ARO–DM1 showed much higher *DMPK* knockdown (~80%) in NHP triceps compared with a single 22 mg/kg dose of TfR–*DMPK*–siRNA conjugate.^{161,162} Proof-of-concept studies in DM1-like mice confirmed *DMPK1* silencing and attenuation in splicing

errors following systemic delivery of species-specific ARO–DM1.^{161,162} Arrowhead recently reported that the ARO–DM1–1001 conjugate has entered phase 1 clinical trials [NCT06138743] for type 1 myotonic dystrophy (DM1) (Table 4).¹⁶²

Endosomal Escape Enhancers. As discussed earlier, the entrapment of oligonucleotides into endosomes is a major therapeutic challenge for extrahepatic tissue delivery. The endosomal escape problem may be considered a silver lining for liver delivery since it enables the formation of a long-term depot, and siRNA molecules are slowly released over a period of several months. However, unlike ASGPRs that are expressed abundantly on hepatocytes and have a robust recycling mechanism, most receptors that are expressed on extrahepatic cells likely do not have such high levels of expression or similar recycling kinetics. Furthermore, receptor availability and trafficking may be cell-type-dependent and may limit the amounts of oligonucleotides delivered. Chloroquine (an endolytic agent) can enhance endosomal escape as it is membrane permeable and is protonated in the acidic pH of endosomes and is therefore retained in that location.¹⁶³ Subsequently, the hydrophobic motif of chloroquinone inserts into the endosomal bilayer, thereby disrupting the endosomes.¹⁶³ However, chloroquinone operates nonspecifically across all endosomes and can cause toxicity; it is therefore not a viable option for endosomal escape of oligonucleotides.¹⁶³ More recently, a synthetic lipid analog, SH-BC-893, was shown to increase delivery of oligonucleotides to the cytosol.¹⁶⁴ Compared to treatment of HeLa cells with ASO (3-10-3 cEt gapmer) alone, cotreatment with ASO and SH-BC-893 led to an almost 10-fold increase in the amount of ASO that escaped into the cytosol.¹⁶⁴ Enhanced accumulation of ASO in HeLa cells was associated with a significant increase (approximately 100-fold) in ASO activity, as evidenced by the knockdown of the ASO target, *MALAT1*.¹⁶⁴ This knockdown effect was observed to varying degrees across multiple cell lines.¹⁶⁴ Additionally, mice were systemically treated with an ASO targeting *Scnn1a* mRNA, which encodes a subunit of the ENaC sodium channel, a target for cystic fibrosis.¹⁶⁴ Compared to saline-treated mice receiving ASO alone, coadministration of SH-BC-893 improved *Scnn1a* knockdown from 17% to 59%. Additionally, the authors demonstrated that oral delivery of SH-BC-893 enhanced the activity of systemically delivered ASOs.¹⁶⁴ These data provide evidence that molecules such as SH-BC-893, which modulate cellular trafficking, may enhance oligonucleotide activity.

Sapreme Technologies has also developed a platform to enhance the release of oligonucleotides from endosomes.¹⁶⁵ Specifically, their team is developing an endosomal-specific escape enhance platform that consists of an endosomal escape module with a linker (dubbed SPT).¹⁶⁵ SPT molecules belong to a class of triterpenoids (saponins) that are endosomal selective and require endocytic machinery, cholesterol, and acidic pH to be active.¹⁶⁵ Sapreme's team screened for triterpenoid molecules in an in vitro system, and they identified several candidates with increased efficacy (for endosomal escape) and reduced toxicity.¹⁶⁵ SPT may be conjugated to various targeting ligands, and the targeted delivery may result in a lower effective dose requirement for SPT.¹⁶⁵ In their proof-of-concept experiments, they conjugated an oligonucleotide to GalNAc and SPT and showed that the resultant conjugate was highly effective in reducing ApoB expression in ASGPR1-expressing cells.¹⁶⁵ In addition, the silencing potency with GalNAc–ApoB siRNA–SPT conjugates was substantially

higher (by a factor of 100) compared with conjugates lacking SPT, indicating that SPT is fundamental to the enhancement of endosomal escape.¹⁶⁵ Systemic administration of 1 mg/kg of a trivalent GalNAc–SPT–ApoB siRNA conjugate displayed greater than 75% reduction (compared with a conjugate lacking SPT) in *APOB* gene expression 24 h postdosing.¹⁶⁵ These data suggest that Sapreme's technology in leveraging endosomal escape enhancement may potentially improve the efficacy of siRNAs (or ASOs) within extrahepatic tissues where endosomal escape is a major rate-limiting step.

■ PRECLINICAL AND CLINICAL STUDIES

Muscle Indications. In preclinical studies, Avidity Biosciences demonstrated that a single systemic dose of an antibody–siRNA conjugate in animals resulted in >75% mRNA knockdown in skeletal and cardiac muscles.¹⁶⁶ Avidity Biosciences is currently pursuing clinical trials to evaluate the potential of an antibody–siRNA conjugate for the treatment of myotonic dystrophy (Marina trial: NCT05027269).¹⁶⁶ Avidity's investigational drug AOC 1001 consists of a humanized anti-TfR1 mAb conjugated to DMPK siRNA (via a noncleavable linker), and the OAR is 1 (Figure 4).¹⁶⁷ DM1 is a commonly inherited rare multisystem disorder that predominantly affects smooth and skeletal muscles but may also have an adverse effect on the patient's eyes, heart, endocrine system, and CNS.¹⁶⁸ One of the typical manifestations of this disease is myotonia, which is characterized by difficulty opening and closing the hand. The prevalence of myotonic dystrophy is estimated to be 1 in 20,000 people,¹⁶⁹ and it is caused by a toxic gain-of-function mutation in *DMPK* mRNA that results in (CUG)*n* repeats. Patients with DM1 have CUG repeats ranging from the hundreds to the thousands, and this correlates with the severity of the disease (Figure 4). These repeats form hairpin structures that bind and sequester muscle blind-like protein (MBNL), which is essential for splicing (Figure 4). Sequestration of MBNL causes aberrant splicing of multiple mRNAs in myotubes, thus driving the muscular disease condition in patients with DM1. In NHPs, Avidity's team estimated that their siRNA for *DMPK* has an EC₅₀ of 100 pM, suggestive of a robust potency of siRNA at low doses. Drugs in phase 1/2 clinical studies from Avidity displayed favorable safety and tolerability profiles in patients with DM1.¹⁷⁰ Furthermore, administration of AOC 1001 in patients resulted in a dose-dependent increase of siRNA in muscles, which ultimately led to a mean reduction of 42% of *DMPK* mRNA (after 6 weeks) following a single dose of 1 mg/kg or two doses of 2 mg/kg or 4 mg/kg.¹⁷⁰ It has been speculated that AOC 1001 operates in the nucleus, which may be supported by recent results from the Corey laboratory that showed the presence of Argonaut 2 and RISC factors in the cell nuclei.⁷ In addition, Avidity's data revealed that *DMPK* knockdown is associated with normalization of splicing¹⁷¹ and subsequent enhancement of clinical activity.¹⁷⁰ Remarkably, multiple dosing of 4 mg/kg of AOC 1001 led to improvements in myotonia and total muscle strength that translated into coordinated functioning of muscles in patients with DM1.¹⁷⁰ Because of these promising clinical data, AOC 1001 has advanced to phase 3 clinical trials.

PepGen has developed an enhanced delivery oligonucleotide (EDO) peptide platform for in vivo delivery of ASOs in patients with DMD. These linear cell-penetrating peptides are short in length (<20 amino acids).^{172,173} Additionally, these non-immunogenic peptides feature two flanking polyarginine domains interspersed with non-natural amino acids to enhance stability.^{172,173} Furthermore, a hydrophobic core exists between

the arginine-rich (cationic) domains, likely aiding in endosomal escape and nuclear delivery.^{172,173} They also established proof of concept using enhanced nuclear delivery of PMO23 (an ASO targeting exon 23 of dystrophin) in mice.^{173,174} More specifically, in mice myotubes they showed a 25-fold increase in nuclear delivery of PMO with EDO-conjugated PMO23 vs naked PMO23.¹⁷³ Note that PGN–PMO23 is a murine analogue of the clinical candidate PGN–PMO51. In addition, iv delivery of two doses of 30 mg/kg (day 1 and day 15) of EDO-conjugated PMO51 in NHPs showed 72% ASO distribution in the nucleus of skeletal muscle (quadriceps) cells.^{173,174} A single dose of 10 mg/kg of EDO-conjugated PMO23 resulted in substantial exon skipping activity in the biceps and quadriceps of mice.^{173,174} Interestingly, multiple systemic dosing (Q4W at 30 mg/kg) of EDO-PMO23 in an mdx mice model (mice with a single-point mutation in exon 23 of the dystrophin gene) led to enhanced exon skipping (~90%) and increased dystrophin protein expression (up to 80% of normal), which was uniformly distributed across muscle.¹⁷³ Significantly, three doses (Q2W) at 30 mg/kg of the clinical candidate (PGN-EDO51) in NHPs resulted in a >70% increase of exon-skipping activity across different skeletal muscle tissues, including the diaphragm.^{173,174} In a phase 1 clinical trial (NCT06079736), healthy volunteers were administered single ascending iv doses (5, 10, and 15 mg/kg) of PGN-EDO51. There was a dose-dependent increase (albeit a modest one) in exon 51 skipping in muscle biopsies; this increase was sustained over a period of 28 days.^{175,176} Further, the increase in exon skipping correlated with enhanced accumulation of the drug in the biceps.^{175,176} The safety and tolerability profile of PGN-EDO51 was favorable, and only mild-to-moderate adverse events were observed.^{175,176} Based on the data from preclinical and phase 1 clinical studies, their phase 2 studies will involve four iv multiple ascending doses (Q4W) in a DMD population responsive to exon 51 skipping.^{175,176}

Dyne recently reported clinical data on their investigational drug, DYNE-251. Dyne's technology is based on the FORCE platform, which consists of three components: a Fab that targets TfR1; a cleavable linker (valine-citrulline); and an exon-skipping PMO.^{167,177} The oligonucleotide-to-antibody ratio is 2. DYNE-251 uses a PMO that modulates exon 51 skipping of dystrophin mRNA, with the aim of generating shorter but functional dystrophin protein.¹⁷⁷ Since TfR1 is expressed on muscle cells, DYNE-251 is designed to provide targeted delivery of PMO to cardiac and skeletal muscles, including the diaphragm.^{177–180} Their phase 1/2 clinical trial (termed DELIVER; NCT05524883) enrolled male patients aged 4–16 years who were amenable to exon 51 skipping.¹⁸¹ The DELIVER trial was initiated with multiple ascending doses (Q4W for 6 months) of DYNE-251 or placebo via iv delivery.¹⁸¹ The safety profile of DYNE-251 was generally favorable, with mild-to-moderate TEAEs with doses ranging from 0.7 mg/kg to 20 mg/kg.¹⁸² Impressively, patients treated monthly for 6 months with DYNE-251 (5 mg/kg of PMO) showed an increase in dystrophin levels (from 0.6% to 0.88% of normal) compared with no meaningful dystrophin protein increase in patients treated with placebo.¹⁸² These results were superior to the data shown with eteplirsén, where 30 mg/kg QW treatment for 6 months only led to a modest increase in dystrophin protein levels (from 0.24% to 0.30% of normal).¹⁸³ In addition, there was a significantly higher increase in the percentage of dystrophin fibers in patients treated with DYNE-251 (5 mg/kg; Q4W) vs eteplirsén (30 mg/kg; QW) over a period of 6 months.¹⁸³ In their more recently released clinical data, Dyne

showed that treatment of patients with 20 mg/kg PMO equivalent of Dyne-251 (Q4W for 6 months) led to mean absolute dystrophin expression of 3.71% of normal, which is 10-fold higher than current standard of care (eteplirsen).¹⁸⁴ Across both the 20 mg/kg and 10 mg/kg PMO equivalent DYNE-251 treated patients, there were notable functional improvements over time.¹⁸⁴ While the safety data from 10 mg/kg and 20 mg/kg Dyne-251 was favorable, administration with 40 mg/kg Q4W of Dyne-251 showed three serious TEAEs in two patients likely resulting from study drug.¹⁸⁴ Therefore, Dyne is progressing with dose escalation of up to a 20 mg/kg dosage for all cohorts.

Targeting of oligonucleotides solely to skeletal muscle is garnering attention, especially for conditions such as fascioscapulohumeral muscular dystrophy (FSHD), a condition that primarily affects the face, shoulders, and upper arms. Recently, Avidity Biosciences released interim phase 1/2 data for their investigational drug, AOC 1020, for FSHD. AOC 1020, also termed *del-brax*, is an antibody–oligonucleotide conjugate, where the antibody part targets TfR1 and the conjugated siRNA targets *DUX4*, a transcription factor.^{185,186} *del-brax* is administered systemically (iv), and it efficiently delivers *DUX4* siRNA to muscle.¹⁸⁵ Since *DUX4* is expressed at very low levels in muscle cells, the read-out for *DUX4* silencing is indirectly assessed by measuring the levels of *DUX4*-regulated gene expression. Patients treated with 2 mg/kg *del-brax* displayed a > 50% reduction in *DUX4*-regulated genes as measured using patient muscle biopsies.¹⁸⁵ Furthermore, *del-brax* treatment led to a significant decrease in the *DUX4*-regulated novel circulating biomarker (undisclosed) and in the levels of creatinine kinase for up to 4 months.¹⁸⁵ Notably, there were improvements in patients' range of motion and function, and the overall safety and tolerability profiles were also favorable.

These encouraging results demonstrate the potential for this first-in-class *DUX4* silencing drug in the treatment of FSHD, and in general, broadly support the targeted delivery of therapeutic oligonucleotides to muscles.

CNS Indications. In the CNS space, there is a major therapeutic need for many diseases, including Parkinson's disease, Alzheimer's disease, Huntington's disease, and other genetic-based neurodegenerative diseases. However, the biggest challenge by far in the research and development for CNS therapeutics is overcoming the BBB. Alnylam recently reported successful local delivery of siRNA to the CNS.¹⁸⁷ Their team conjugated a fully stabilized *SOD1* siRNA (containing 5' vinyl phosphonate) to 2'-O-hexadecyl (C16) and showed that a single IT injection of the subsequent conjugate in rats resulted in broad biodistribution of siRNA throughout the CNS.¹⁸⁷ The authors used the same conjugation platform to generate ALN-APP, which is an siRNA–C16 conjugate targeting APP mRNA.¹⁸⁷ APP is a membrane-bound protein whose proteolytic cleavage generates many fragments, including amyloid beta ($A\beta$) peptide, which enters the brain parenchyma or microvasculature, thereby leading to pathologies associated with Alzheimer's disease or cerebral amyloid angiopathy, respectively.¹⁸⁸ In a Tg-hAPPsw-DI/mNos2–/– (CVN) model of disease, the Alnylam team found that a single icv dose (120 mg) of ALN-APP resulted in 75% knockdown of APP mRNA within the cortex 30 days postdose.¹⁸⁷ Importantly, there was a significant reduction in the number of $A\beta$ 40 plaques in the parenchyma of whole brain, hippocampus, and cortex after 3 or 6 months postdosage.¹⁸⁷ Further, Alnylam recently demonstrated in NHPs that a single IT dose of 60 mg led to a ~ 50% reduction in protein knockdown of sAPP α and sAPP β (biomarkers for Alzheimer's

disease) in the CSF after 6 months, which supports the long durability of APP knockdown with ALN-APP. Alnylam's phase 1 clinical study (NCT05231785) showed that ALN-APP had a favorable safety profile at doses of 25 mg, 50 mg, and 75 mg in patients with early onset Alzheimer's disease.^{189,190} Remarkably, a robust reduction (>50% mean knockdown up to 6 months) of sAPP α and sAPP β was observed with the 75 mg dose.^{189,190} It is important to emphasize that these profound findings are the first to be reported for CNS conditions using siRNA therapeutics.

Another company, Vect-Horus, developed targeted TfR1 VHH ligands (~14 kDa) that deliver siRNA to the CNS. VHH are variable-region domains of heavy-chain antibodies found in *Camelidae*, and these VHH domains have deep tissue penetration properties.¹⁹¹ In addition, their team has created a PK-enhanced conjugate, which is made up of the VHH domain that is fused to the Fc domain of an IgG1 antibody.¹⁹² VHH–siRNA conjugates were generated through a copper-free click reaction, with a site-specific azido-linker on the VHH molecule and a complementary constrained alkyne group on the siRNA.¹⁹² Further, Vect-Horus has developed novel VHH ligands with varying affinities targeting TfR1 in rodents and humans. Following local icv administration of TfR-binding VHH–siRNA conjugates in WT mice, robust knockdown (80–90%) was observed in the CNS and spine, indicating retention and uptake of TfR-binding conjugates in the brain parenchymal cells and the spinal cord.¹⁹² Treatment of cultured murine neuroblastoma cells with TfR-binding VHH–siSOD1m-5'VP and VHH–hFc-siSOD1m-5'VP conjugates showed potent and dose-dependent silencing, with mean IC₅₀ values of 7.4 nM and 8.7 nM, respectively.¹⁹² In contrast, the incubation of cells with siSOD1m-5'VP alone showed no effect on mSOD1 mRNA levels. As such, PK-enhanced VHH-conjugated siRNA binds efficiently to the TfR and is endocytosed, followed by endosomal escape and delivery of released siRNA to the cytoplasm or nucleus to mediate gene silencing.¹⁹² In general, these data suggest that VHH domains can potentially be optimized to mediate the systemic delivery of oligonucleotides to the CNS.

Recently, a branched siRNA modality termed divalent siRNA (di-siRNA) was introduced, which consists of two monovalent chemically stabilized siRNAs connected via a triethylene glycol linker at the 3' end of the sense strands. Local administration (icv) in mice of 475 μ g of di-siRNA was shown to elicit robust and durable knockdown (up to 6 months) of *HTT* mRNA in multiple regions of the brain.¹⁹³ A similar knockdown pattern was observed with a separate target (*APOE* mRNA) in mice after icv injection with 238 μ g of di-siRNA scaffold.¹⁹³ To translate these findings from mice to larger mammals, the authors employed various local administration methods for delivery of di-siRNA to the sheep CNS and found that there was an efficient distribution of di-siRNA throughout the brain including the cortex, caudate, and putamen.¹⁹⁴ In addition, icv administration of di-siRNA displayed broad distribution of this branched siRNA within the spinal cord of sheep.¹⁹⁴ Consistent with these findings, the authors also demonstrated effective biodistribution in the CNS and spinal cord of cynomolgous macaques after icv administration of di-siRNA (25 mg), and the increase in di-siRNA accumulation in the brain and spinal cord resulted in potent silencing of *HTT* for up to 1 month,¹⁹³ showing that there was a strong correlation between PK and PD properties. Importantly, the safety profile of di-siRNA was favorable in NHPs, showing no significant increase in the level of neuroinflammatory markers. Atalanta Therapeutics, a spin-off

company based on this technology, is further advancing the di-siRNA platform.

Ionis has two FDA-approved ASO drugs for CNS indications: Spinraza (nusinersen) and Qalsody (toferson; Table 2). Nusinersen was granted FDA approval in 2016 for spinal muscular dystrophy, which is caused by a loss-of-function mutation of the *SMN1* gene.¹⁹⁵ Nusinersen operates via splice modulation of the *SMN2* gene (a closely related ortholog of the *SMN1* gene but with a point mutation in intron 7) and leads to functional survival motor neuron protein with exon 7 inclusion.¹⁹⁵ Notably, nusinersen belongs to the class of splice-switching ASOs, and contains a sequence of RNA oligonucleotides (with a PS backbone) modified with 2'MOE.¹ The approval of Spinraza was regarded as a significant scientific breakthrough, as it significantly alleviated the suffering of children with spinal muscular atrophy. IT administration of nusinersen resulted in widespread distribution of the drug throughout the CNS.¹⁹⁶

Pulmonary Indications. Hariharan and colleagues recently reported that a di-siRNA molecule can be delivered locally to the lungs in a mouse model of SARS-CoV-2.¹⁹⁷ The authors chemically stabilized the di-siRNA molecule, which targeted the mRNA sequence found in various variants of SARS-CoV-2 (including delta and omicron), and found that the di-siRNA molecule mediated degradation of viral mRNA inside the cells.¹⁹⁷ Subsequently, they showed that viral protein translation was attenuated in animal models by 60–80%, thereby preventing infection.¹⁹⁷ This study represents the first example of intranasal delivery of multimeric siRNA in an animal model of SARS-CoV-2 that is well-tolerated and induces robust viral gene silencing.¹⁶²

Arrowhead is using its TRiM platform to deliver siRNA to the pulmonary epithelium. Their molecule is dubbed “ARO-RAGE”, which is a conjugate of chemically stabilized siRNA (specific for the receptor for advanced glycation end products [RAGE] mRNA transcript) and a targeting small molecule (specific for integrin α -v β -6 expressed on pulmonary epithelial cells).^{178,198} RAGE is an upstream mediator of the Type 2 inflammatory pathway and is abundantly expressed in the lung epithelium compared to other tissues.^{199,200} Further, RAGE can be cleaved to generate soluble RAGE (sRAGE), which is secreted in the serum and airways. In a rat model, their team demonstrated robust knockdown (~90%) of lung RAGE mRNA following a single inhalation of 0.5 mg/kg of a rat-specific RAGE–siRNA conjugate.²⁰¹ It is noteworthy that the reduction in RAGE mRNA expression was sustained for up to 2 months in rats.²⁰¹ Furthermore, pretreatment in a rat asthma model with a RAGE-directed siRNA conjugate led to reduced pulmonary inflammation, characterized by decreased recruitment of eosinophils, neutrophils, and inflammatory cytokines.²⁰¹ Importantly, there was robust reduction of RAGE mRNA expression (~90%) in the lungs of NHPs that inhaled a single dose of 1 mg/kg of ARO-RAGE vs saline.²⁰¹ In addition, weekly sc dosing led to potent and durable lung RAGE and sRAGE silencing in both rats and NHPs.²⁰² Arrowhead has recently initiated a phase 1/2 clinical study (NCT05276570) with ARO-RAGE in individuals with and without asthma.²⁰³ The study design included single or multiple ascending doses in healthy volunteers and multiple ascending doses in patients with mild-to-moderate asthma.²⁰³ No serious drug-associated adverse effects were seen with the different cohorts.²⁰³ In healthy volunteers, bronchoalveolar lavage fluid sRAGE decreased by ~94% and serum sRAGE decreased by ~88% at 4 weeks

following two doses of 184 mg of ARO-RAGE; notably, similar effects on serum sRAGE reduction were observed in patients with asthma that were administered with the same dose regimen.²⁰³ It should be noted that this is the first-in-human clinical study with a potential siRNA therapeutic for a pulmonary indication.

Skin Indications. A growing body of evidence suggests that Janus kinase 1 (JAK1) is involved in multiple inflammatory diseases. Recent studies in human skin explants demonstrated that intradermal administration of a DCA conjugate of JAK1-specific (chemically stabilized) siRNA resulted in knockdown of JAK1 mRNA transcript, and this reduction in JAK1 expression was associated with attenuation of interferon γ -induced JAK1-mediated upregulation of the chemokines CXL9, CXL10, and CXL11.²⁰⁴ Further, the authors found that a single sc dose of 20 mg/kg of their lead JAK1 siRNA (si3033) molecule in mice led to a ~50% reduction in JAK1 mRNA transcript in tail skin following 1 week of administration, and the reduction in JAK1 expression was observed (to a lesser degree) for up to 5 weeks.²⁰⁴ Remarkably, in a mouse model of vitiligo, prophylactic intradermal administration of two 0.08 mg/kg doses of si3033 had an ameliorative effect, as demonstrated by decreased skin infiltration of melanocyte-specific premelanosome protein CD8⁺ T cells and reduced depigmentation of the skin.

Cardiovascular Indications. Accumulating studies highlight the applicability of siRNA therapeutics for cardiovascular indications. An example is the FDA-approved drug inclisiran, which was discovered by Alnylam and developed by Novartis to treat hypercholesterolemia.²⁰⁵ Inclisiran is a chemically stabilized siRNA that is conjugated to trivalent GalNAc and is used to target proprotein convertase subtilisin/kexin type 9 (PCSK9) mRNA.²⁰⁵ PCSK9 degrades the LDLR, and silencing of the PCSK9 gene by inclisiran can, therefore, lead to a reduction in LDL-C levels. Indeed, phase 3 clinical data revealed that systemic administration (iv) of inclisiran every 6 months led to a ~50% reduction in LDL-C levels, thereby demonstrating a potent and durable effect of this drug in treating hypercholesterolemia.²⁰⁶ In the phase 3 clinical trials of inclisiran, approximately 3,200 patients were treated. These trials reported mild adverse events at the injection site in some inclisiran-treated patients over an 18-month period.²⁰⁶ The approval of inclisiran for hypercholesterolemia was a significant milestone, as oligonucleotides were previously considered as therapeutics for only rare diseases. Another therapeutic oligonucleotide, mipomersen, a PS 2'MOE ASO that degrades the mRNA for apoB-100, was initially approved for the treatment of homozygous familial hypercholesterolemia (HoFH).²⁰⁷ LDL-C has a core apolipoprotein, apoB-100, which is produced in hepatocytes and binds to the LDLR. Depletion of apoB-100 mRNA by mipomersen led to reductions in LDL-C levels in patients with HoFH.²⁰⁷ However, this drug was discontinued due to significant liver toxicity.²⁰⁸ Nonetheless, numerous clinical trials are currently underway for ASO therapeutics in the cardiovascular space.⁵⁴ While oligonucleotides are mostly delivered to liver targets for cardiovascular indications, recent studies support the role of oligonucleotides in directly targeting vascular tissues. Comanche Biopharma just reported FDA clearance of its siRNA-based investigation drug (CBP-4888) application for the treatment of preeclampsia.²⁰⁹ CBP-4888 is formulated as a combination of two chemically synthesized, lipid-conjugated siRNA duplex oligonucleotides that target two soluble fms-like tyrosine kinase-1-like mRNA isoforms.²⁰⁹ In preeclampsia, sFLT1 is excessively produced by the placenta and

is associated with pathological conditions of this severe vascular disease. Systemic administration (sc) of CBP-488 inhibited *sFLT1* expression in the placenta and may therefore serve as a potential therapeutic for treatment of preeclampsia.

Adipose Indications. Arrowhead Pharmaceuticals recently released data on their obesity programs, ARO-INHBE and ARO-ALK7.²¹⁰ Inhibin subunit beta E (INHBE) is expressed in hepatocytes and produces activin E, a dimeric INHBE protein.²¹¹ Activin E is secreted by hepatocytes and binds to the ALK7 (activin receptor-like kinase 7) receptor, a TGF- β superfamily member, which is expressed on adipose tissue.²¹² This binding ultimately suppresses lipolysis and promotes fat storage. ARO-INHBE consists of a GalNAc-conjugated siRNA to silence INHBE in the liver.²¹⁰ Consequently, systemic delivery of ARO-INHBE should result in inhibition of the production of activin E. In preclinical studies, the Arrowhead team showed that treatment of mice (on a high-fat diet) with a species specific version of ARO-INHBE resulted in ~95% suppression of INHBE mRNA and 19% suppression in body weight gain relative to animals that were treated with saline.²¹⁰ Importantly, INHBE silencing reduced fat mass but preserved lean muscle mass in a diet-induced mouse obesity model.²¹⁰ It should be noted that INHBE siRNA targets the liver but affects fat storage in adipose tissues.²¹³ In contrast, ALK7 is preferentially expressed on adipose tissue, and ARO-ALK7 targets ALK7 mRNA. ARO-ALK7 is a dual-lipid siRNA conjugate, where a lipid is linked to the 3' and 5' ends of the sense strand within the siRNA targeting ALK7.²¹⁰ Weekly 3 mg/kg systemic dosing (sc) of mice with ALK7 siRNA resulted in an 80% mRNA reduction in inguinal white adipose tissue and a 40% mRNA reduction in perigonadal white adipose tissue depots, and these findings correlated with a 37% decreased body weight gain in ALK7 siRNA vs saline-treated mice.²¹⁰ Further, ALK7 siRNA treatment reduced fat mass by 50% but not lean muscle mass in a diet-induced model of obesity.²¹⁰ A single sc administration of 3 mg/kg of ARO-ALK7 in NHPs caused robust and durable ALK7 mRNA knockdown in adipose tissue.²¹⁰ Of note, phase 1/2a clinical trials are being initiated separately for both ARO-INHBE and ARO-ALK7 in volunteers with obesity.²¹⁰

Eye Indications. Several studies are currently underway to optimize the delivery and formulation of oligonucleotides for eye diseases. Cheng et al. recently unveiled tetra siRNA and reported that a single ivt administration in mice of tetra siRNA (targeting Huntington mRNA) resulted in robust (~75% protein knockdown) and durable (lasting up to 6 months) gene silencing in photoreceptors.²¹⁴ The authors translated these findings in pigs and found that a single ivt dose of 300 μ g led to a reduction in Huntington protein of ~80%, and the reduction was maintained over a period of 4 months.²¹⁴ As previously discussed, Alnylam used a C16-siRNA conjugate for the delivery of siRNA to the CNS. In the same publication, they also showed that the C16-siRNA conjugate had a functional effect in the eye, demonstrating that a single ivt administration of *TTR*-targeting C16-siRNA with vinyl phosphonate in NHPs led to potent and durable *TTR* knockdown in the eye.¹⁸⁷ Another method for delivering siRNA into the eye is topically, which is less invasive compared with ivt delivery. Many companies are investigating topical administration for eye indications. Notably, Amber Ophthalmics has achieved success with their clinical investigative drug, Nexagon (iufepirsén), an unmodified ASO formulated into a gel.²¹⁵ This ASO is targeted

to connexin-43 and is currently being used in phase 2 clinical trials for the treatment of persistent corneal epithelial defect.²¹⁵

■ CHALLENGES AND PERSPECTIVES

We have provided a comprehensive review delineating the progress made in the design and delivery of oligonucleotides, and we have presented examples from preclinical and clinical studies that exemplify the therapeutic potential of these medicines. However, there remain challenges in the field that must be addressed.

Route of Administration for Oligonucleotides. While systemic administration is employed for hepatic indications, local administration is often the preferred route for delivery of oligonucleotides to other tissues. In this context, conjugation of siRNA to target ligands appears to be a commonly applied approach for the delivery of siRNA. However, ASOs are single strand vs the duplex for siRNAs (less overall charge), and the greater use of phosphorothioates makes them more hydrophobic (and prone to protein-binding) than siRNAs; both these features contribute to the cell permeability that ASOs possess. Therefore ASOs, unlike siRNAs, do not have a strict requirement for a ligand for cell permeability. The focus of future oligonucleotide research is to target extrahepatic tissues via systemic administration while minimizing biodistribution to clearance organs (liver and kidney) and muscles. Single vs multiple dose regimens may also influence the PK/PD response of oligonucleotides. Furthermore, oral delivery of oligonucleotides remains a challenge since these molecules are sparingly absorbed, and a recent study revealed that low intestinal permeability is a major hurdle in the delivery of GalNAc-siRNA conjugates.²¹⁶ Nonetheless, oral dosing may be beneficial for GI indications such as Crohn's disease. This would require a complex enteric coating and an optimized formulation approach to achieve oligonucleotide stability in the extreme acidic pH of the stomach.

Criteria for Selection of ASOs vs siRNA as a Therapeutic Modality. The choice of therapeutic oligonucleotide, siRNA or ASO, may be an important consideration and may depend on where in the cell the mRNA being targeted is located. ASO may be used for nuclear mRNA targets, whereas siRNA may provide a more durable effect for cytoplasmic mRNA targets. Further, the size of the targeted delivery vehicles (mAbs (or their various formats) vs peptides) may play an important role in modulating the PK properties and toxicology profile of oligonucleotide conjugates following systemic administration.

The Potential Role of Receptor Availability and Trafficking for Delivering Oligonucleotides. Identifying high copy number receptors with favorable intracellular trafficking is crucial for achieving an optimal therapeutic index of oligonucleotides in extrahepatic tissues. This approach enables the administration of more potent molecules at significantly lower doses, thereby enhancing the overall likelihood of reaching the optimal pharmacological dose in extrahepatic tissues.

Bypassing the BBB for CNS Indications. Apart from the liver, most preclinical and clinical efforts in the oligonucleotide field are currently focused on delivering oligonucleotides systemically to muscle or CNS tissues. However, targeted delivery will be key for systemic coverage, especially for CNS indications, and we know that the clinical success of nusinersen for spinal muscular atrophy required IT (local) delivery to bypass the BBB. It will be a *tour de force* in the CNS space to

develop oligonucleotide therapeutics that can be administered systemically while overcoming the BBB and displaying favorable PK properties and efficacy. The TfR1 receptor expressed on endothelial cells of the brain is mostly targeted to systemically deliver oligonucleotides to the CNS. TfR1 receptors are abundantly expressed on brain endothelial cells but are not found on liver hepatocytes, and therefore, TfR1-oligo conjugates can circumvent the first-pass metabolism effect.

Influence of OAR and Receptor Affinity on the Therapeutic Potential of AOCs. Ongoing clinical studies indicate that conjugating oligonucleotides to antibodies (and their various formats) to create AOCs enhances the targeted delivery of oligonucleotides to extrahepatic tissues.¹⁶⁷ In addition, an OAR of 1 for AOCs appears to convey favorable PK properties and enhanced efficacy in vivo.²¹⁷ Higher OARs may result in decreased PK, likely due to clearance through scavenger receptors expressed on the liver. Optimization of the antibody format, site, linker, conjugation chemistry, and receptor affinity will need to be carefully examined to achieve a favorable therapeutic index for an AOC. Notably, in the CNS space, a consensus is emerging that lower receptor affinity (nanomolar range) may be advantageous to facilitate transcytosis of an AOC across the BBB. AOCs can apply some learnings from the well-established field of ADCs,²¹⁸ with a few caveats. The main difference between an AOC and a classic ADC is the physiochemical property of the payload. An ADC generally contains a small-sized (<1 kDa) and hydrophobic payload whereas an AOC contains a larger sized (>14 kDa) and highly polar payload. Therefore, the methods for analytical characterizations for an AOC would differ from those employed by an ADC, and multiple orthogonal methods would need to be developed and compared to obtain accurate OAR and AOC concentrations.

Antibody Engineering Approaches to Diminish FcγR Effector Function. Engineering of an antibody to generate Fc variants that inhibit FcγR, but not FcRn, binding may be pursued to prevent the induction of an unwarranted immune response. Selecting the right IgG isotype may also be an important consideration in the design of an AOC. The IgG4 isotype has a lower binding affinity to most FcγRs compared to IgG1, resulting in a reduced likelihood of activating immune responses.²¹⁹ In addition, mAbs can be deglycosylated or engineered (e.g., N297A mutation) to decrease their affinity for FcγRs.²²⁰ Another commonly applied engineering approach to silence the Fc effector function is through the introduction of the LALA (L234A/L235A) mutation for the IgG1 subclass. However, it is worth pointing out that these alterations do not completely ablate the Fc effector function. In addition, it is crucial for mAbs to retain FcRn binding since this interaction sequesters the mAb in the acidic pH of endosomes, thereby preventing catabolism of the antibody in the lysosomes.²²¹ Subsequently, the antibody is recycled back to the surface of the cell and into circulation, which ultimately results in an increase in its half-life.²²¹ To further increase the PK property of mAbs, introduction of a triple substitution M252Y/S254T/T256E (YTE) into the Fc region has been proposed to enhance the affinity of mAbs for FcRn at pH 6.0.²¹⁹ Taken together, the aim of these antibody-engineering approaches for oligonucleotide delivery is to attenuate the binding of mAb to FcγR without adversely affecting FcRn binding.

Improvements in Oligonucleotide Delivery via LNPs. Delivery of oligonucleotides via LNPs also requires further refinement. Due to the relatively large size of LNPs (>50 nm)

and the lack of a transcytosis pathway for overcoming the endothelial barrier, the targeting sites for LNPs are mainly constrained to the intravascular compartments and lymphatic systems, extravascular compartments with leaky vascular structures, or cells within the permeation range of LNPs that are close to the administration site.^{111,222–225} Therefore, some extravasation and transcytosis mechanisms need to be introduced to the LNP delivery system to broaden its application.^{226–229} In addition, due to their physiochemical properties, LNPs tend to be taken up by the liver,^{108,230} meaning there is a need for effective detargeting strategies.^{107,231,232} Detargeting is characterized as the inhibition of LNP delivery to the liver, whereas retargeting is defined as the furthering of LNP delivery to the desired extrahepatic tissue. Some detargeting approaches that have shown success include coating of nonfouling materials (to resist APOE binding)^{233–235} or the addition of SORT molecules (to facilitate the binding of LNPs to other serum proteins).^{118,119} However, these technologies have their own limitations since nonfouling materials may inhibit endosomal escape^{112,113} and SORT molecule composition may need to be optimized across different species and populations for appropriate serum protein binding.^{236,237} Approaches for the improvement of detargeting and retargeting may involve innovations in lipid chemistry for lipid components in LNPs or LNP formulation, or the engineering of appropriate protein corona on the LNP surface. For retargeting strategies, engineered LNPs may be modified with various targeted ligands (e.g., antibodies). Furthermore, various conjugation chemistries can be optimized for both the detargeting and retargeting of LNPs. Separately, lipids have also shown great promise as delivery vehicles for oligonucleotides; however, lipid-oligonucleotide conjugates display nonspecific targeting, and therefore, lipids are better suited for local than systemic delivery.

Identifying the Right Delivery Ligand for Oligonucleotides. In instances where the conjugated oligonucleotide is highly potent but shows nontargeted biodistribution to peripheral tissues, a shorter circulation time of the oligonucleotide conjugate may be advantageous in reducing toxicity. Under such scenarios, peptides (or smaller antibody formats) provide an advantage over bivalent antibodies as delivery vehicles. Furthermore, cyclic CPPs exhibit greater protease stability and thus may be favored over linear CPPs for the systemic delivery of oligonucleotides to extrahepatic tissues. Lastly, identifying nontoxic endosomal escape enhancers that, when conjugated to oligonucleotides, specifically target oligonucleotide-containing endosomes may improve the therapeutic index for oligonucleotides.

Evaluating the Role of Oligonucleotides in Other Common Diseases. Other areas of investigation where oligonucleotides are gaining traction is cancer, with several clinical trials ongoing;² however, there is yet to be an approved oligonucleotide therapy in this area. In addition, kidney indications, including acute and chronic kidney diseases, are topics of research, where oligonucleotides have yet to make their mark.

Future Advances in Chemistry to Generate Best-in-Class Oligonucleotide Therapeutics. Chemistry has largely played a critical role over the past decade in the discovery of therapeutic oligonucleotides for liver indications. Although numerous chemical modifications have been incorporated into oligonucleotides, only a few of them have had clinical success. Additional novel modifications will need to be introduced to further advance the metabolic stability, safety, and efficacy of

oligonucleotides. As previously described, unmodified siRNA is labile and can activate the immune responses. siRNA can be stabilized through encapsulation into lipid nanoparticles (LNPs). Furthermore, extensive chemical modifications of siRNA, such as enhanced stabilization chemistry (ESC) and ESC+ designs, significantly enhance metabolic stability, improve pharmacokinetic (PK) properties, and reduce off-target effects mediated by the seed region. These examples illustrate how innovations in chemistry have led to significant advancements in the field.

In this context, a very recent study reported a novel backbone modification in siRNA, termed extended nucleic acid (exNA), which contained an inserted methylene group.²³⁸ The presence of exNA and PS modifications at the 3' position of the antisense strand did not interfere with RISC loading but, importantly, did enhance resistance to in vivo 3' exonuclease activity when compared to the same siRNA-containing 3' PS modification with no exNA modification of the antisense strand. This study thus provides another example of how new chemical approaches can enhance the stability of oligonucleotides.

In conclusion, we believe that future progress in the oligonucleotide field will be led by continued advances in innovative chemical designs and delivery platforms, in combination with novel administration routes and optimized dose regimens, which will bring to fruition the expansion of targeting oligonucleotides to extrahepatic tissues.

AUTHOR INFORMATION

Corresponding Author

Puneet Anand – Regeneron Genetic Medicines, Regeneron Pharmaceuticals, Inc., Tarrytown, New York 10591, United States; orcid.org/0009-0002-1112-0924; Email: puneet.anand@regeneron.com

Authors

Yu Zhang – Regeneron Genetic Medicines, Regeneron Pharmaceuticals, Inc., Tarrytown, New York 10591, United States

Spoorthi Patil – Regeneron Genetic Medicines, Regeneron Pharmaceuticals, Inc., Tarrytown, New York 10591, United States

Keerat Kaur – Regeneron Genetic Medicines, Regeneron Pharmaceuticals, Inc., Tarrytown, New York 10591, United States

Complete contact information is available at:

<https://pubs.acs.org/10.1021/acs.jmedchem.4c02528>

Notes

The authors declare no competing financial interest.

Biographies

Puneet Anand obtained his PhD in Biochemistry from Duke University, followed by postdoctoral studies at Case Western Reserve University. He is currently working as a Senior Principal Scientist at Regeneron Pharmaceuticals and his research is focused on the targeted delivery of oligonucleotides to extrahepatic tissues for various therapeutics areas, including the CNS, skeletal muscle, and the ocular and pulmonary systems.

Yu Zhang is a Principal Scientist at Regeneron Pharmaceuticals. He holds a PhD in Polymer Chemistry from the University of Chinese Academy of Sciences. Following his postdoctoral training at Cornell University, he now focuses on developing ligand-conjugated lipid

nanoparticles and oligonucleotides for the extrahepatic delivery of genetic medicines.

Spoorthi Patil is an undergraduate student at Cornell University College of Engineering pursuing a degree in Biomedical Engineering with a concentration in Biomaterials and Drug Delivery. She is a co-op at Regeneron Pharmaceuticals and is working to develop conjugated lipid nanoparticles and oligonucleotides for extrahepatic delivery.

Keerat Kaur earned her PhD in Physiology from New York Medical College, and completed her postdoctoral research at Mount Sinai, specializing in mRNA technology for cardiac reprogramming. She is now a Principal Scientist at Regeneron, where she focuses on utilizing mRNA platform for gene editing and gene therapy.

ACKNOWLEDGMENTS

The authors acknowledge Christos Kyratsous and Olivier Harari for valuable feedback.

ABBREVIATIONS

2'-cEt, 2'-O-(2-methoxyethyl) ethyl; 2'F, 2'-fluoro; 2'MOE, 2'-O-methoxyethyl; 2'OMe, 2' O-methyl; AOC, antibody-oligonucleotide conjugate; ADC, antibody-drug conjugate; ASGPR, asialoglycoprotein receptor; ATTR, transthyretin-mediated; BTC, bicycle-toxin conjugate; CPP, cell penetrating peptide; DCA, docosanoic acid; DHA, docosahexaenoic acid; di-siRNA, divalent small interfering RNA; DM1, type 1 myotonic dystrophy; DMD, Duchenne muscular dystrophy; DSPC, distearoylphosphatidylcholine; EDO, enhanced delivery oligonucleotide; exNA, extended nucleic acid; FSHD, fascio-scapulohumeral muscular dystrophy; GalNAc, N-acetylgalactosamine; GNA, glycol nucleic acid; HoFH, homozygous familial hypercholesterolemia; HPRT, hypoxanthine-guanine phosphoribosyltransferase; icv, intracerebroventricular; ivt, intravitreal; JAK1, Janus kinase 1; LDLR, low-density lipoprotein receptor; LNA, locked nucleic acid; LNP, lipid nanoparticle; mAb, monoclonal antibody; MBNL, muscle blind-like protein; MsPA, mesyl-phosphoramidate; NAT, nucleic acid therapeutic; NHP, nonhuman primate; NP, nanoparticle; OAR, oligonucleotide-to-antibody ratio; OTV, oligonucleotide transport vehicle; PC, phosphatidyl choline; PCSK9, proprotein convertase subtilisin/kexin type 9; PMO, phosphorodiamidate morpholino oligonucleotide; PN, phosphoramidate; PO, phosphodiester; PS, phosphorothioate; RAGE, receptor for advanced glycation end products; RISC, RNA-induced silencing complex; siRNA, small interfering RNA; RNase, ribonuclease; Rp, right-handed; SORT, selective organ targeting; Sp, left-handed; sRAGE, soluble RAGE; TfR, transferrin receptor; TMO, thiomorpholino oligonucleotide; TRiM, targeted RNA interference molecule; TTR, transthyretin; VP, vinyl phosphonate

REFERENCES

- (1) Egli, M.; Manoharan, M. Chemistry, structure and function of approved oligonucleotide therapeutics. *Nucleic Acids Res.* **2023**, *51*, 2529–2573.
- (2) Belgrad, J.; Fakhri, H. H.; Khvorova, A. Nucleic acid therapeutics: Successes, milestones, and upcoming innovation. *Nucleic Acid Ther.* **2024**, *34*, 52–72.
- (3) Fire, A.; Xu, S.; Montgomery, M. K.; Kostas, S. A.; Driver, S. E.; Mello, C. C. Potent and specific genetic interference by double-stranded RNA in *Caenorhabditis elegans*. *Nature*. **1998**, *391*, 806–811.
- (4) Elbashir, S. M.; Harborth, J.; Lendeckel, W.; Yalcin, A.; Weber, K.; Tuschl, T. Duplexes of 21-nucleotide RNAs mediate RNA interference in cultured mammalian cells. *Nature*. **2001**, *411*, 494–498.

- (5) Soutschek, J.; Akinc, A.; Bramlage, B.; Charisse, K.; Constien, R.; Donoghue, M.; Elbashir, S.; Geick, A.; Hadwiger, P.; Harborth, J.; John, M.; Kesavan, V.; Lavine, G.; Pandey, R. K.; Racie, T.; Rajeev, K. G.; Rohl, I.; Toudjarska, I.; Wang, G.; Wuschko, S.; Bumcrot, D.; Koteliansky, V.; Limmer, S.; Manoharan, M.; Vornlocher, H. P. Therapeutic silencing of an endogenous gene by systemic administration of modified siRNAs. *Nature*. **2004**, *432*, 173–178.
- (6) Davis, M. E.; Zuckerman, J. E.; Choi, C. H.; Seligson, D.; Tolcher, A.; Alabi, C. A.; Yen, Y.; Heidel, J. D.; Ribas, A. Evidence of RNAi in humans from systemically administered siRNA via targeted nanoparticles. *Nature*. **2010**, *464*, 1067–1070.
- (7) Johnson, K. C.; Kilikevicius, A.; Hofman, C.; Hu, J.; Liu, Y.; Aguilar, S.; Graswich, J.; Han, Y.; Wang, T.; Westcott, J. M.; Brekken, R. A.; Peng, L.; Karagkounis, G.; Corey, D. R. Nuclear localization of Argonaute 2 is affected by cell density and may relieve repression by microRNAs. *Nucleic Acids Res.* **2024**, *52*, 1930–1952.
- (8) Urits, I.; Swanson, D.; Swett, M. C.; Patel, A.; Berardino, K.; Amgalan, A.; Berger, A. A.; Kassem, H.; Kaye, A. D.; Viswanath, O. A review of patisiran (ONPATTRA®) for the treatment of polyneuropathy in people with hereditary transthyretin amyloidosis. *Neurol Ther.* **2020**, *9*, 301–315.
- (9) Zamecnik, P. C.; Stephenson, M. L. Inhibition of Rous sarcoma virus replication and cell transformation by a specific oligodeoxynucleotide. *Proc. Natl. Acad. Sci. U. S. A.* **1978**, *75*, 280–284.
- (10) Stephenson, M. L.; Zamecnik, P. C. Inhibition of Rous sarcoma viral RNA translation by a specific oligodeoxyribonucleotide. *Proc. Natl. Acad. Sci. U. S. A.* **1978**, *75*, 285–288.
- (11) Juliano, R. L.; Ming, X.; Nakagawa, O. Cellular uptake and intracellular trafficking of antisense and siRNA oligonucleotides. *Bioconjug Chem.* **2012**, *23*, 147–157.
- (12) Izant, J. G.; Weintraub, H. Constitutive and conditional suppression of exogenous and endogenous genes by anti-sense RNA. *Science*. **1985**, *229*, 345–352.
- (13) Dhuri, K.; Bechtold, C.; Quijano, E.; Pham, H.; Gupta, A.; Vikram, A.; Bahal, R. Antisense oligonucleotides: An emerging area in drug discovery and development. *J. Clin Med.* **2020**, *9* (6), 2004.
- (14) Khvorova, A.; Watts, J. K. The chemical evolution of oligonucleotide therapies of clinical utility. *Nat. Biotechnol.* **2017**, *35*, 238–248.
- (15) Qin, S.; Tang, X.; Chen, Y.; Chen, K.; Fan, N.; Xiao, W.; Zheng, Q.; Li, G.; Teng, Y.; Wu, M.; Song, X. mRNA-based therapeutics: Powerful and versatile tools to combat diseases. *Signal Transduct Target Ther.* **2022**, *7*, 166.
- (16) Juliano, R. L. The delivery of therapeutic oligonucleotides. *Nucleic Acids Res.* **2016**, *44*, 6518–6548.
- (17) Sayers, E. J.; Peel, S. E.; Schantz, A.; England, R. M.; Beano, M.; Bates, S. M.; Desai, A. S.; Puri, S.; Ashford, M. B.; Jones, A. T. Endocytic profiling of cancer cell models reveals critical factors influencing LNP-mediated mRNA delivery and protein expression. *Mol. Ther.* **2019**, *27*, 1950–1962.
- (18) Zhdanov, V. P. Intracellular RNA delivery by lipid nanoparticles: Diffusion, degradation, and release. *Biosystems*. **2019**, *185*, No. 104032.
- (19) Yanez Arteta, M.; Kjellman, T.; Bartesaghi, S.; Wallin, S.; Wu, X.; Kvist, A. J.; Dabkowska, A.; Szekely, N.; Radulescu, A.; Bergenholtz, J.; Lindfors, L. Successful reprogramming of cellular protein production through mRNA delivered by functionalized lipid nanoparticles. *Proc. Natl. Acad. Sci. U. S. A.* **2018**, *115*, E3351–E3360.
- (20) Moazzam, M.; Zhang, M.; Hussain, A.; Yu, X.; Huang, J.; Huang, Y. The landscape of nanoparticle-based siRNA delivery and therapeutic development. *Mol. Ther.* **2024**, *32*, 284–312.
- (21) Brown, J. M.; Dahlman, J. E.; Neuman, K. K.; Prata, C. A. H.; Krampert, M. C.; Hadwiger, P. M.; Vornlocher, H. P. Ligand conjugated multimeric siRNAs enable enhanced uptake and multiplexed gene silencing. *Nucleic Acid Ther.* **2019**, *29*, 231–244.
- (22) Damase, T. R.; Sukhovshin, R.; Boada, C.; Taraballi, F.; Pettigrew, R. I.; Cooke, J. P. The limitless future of RNA therapeutics. *Front Bioeng Biotechnol.* **2021**, *9*, No. 628137.
- (23) Agrawal, N.; Dasaradhi, P. V.; Mohammed, A.; Malhotra, P.; Bhatnagar, R. K.; Mukherjee, S. K. RNA interference: Biology, mechanism, and applications. *Microbiol. Mol. Biol. Rev.* **2003**, *67*, 657–685.
- (24) Matranga, C.; Tomari, Y.; Shin, C.; Bartel, D. P.; Zamore, P. D. Passenger-strand cleavage facilitates assembly of siRNA into Ago2-containing RNAi enzyme complexes. *Cell*. **2005**, *123*, 607–620.
- (25) Liu, J.; Carmell, M. A.; Rivas, F. V.; Marsden, C. G.; Thomson, J. M.; Song, J. J.; Hammond, S. M.; Joshua-Tor, L.; Hannon, G. J. Argonaute2 is the catalytic engine of mammalian RNAi. *Science*. **2004**, *305*, 1437–1441.
- (26) Elkayam, E.; Kuhn, C. D.; Tocilj, A.; Haase, A. D.; Greene, E. M.; Hannon, G. J.; Joshua-Tor, L. The structure of human argonaute-2 in complex with miR-20a. *Cell*. **2012**, *150*, 100–110.
- (27) Foster, D. J.; Brown, C. R.; Shaikh, S.; Trapp, C.; Schlegel, M. K.; Qian, K.; Sehgal, A.; Rajeev, K. G.; Jadhav, V.; Manoharan, M.; Kuchimanchi, S.; Maier, M. A.; Milstein, S. Advanced siRNA designs further improve in vivo performance of GalNAc-siRNA conjugates. *Mol. Ther.* **2018**, *26*, 708–717.
- (28) Witttrup, A.; Ai, A.; Liu, X.; Hamar, P.; Trifonova, R.; Charisse, K.; Manoharan, M.; Kirchhausen, T.; Lieberman, J. Visualizing lipid-formulated siRNA release from endosomes and target gene knock-down. *Nat. Biotechnol.* **2015**, *33*, 870–876.
- (29) Dowdy, S. F.; Setten, R. L.; Cui, X. S.; Jadhav, S. G. Delivery of RNA therapeutics: The great endosomal escape! *Nucleic Acid Ther.* **2022**, *32*, 361–368.
- (30) Brown, C. R.; Gupta, S.; Qin, J.; Racie, T.; He, G.; Lentini, S.; Malone, R.; Yu, M.; Matsuda, S.; Shulga-Morskaya, S.; Nair, A. V.; Theile, C. S.; Schmidt, K.; Shahrzad, A.; Goel, V.; Parmar, R. G.; Zlatev, I.; Schlegel, M. K.; Nair, J. K.; Jayaraman, M.; Manoharan, M.; Brown, D.; Maier, M. A.; Jadhav, V. Investigating the pharmacodynamic durability of GalNAc-siRNA conjugates. *Nucleic Acids Res.* **2020**, *48*, 11827–11844.
- (31) Dowdy, S. F. Overcoming cellular barriers for RNA therapeutics. *Nat. Biotechnol.* **2017**, *35*, 222–229.
- (32) Liu, L.; Botos, I.; Wang, Y.; Leonard, J. N.; Shiloach, J.; Segal, D. M.; Davies, D. R. Structural basis of toll-like receptor 3 signaling with double-stranded RNA. *Science*. **2008**, *320*, 379–381.
- (33) Kariko, K.; Bhuyan, P.; Capodici, J.; Weissman, D. Small interfering RNAs mediate sequence-independent gene suppression and induce immune activation by signaling through toll-like receptor 3. *J. Immunol.* **2004**, *172*, 6545–6549.
- (34) Jadhav, V.; Vaishnav, A.; Fitzgerald, K.; Maier, M. A. RNA interference in the era of nucleic acid therapeutics. *Nat. Biotechnol.* **2024**, *42*, 394–405.
- (35) Debacker, A. J.; Voutilainen, J.; Catley, M.; Blakey, D.; Habib, N. Delivery of Oligonucleotides to the Liver with GalNAc: From Research to Registered Therapeutic Drug. *Mol. Ther.* **2020**, *28*, 1759–1771.
- (36) Hu, B.; Zhong, L.; Weng, Y.; Peng, L.; Huang, Y.; Zhao, Y.; Liang, X. J. Therapeutic siRNA: state of the art. *Signal Transduct Target Ther.* **2020**, *5*, 101.
- (37) Bost, J. P.; Barriga, H.; Holme, M. N.; Gallud, A.; Maugeri, M.; Gupta, D.; Lehto, T.; Valadi, H.; Esbjorn, E. K.; Stevens, M. M.; El-Andaloussi, S. Delivery of oligonucleotide therapeutics: Chemical modifications, lipid nanoparticles, and extracellular vesicles. *ACS Nano* **2021**, *15*, 13993–14021.
- (38) Roberts, T. C.; Langer, R.; Wood, M. J. A. Advances in oligonucleotide drug delivery. *Nat. Rev. Drug Discovery* **2020**, *19*, 673–694.
- (39) Eckstein, F. Nucleoside phosphorothioates. *J. Am. Chem. Soc.* **1966**, *88*, 4292–4294.
- (40) Stec, W. J.; Zon, G.; Egan, W. Automated solid-phase synthesis, separation, and stereochemistry of phosphorothioate analogs of oligodeoxyribonucleotides. *J. Am. Chem. Soc.* **1984**, *106*, 6077–6079.
- (41) Nair, J. K.; Attarwala, H.; Sehgal, A.; Wang, Q.; Aluri, K.; Zhang, X.; Gao, M.; Liu, J.; Indrakanti, R.; Schofield, S.; Kretschmer, P.; Brown, C. R.; Gupta, S.; Willoughby, J. L. S.; Boshar, J. A.; Jadhav, V.; Charisse, K.; Zimmermann, T.; Fitzgerald, K.; Manoharan, M.; Rajeev, K. G.; Akinc, A.; Hutabarat, R.; Maier, M. A. Impact of enhanced metabolic stability on pharmacokinetics and pharmacodynamics of GalNAc-siRNA conjugates. *Nucleic Acids Res.* **2017**, *45*, 10969–10977.

- (42) Jahns, H.; Taneja, N.; Willoughby, J. L. S.; Akabane-Nakata, M.; Brown, C. R.; Nguyen, T.; Bisbe, A.; Matsuda, S.; Hettinger, M.; Manoharan, R. M.; Rajeev, K. G.; Maier, M. A.; Zlatev, I.; Charisse, K.; Egli, M.; Manoharan, M. Chirality matters: stereo-defined phosphorothioate linkages at the termini of small interfering RNAs improve pharmacology in vivo. *Nucleic Acids Res.* **2022**, *50*, 1221–1240.
- (43) Biscans, A.; Caiazza, J.; Davis, S.; McHugh, N.; Sousa, J.; Khvorova, A. The chemical structure and phosphorothioate content of hydrophobically modified siRNAs impact extrahepatic distribution and efficacy. *Nucleic Acids Res.* **2020**, *48*, 7665–7680.
- (44) Parmar, R.; Willoughby, J. L.; Liu, J.; Foster, D. J.; Brigham, B.; Theile, C. S.; Charisse, K.; Akinc, A.; Guidry, E.; Pei, Y.; Strapps, W.; Cancilla, M.; Stanton, M. G.; Rajeev, K. G.; Sepp-Lorenzino, L.; Manoharan, M.; Meyers, R.; Maier, M. A.; Jadhav, V. 5'-(E)-vinylphosphonate: A stable phosphate mimic can improve the RNAi activity of siRNA-GalNAc conjugates. *Chembiochem.* **2016**, *17*, 985–989.
- (45) Hassler, M. R.; Turanov, A. A.; Alterman, J. F.; Haraszti, R. A.; Coles, A. H.; Osborn, M. F.; Echeverria, D.; Nikan, M.; Salomon, W. E.; Roux, L.; Godinho, B.; Davis, S. M.; Morrissey, D. V.; Zamore, P. D.; Karumanchi, S. A.; Moore, M. J.; Aronin, N.; Khvorova, A. Comparison of partially and fully chemically-modified siRNA in conjugate-mediated delivery in vivo. *Nucleic Acids Res.* **2018**, *46*, 2185–2196.
- (46) Honor, A.; Rudnick, S. R.; Bonkovsky, H. L. Givosiran to treat acute porphyria. *Drugs Today (Barc.)* **2021**, *57*, 47–59.
- (47) Judge, D. P.; Kristen, A. V.; Grogan, M.; Maurer, M. S.; Falk, R. H.; Hanna, M.; Gillmore, J.; Garg, P.; Vaishnav, A. K.; Harrop, J.; Powell, C.; Karsten, V.; Zhang, X.; Sweetser, M. T.; Vest, J.; Hawkins, P. N. Phase 3 multicenter study of revusiran in patients with hereditary transthyretin-mediated (hATTR) amyloidosis with cardiomyopathy (ENDEAVOUR). *Cardiovasc Drugs Ther.* **2020**, *34*, 357–370.
- (48) Garber, K. Alnylam terminates revusiran program, stock plunges. *Nat. Biotechnol.* **2016**, *34*, 1213–1214.
- (49) Alagia, A.; Jorge, A. F.; Avino, A.; Cova, T.; Crehuet, R.; Grijalvo, S.; Pais, A.; Eritja, R. Exploring PAZ/3'-overhang interaction to improve siRNA specificity. A combined experimental and modeling study. *Chem. Sci.* **2018**, *9*, 2074–2086.
- (50) Jackson, A. L.; Bartz, S. R.; Schelter, J.; Kobayashi, S. V.; Burchard, J.; Mao, M.; Li, B.; Cavet, G.; Linsley, P. S. Expression profiling reveals off-target gene regulation by RNAi. *Nat. Biotechnol.* **2003**, *21*, 635–637.
- (51) Egli, M.; Schlegel, M. K.; Manoharan, M. Acyclic (S)-glycol nucleic acid (S-GNA) modification of siRNAs improves the safety of RNAi therapeutics while maintaining potency. *RNA* **2023**, *29*, 402–414.
- (52) Schlegel, M. K.; Janas, M. M.; Jiang, Y.; Barry, J. D.; Davis, W.; Agarwal, S.; Berman, D.; Brown, C. R.; Castoreno, A.; LeBlanc, S.; Liebow, A.; Mayo, T.; Milstein, S.; Nguyen, T.; Shulga-Morskaya, S.; Hyde, S.; Schofield, S.; Szeto, J.; Woods, L. B.; Yilmaz, V. O.; Manoharan, M.; Egli, M.; Charisse, K.; Sepp-Lorenzino, L.; Haslett, P.; Fitzgerald, K.; Jadhav, V.; Maier, M. A. From bench to bedside: Improving the clinical safety of GalNAc-siRNA conjugates using seed-pairing destabilization. *Nucleic Acids Res.* **2022**, *50*, 6656–6670.
- (53) Kumar, P.; Parmar, R. G.; Brown, C. R.; Willoughby, J. L. S.; Foster, D. J.; Babu, I. R.; Schofield, S.; Jadhav, V.; Charisse, K.; Nair, J. K.; Rajeev, K. G.; Maier, M. A.; Egli, M.; Manoharan, M. 5'-Morpholino modification of the sense strand of an siRNA makes it a more effective passenger. *Chem. Commun. (Camb.)* **2019**, *55*, 5139–5142.
- (54) Crooke, S. T.; Baker, B. F.; Crooke, R. M.; Liang, X. H. Antisense technology: an overview and prospectus. *Nat. Rev. Drug Discovery* **2021**, *20*, 427–453.
- (55) Crooke, S. T. Molecular mechanisms of antisense oligonucleotides. *Nucleic Acid Ther.* **2017**, *27*, 70–77.
- (56) Wu, H.; Lima, W. F.; Zhang, H.; Fan, A.; Sun, H.; Crooke, S. T. Determination of the role of the human RNase H1 in the pharmacology of DNA-like antisense drugs. *J. Biol. Chem.* **2004**, *279*, 17181–17189.
- (57) Liang, X. H.; Sun, H.; Nichols, J. G.; Crooke, S. T. RNase H1-dependent antisense oligonucleotides are robustly active in directing RNA cleavage in both the cytoplasm and the nucleus. *Mol. Ther.* **2017**, *25*, 2075–2092.
- (58) Crooke, S. T.; Vickers, T. A.; Lima, W. F.; Wu, H.-J. *Antisense Drug Technology: Principles, Strategies, and Applications*; CRC Press, 2008.
- (59) Agrawal, S.; Goodchild, J.; Civeira, M. P.; Thornton, A. H.; Sarin, P. S.; Zamecnik, P. C. Oligodeoxynucleoside phosphoramidates and phosphorothioates as inhibitors of human immunodeficiency virus. *Proc. Natl. Acad. Sci. U. S. A.* **1988**, *85*, 7079–7083.
- (60) Sands, H.; Gorey-Feret, L. J.; Cocuzza, A. J.; Hobbs, F. W.; Chidester, D.; Trainor, G. L. Biodistribution and metabolism of internally 3H-labeled oligonucleotides. I. Comparison of a phosphodiester and a phosphorothioate. *Mol. Pharmacol.* **1994**, *45*, 932–943.
- (61) Geary, R. S.; Watanabe, T. A.; Truong, L.; Freier, S.; Lesnik, E. A.; Sioufi, N. B.; Sasmor, H.; Manoharan, M.; Levin, A. A. Pharmacokinetic properties of 2'-O-(2-methoxyethyl)-modified oligonucleotide analogs in rats. *J. Pharmacol. Exp. Ther.* **2001**, *296*, 890–897.
- (62) Levin, A. A.; Yu, R. Z.; Geary, R. S. Basic Principles of the Pharmacokinetics of Antisense Oligonucleotide Drugs. In *Antisense Drug Technology: Principles, Strategies, and Applications*, 2nd ed.; Crooke, S. T., Ed.; CRC Press, 2008; pp 183–216.
- (63) Geary, R. S.; Norris, D.; Yu, R.; Bennett, C. F. Pharmacokinetics, biodistribution and cell uptake of antisense oligonucleotides. *Adv. Drug Deliv. Rev.* **2015**, *87*, 46–51.
- (64) Teplova, M.; Minasov, G.; Tereshko, V.; Inamati, G. B.; Cook, P. D.; Manoharan, M.; Egli, M. Crystal structure and improved antisense properties of 2'-O-(2-methoxyethyl)-RNA. *Nat. Struct. Biol.* **1999**, *6*, 535–539.
- (65) Geary, R. S.; Baker, B. F.; Crooke, S. T. Clinical and preclinical pharmacokinetics and pharmacodynamics of mipomersen (kynamro®): a second-generation antisense oligonucleotide inhibitor of apolipoprotein B. *Clin Pharmacokinet.* **2015**, *54*, 133–146.
- (66) Tillman, L. G.; Geary, R. S.; Hardee, G. E. Oral delivery of antisense oligonucleotides in man. *J. Pharm. Sci.* **2008**, *97*, 225–236.
- (67) Benson, M. D.; Waddington-Cruz, M.; Berk, J. L.; Polydefkis, M.; Dyck, P. J.; Wang, A. K.; Plante-Bordeneuve, V.; Barroso, F. A.; Merlini, G.; Obici, L.; Scheinberg, M.; Brannagan, T. H., 3rd; Litchy, W. J.; Whelan, C.; Drachman, B. M.; Adams, D.; Heitner, S. B.; Conceicao, I.; Schmidt, H. H.; Vita, G.; Campistol, J. M.; Gamez, J.; Gorevic, P. D.; Gane, E.; Shah, A. M.; Solomon, S. D.; Monia, B. P.; Hughes, S. G.; Kwoh, T. J.; McEvoy, B. W.; Jung, S. W.; Baker, B. F.; Ackermann, E. J.; Gertz, M. A.; Coelho, T. Inotersen treatment for patients with hereditary transthyretin amyloidosis. *N Engl J. Med.* **2018**, *379*, 22–31.
- (68) Witztum, J. L.; Gaudet, D.; Freedman, S. D.; Alexander, V. J.; Digenio, A.; Williams, K. R.; Yang, Q.; Hughes, S. G.; Geary, R. S.; Arca, M.; Strees, E. S. G.; Bergeron, J.; Soran, H.; Civeira, F.; Hemphill, L.; Tsimikas, S.; Blom, D. J.; O'Dea, L.; Bruckert, E. Volanesorsen and triglyceride levels in familial chylomicronemia syndrome. *N Engl J. Med.* **2019**, *381*, 531–542.
- (69) Seth, P. P.; Siwkowski, A.; Allerson, C. R.; Vasquez, G.; Lee, S.; Prakash, T. P.; Kinberger, G.; Migawa, M. T.; Gaus, H.; Bhat, B.; Swayze, E. E. Design, synthesis and evaluation of constrained methoxyethyl (cMOE) and constrained ethyl (cEt) nucleoside analogs. *Nucleic Acids Symp. Ser. (Oxf.)* **2008**, *52*, 553–554.
- (70) Bennett, C. F. Therapeutic antisense oligonucleotides are coming of age. *Annu. Rev. Med.* **2019**, *70*, 307–321.
- (71) Crooke, S. T.; Baker, B. F.; Pham, N. C.; Hughes, S. G.; Kwoh, T. J.; Cai, D.; Tsimikas, S.; Geary, R. S.; Bhanot, S. The effects of 2'-O-methoxyethyl oligonucleotides on renal function in humans. *Nucleic Acid Ther.* **2018**, *28*, 10–22.
- (72) Crooke, S. T.; Baker, B. F.; Xia, S.; Yu, R. Z.; Viney, N. J.; Wang, Y.; Tsimikas, S.; Geary, R. S. Integrated assessment of the clinical performance of GalNAc(3)-conjugated 2'-O-methoxyethyl chimeric antisense oligonucleotides: I. Human volunteer experience. *Nucleic Acid Ther.* **2019**, *29*, 16–32.
- (73) Janssen, H. L.; Reesink, H. W.; Lawitz, E. J.; Zeuzem, S.; Rodriguez-Torres, M.; Patel, K.; van der Meer, A. J.; Patick, A. K.; Chen, A.; Zhou, Y.; Persson, R.; King, B. D.; Kauppinen, S.; Levin, A. A.

Hodges, M. R. Treatment of HCV infection by targeting microRNA. *N Engl J. Med.* **2013**, *368*, 1685–1694.

(74) Hagedorn, P. H.; Persson, R.; Funder, E. D.; Albæk, N.; Diemer, S. L.; Hansen, D. J.; Möller, M. R.; Papargyri, N.; Christiansen, H.; Hansen, B. R.; Hansen, H. F.; Jensen, M. A.; Koch, T. Locked nucleic acid: modality, diversity, and drug discovery. *Drug Discov Today*. **2018**, *23*, 101–114.

(75) Viney, N. J.; van Capelleveen, J. C.; Geary, R. S.; Xia, S.; Tami, J. A.; Yu, R. Z.; Marcovina, S. M.; Hughes, S. G.; Graham, M. J.; Crooke, R. M.; Crooke, S. T.; Witztum, J. L.; Stroes, E. S.; Tsimikas, S. Antisense oligonucleotides targeting apolipoprotein(a) in people with raised lipoprotein(a): Two randomised, double-blind, placebo-controlled, dose-ranging trials. *Lancet*. **2016**, *388*, 2239–2253.

(76) Nie, T. Eplontersen: First approval. *Drugs*. **2024**, *84*, 473–478.

(77) Zhang, L.; Vickers, T. A.; Sun, H.; Liang, X. H.; Crooke, S. T. Binding of phosphorothioate oligonucleotides with RNase H1 can cause conformational changes in the protein and alter the interactions of RNase H1 with other proteins. *Nucleic Acids Res.* **2021**, *49*, 2721–2739.

(78) Shen, W.; De Hoyos, C. L.; Migawa, M. T.; Vickers, T. A.; Sun, H.; Low, A.; Bell, T. A., 3rd; Rahdar, M.; Mukhopadhyay, S.; Hart, C. E.; Bell, M.; Riney, S.; Murray, S. F.; Greenlee, S.; Crooke, R. M.; Liang, X. H.; Seth, P. P.; Crooke, S. T. Chemical modification of PS-ASO therapeutics reduces cellular protein-binding and improves the therapeutic index. *Nat. Biotechnol.* **2019**, *37*, 640–650.

(79) Anderson, B. A.; Freestone, G. C.; Low, A.; De-Hoyos, C. L.; Drury, W. J., III; Oestergaard, M. E.; Migawa, M. T.; Fazio, M.; Wan, W. B.; Berdeja, A.; Scandalis, E.; Burel, S. A.; Vickers, T. A.; Crooke, S. T.; Swayze, E. E.; Liang, X.; Seth, P. P. Towards next generation antisense oligonucleotides: Methylphosphoramidate modification improves therapeutic index and duration of effect of gapmer antisense oligonucleotides. *Nucleic Acids Res.* **2021**, *49*, 9026–9041.

(80) Zhang, L.; Liang, X. H.; De Hoyos, C. L.; Migawa, M.; Nichols, J. G.; Freestone, G.; Tian, J.; Seth, P. P.; Crooke, S. T. The combination of methyl-phosphoramidate inter-nucleotide linkages and 2'-O-methyl in selected positions in the antisense oligonucleotide enhances the performance of RNaseH1 active PS-ASOs. *Nucleic Acid Ther.* **2022**, *32*, 401–411.

(81) Iversen, P. L. Phosphorodiamidate morpholino oligomers: favorable properties for sequence-specific gene inactivation. *Curr. Opin Mol. Ther.* **2001**, *3*, 235–238.

(82) Matsuo, M. Antisense oligonucleotide-mediated exon-skipping therapies: Precision medicine spreading from Duchenne muscular dystrophy. *JMA J.* **2021**, *4*, 232–240.

(83) Nowak, K. J.; Davies, K. E. Duchenne muscular dystrophy and dystrophin: pathogenesis and opportunities for treatment. *EMBO Rep.* **2004**, *5*, 872–876.

(84) Cirak, S.; Arechavala-Gomez, V.; Guglieri, M.; Feng, L.; Torelli, S.; Anthony, K.; Abbs, S.; Garralda, M. E.; Bourke, J.; Wells, D. J.; Dickson, G.; Wood, M. J.; Wilton, S. D.; Straub, V.; Kole, R.; Shrewsbury, S. B.; Sewry, C.; Morgan, J. E.; Bushby, K.; Muntoni, F. Exon skipping and dystrophin restoration in patients with Duchenne muscular dystrophy after systemic phosphorodiamidate morpholino oligomer treatment: an open-label, phase 2, dose-escalation study. *Lancet*. **2011**, *378*, 595–605.

(85) Le, B. T.; Paul, S.; Jastrzebska, K.; Langer, H.; Caruthers, M. H.; Veedu, R. N. Thiomorpholino oligonucleotides as a robust class of next generation platforms for alternate mRNA splicing. *Proc. Natl. Acad. Sci. U. S. A.* **2022**, *119*, No. e2207956119.

(86) Iwamoto, N.; Butler, D. C. D.; Svrzikapa, N.; Mohapatra, S.; Zlatev, I.; Sah, D. W. Y.; Meena; Standley, S. M.; Lu, G.; Apponi, L. H.; Frank-Kamenetsky, M.; Zhang, J. J.; Vargeese, C.; Verdine, G. L. Control of phosphorothioate stereochemistry substantially increases the efficacy of antisense oligonucleotides. *Nat. Biotechnol.* **2017**, *35*, 845–851.

(87) Vargeese, C. In *Exploring new oligonucleotide backbone chemistries and their deployment to improve the properties of stereopure oligonucleotides*, 2021; Wave Life Sciences, 2021.

(88) Crooke, S. T.; Liang, X. H.; Crooke, R. M.; Baker, B. F.; Geary, R. S. Antisense drug discovery and development technology considered in a pharmacological context. *Biochem. Pharmacol.* **2021**, *189*, No. 114196.

(89) Scharner, J.; Ma, W. K.; Zhang, Q.; Lin, K. T.; Rigo, F.; Bennett, C. F.; Krainer, A. R. Hybridization-mediated off-target effects of splice-switching antisense oligonucleotides. *Nucleic Acids Res.* **2020**, *48*, 802–816.

(90) Migawa, M. T.; Shen, W.; Wan, W. B.; Vasquez, G.; Oestergaard, M. E.; Low, A.; De Hoyos, C. L.; Gupta, R.; Murray, S.; Tanowitz, M.; Bell, M.; Nichols, J. G.; Gaus, H.; Liang, X. H.; Swayze, E. E.; Crooke, S. T.; Seth, P. P. Site-specific replacement of phosphorothioate with alkyl phosphonate linkages enhances the therapeutic profile of gapmer ASOs by modulating interactions with cellular proteins. *Nucleic Acids Res.* **2019**, *47*, S465–S479.

(91) Chen, Y.; Chu, C.; Zhou, Y.; Ru, Y.; Chen, H.; Chen, F.; He, Q.; Zhang, Y.; Zhang, L.; Shi, J. Reversible pore-structure evolution in hollow silica nanocapsules: large pores for siRNA delivery and nanoparticle collecting. *Small*. **2011**, *7*, 2935–2944.

(92) Tao, T.; Rehman, S. U.; Xu, S.; Zhang, J.; Xia, H.; Guo, Z.; Li, Z.; Ma, K.; Wang, J. A biomimetic camouflaged metal organic framework for enhanced siRNA delivery in the tumor environment. *J. Mater. Chem. B* **2024**, *12*, 4080–4096.

(93) Li, Z.; Zhang, X. Q.; Ho, W.; Bai, X.; Jaijyan, D. K.; Li, F.; Kumar, R.; Kolloli, A.; Subbian, S.; Zhu, H.; Xu, X. Lipid-polymer hybrid "particle-in-particle" nanostructure gene delivery platform explored for lyophilizable DNA and mRNA COVID-19 vaccines. *Adv. Funct. Mater.* **2022**, *32*, No. 2204462.

(94) Cullis, P. R.; Hope, M. J. Lipid nanoparticle systems for enabling gene therapies. *Mol. Ther.* **2017**, *25*, 1467–1475.

(95) Hou, X.; Zaks, T.; Langer, R.; Dong, Y. Lipid nanoparticles for mRNA delivery. *Nat. Rev. Mater.* **2021**, *6*, 1078–1094.

(96) Li, H.; Yang, X.; Gao, F.; Qian, C.; Li, C.; Oupicky, D.; Sun, M. Bioreduction-ruptured nanogel for switch on/off release of Bcl2 siRNA in breast tumor therapy. *J. Controlled Release* **2018**, *292*, 78–90.

(97) Li, J.; Liang, H.; Liu, J.; Wang, Z. Poly (amidoamine) (PAMAM) dendrimer mediated delivery of drug and pDNA/siRNA for cancer therapy. *Int. J. Pharm.* **2018**, *546*, 215–225.

(98) Guo, Y.; Zhang, Q.; Zhu, Q.; Gao, J.; Zhu, X.; Yu, H.; Li, Y.; Zhang, C. Copackaging photosensitizer and PD-L1 siRNA in a nucleic acid nanogel for synergistic cancer photoimmunotherapy. *Sci. Adv.* **2022**, *8*, No. eabn2941.

(99) Schott, J. W.; Morgan, M.; Galla, M.; Schambach, A. Viral and synthetic RNA vector technologies and applications. *Mol. Ther.* **2016**, *24*, 1513–1527.

(100) Zhao, L.; Gu, C.; Gan, Y.; Shao, L.; Chen, H.; Zhu, H. Exosome-mediated siRNA delivery to suppress postoperative breast cancer metastasis. *J. Controlled Release* **2020**, *318*, 1–15.

(101) Huang, X.; Guo, H.; Wang, L.; Zhang, Z.; Zhang, W. Biomimetic cell membrane-coated nanocarriers for targeted siRNA delivery in cancer therapy. *Drug Discov Today*. **2023**, *28*, No. 103514.

(102) Hald Albertsen, C.; Kulkarni, J. A.; Witzigmann, D.; Lind, M.; Petersson, K.; Simonsen, J. B. The role of lipid components in lipid nanoparticles for vaccines and gene therapy. *Adv. Drug Deliv Rev.* **2022**, *188*, No. 114416.

(103) Brailovsky, Y.; Rajapreyar, I.; Alvarez, R. TTR amyloidosis: Current state of affairs and promise for the future. *JACC Case Rep.* **2023**, *10*, No. 101759.

(104) Kulkarni, J. A.; Darjuan, M. M.; Mercer, J. E.; Chen, S.; van der Meel, R.; Thewalt, J. L.; Tam, Y. Y. C.; Cullis, P. R. On the formation and morphology of lipid nanoparticles containing ionizable cationic lipids and siRNA. *ACS Nano* **2018**, *12*, 4787–4795.

(105) Sheridan, C. Why gene therapies must go virus-free. *Nat. Biotechnol.* **2023**, *41*, 737–739.

(106) Witzigmann, D.; Kulkarni, J. A.; Leung, J.; Chen, S.; Cullis, P. R.; van der Meel, R. Lipid nanoparticle technology for therapeutic gene regulation in the liver. *Adv. Drug Deliv Rev.* **2020**, *159*, 344–363.

(107) Loughrey, D.; Dahlman, J. E. Non-liver mRNA delivery. *Acc. Chem. Res.* **2022**, *55*, 13–23.

- (108) Dilliard, S. A.; Siegwart, D. J. Passive, active and endogenous organ-targeted lipid and polymer nanoparticles for delivery of genetic drugs. *Nat. Rev. Mater.* **2023**, *8*, 282–300.
- (109) Kim, J.; Eygeris, Y.; Ryals, R. C.; Jozic, A.; Sahay, G. Strategies for non-viral vectors targeting organs beyond the liver. *Nat. Nanotechnol.* **2024**, *19*, 428–447.
- (110) Whitehead, K. A.; Langer, R.; Anderson, D. G. Knocking down barriers: advances in siRNA delivery. *Nat. Rev. Drug Discovery* **2009**, *8*, 129–138.
- (111) Mitchell, M. J.; Billingsley, M. M.; Haley, R. M.; Wechsler, M. E.; Peppas, N. A.; Langer, R. Engineering precision nanoparticles for drug delivery. *Nat. Rev. Drug Discovery* **2021**, *20*, 101–124.
- (112) Zalba, S.; Ten Hagen, T. L. M.; Burgui, C.; Garrido, M. J. Stealth nanoparticles in oncology: Facing the PEG dilemma. *J. Controlled Release* **2022**, *351*, 22–36.
- (113) Chan, C. L.; Majzoub, R. N.; Shirazi, R. S.; Ewert, K. K.; Chen, Y. J.; Liang, K. S.; Safinya, C. R. Endosomal escape and transfection efficiency of PEGylated cationic liposome-DNA complexes prepared with an acid-labile PEG-lipid. *Biomaterials* **2012**, *33*, 4928–4935.
- (114) Saunders, N. R. M.; Paolini, M. S.; Fenton, O. S.; Poul, L.; Devalliere, J.; Mpambani, F.; Darmon, A.; Bergere, M.; Jibault, O.; Germain, M.; Langer, R. A nanoparticle to improve the systemic delivery of siRNA and mRNA. *Nano Lett.* **2020**, *20*, 4264–4269.
- (115) Ouyang, B.; Poon, W.; Zhang, Y. N.; Lin, Z. P.; Kingston, B. R.; Tavares, A. J.; Zhang, Y.; Chen, J.; Valic, M. S.; Syed, A. M.; MacMillan, P.; Couture-Senecal, J.; Zheng, G.; Chan, W. C. W. The dose threshold for nanoparticle tumour delivery. *Nat. Mater.* **2020**, *19*, 1362–1371.
- (116) Qiu, M.; Tang, Y.; Chen, J.; Muriph, R.; Ye, Z.; Huang, C.; Evans, J.; Henske, E. P.; Xu, Q. Lung-selective mRNA delivery of synthetic lipid nanoparticles for the treatment of pulmonary lymphangioleiomyomatosis. *Proc. Natl. Acad. Sci. U. S. A.* **2022**, *119*, No. e2116271119.
- (117) He, C.; Hu, Y.; Yin, L.; Tang, C.; Yin, C. Effects of particle size and surface charge on cellular uptake and biodistribution of polymeric nanoparticles. *Biomaterials* **2010**, *31*, 3657–3666.
- (118) Cheng, Q.; Wei, T.; Farbiak, L.; Johnson, L. T.; Dilliard, S. A.; Siegwart, D. J. Selective organ targeting (SORT) nanoparticles for tissue-specific mRNA delivery and CRISPR-Cas gene editing. *Nat. Nanotechnol.* **2020**, *15*, 313–320.
- (119) Dilliard, S. A.; Cheng, Q.; Siegwart, D. J. On the mechanism of tissue-specific mRNA delivery by selective organ targeting nanoparticles. *Proc. Natl. Acad. Sci. U. S. A.* **2021**, *118*, No. e2109256118.
- (120) Peer, D.; Park, E. J.; Morishita, Y.; Carman, C. V.; Shimaoka, M. Systemic leukocyte-directed siRNA delivery revealing cyclin D1 as an anti-inflammatory target. *Science* **2008**, *319*, 627–630.
- (121) Kim, S. S.; Peer, D.; Kumar, P.; Subramanya, S.; Wu, H.; Asthana, D.; Habiro, K.; Yang, Y. G.; Manjunath, N.; Shimaoka, M.; Shankar, P. RNAi-mediated CCR5 silencing by LFA-1-targeted nanoparticles prevents HIV infection in BLT mice. *Mol. Ther.* **2010**, *18*, 370–376.
- (122) Hajdu, P.; Chimote, A. A.; Thompson, T. H.; Koo, Y.; Yun, Y.; Conforti, L. Functionalized liposomes loaded with siRNAs targeting ion channels in effector memory T cells as a potential therapy for autoimmunity. *Biomaterials* **2013**, *34*, 10249–10257.
- (123) Meissner, J. M.; Toporkiewicz, M.; Czogalla, A.; Matusiewicz, L.; Kuliczowski, K.; Sikorski, A. F. Novel antisense therapeutics delivery systems: In vitro and in vivo studies of liposomes targeted with anti-CD20 antibody. *J. Controlled Release* **2015**, *220*, 515–528.
- (124) Zheng, X.; Vladau, C.; Zhang, X.; Suzuki, M.; Ichim, T. E.; Zhang, Z. X.; Li, M.; Carrier, E.; Garcia, B.; Jevnikar, A. M.; Min, W. P. A novel in vivo siRNA delivery system specifically targeting dendritic cells and silencing CD40 genes for immunomodulation. *Blood* **2009**, *113*, 2646–2654.
- (125) Weinstein, S.; Toker, I. A.; Emmanuel, R.; Ramishetti, S.; Hazan-Halevy, I.; Rosenblum, D.; Goldsmith, M.; Abraham, A.; Benjamini, O.; Bairey, O.; Raanani, P.; Nagler, A.; Lieberman, J.; Peer, D. Harnessing RNAi-based nanomedicines for therapeutic gene silencing in B-cell malignancies. *Proc. Natl. Acad. Sci. U. S. A.* **2016**, *113*, E16–22.
- (126) Ramishetti, S.; Kedmi, R.; Goldsmith, M.; Leonard, F.; Sprague, A. G.; Godin, B.; Gozin, M.; Cullis, P. R.; Dykxhoorn, D. M.; Peer, D. Systemic gene silencing in primary T lymphocytes using targeted lipid nanoparticles. *ACS Nano* **2015**, *9*, 6706–6716.
- (127) Katakowski, J. A.; Mukherjee, G.; Wilner, S. E.; Maier, K. E.; Harrison, M. T.; DiLorenzo, T. P.; Levy, M.; Palliser, D. Delivery of siRNAs to dendritic cells using DEC205-targeted lipid nanoparticles to inhibit immune responses. *Mol. Ther.* **2016**, *24*, 146–155.
- (128) Kedmi, R.; Veiga, N.; Ramishetti, S.; Goldsmith, M.; Rosenblum, D.; Dammes, N.; Hazan-Halevy, I.; Nahary, L.; Leviatan-Ben-Arye, S.; Harlev, M.; Behlke, M.; Benhar, I.; Lieberman, J.; Peer, D. A modular platform for targeted RNAi therapeutics. *Nat. Nanotechnol.* **2018**, *13*, 214–219.
- (129) Veiga, N.; Goldsmith, M.; Diesendruck, Y.; Ramishetti, S.; Rosenblum, D.; Elinav, E.; Behlke, M. A.; Benhar, I.; Peer, D. Leukocyte-specific siRNA delivery revealing IRF8 as a potential anti-inflammatory target. *J. Controlled Release* **2019**, *313*, 33–41.
- (130) Sakurai, Y.; Abe, N.; Yoshikawa, K.; Oyama, R.; Ogasawara, S.; Murata, T.; Nakai, Y.; Tange, K.; Tanaka, H.; Akita, H. Targeted delivery of lipid nanoparticle to lymphatic endothelial cells via anti-podoplanin antibody. *J. Controlled Release* **2022**, *349*, 379–387.
- (131) Lee, Y. K.; Lee, T. S.; Song, I. H.; Jeong, H. Y.; Kang, S. J.; Kim, M. W.; Ryu, S. H.; Jung, I. H.; Kim, J. S.; Park, Y. S. Inhibition of pulmonary cancer progression by epidermal growth factor receptor-targeted transfection with Bcl-2 and survivin siRNAs. *Cancer Gene Ther.* **2015**, *22*, 335–343.
- (132) Okamoto, A.; Asai, T.; Hirai, Y.; Shimizu, K.; Koide, H.; Minamino, T.; Oku, N. Systemic administration of siRNA with anti-HB-EGF antibody-modified lipid nanoparticles for the treatment of triple-negative breast cancer. *Mol. Pharmaceutics* **2018**, *15*, 1495–1504.
- (133) Shi, D.; Toyonaga, S.; Anderson, D. G. In vivo RNA delivery to hematopoietic stem and progenitor cells via targeted lipid nanoparticles. *Nano Lett.* **2023**, *23*, 2938–2944.
- (134) Nair, J. K.; Willoughby, J. L.; Chan, A.; Charisse, K.; Alam, M. R.; Wang, Q.; Hoekstra, M.; Kandasamy, P.; Kel'in, A. V.; Milstein, S.; Taneja, N.; O'Shea, J.; Shaikh, S.; Zhang, L.; van der Sluis, R. J.; Jung, M. E.; Akinc, A.; Hutabarat, R.; Kuchimanchi, S.; Fitzgerald, K.; Zimmermann, T.; van Berkel, T. J.; Maier, M. A.; Rajeev, K. G.; Manoharan, M. Multivalent N-acetylgalactosamine-conjugated siRNA localizes in hepatocytes and elicits robust RNAi-mediated gene silencing. *J. Am. Chem. Soc.* **2014**, *136*, 16958–16961.
- (135) Song, E.; Zhu, P.; Lee, S. K.; Chowdhury, D.; Kussman, S.; Dykxhoorn, D. M.; Feng, Y.; Palliser, D.; Weiner, D. B.; Shankar, P.; Marasco, W. A.; Lieberman, J. Antibody mediated in vivo delivery of small interfering RNAs via cell-surface receptors. *Nat. Biotechnol.* **2005**, *23*, 709–717.
- (136) Kumar, P.; Ban, H. S.; Kim, S. S.; Wu, H.; Pearson, T.; Greiner, D. L.; Laouar, A.; Yao, J.; Haridas, V.; Habiro, K.; Yang, Y. G.; Jeong, J. H.; Lee, K. Y.; Kim, Y. H.; Kim, S. W.; Peipp, M.; Fey, G. H.; Manjunath, N.; Shultz, L. D.; Lee, S. K.; Shankar, P. T. cell-specific siRNA delivery suppresses HIV-1 infection in humanized mice. *Cell* **2008**, *134*, 577–586.
- (137) Sugo, T.; Terada, M.; Oikawa, T.; Miyata, K.; Nishimura, S.; Kenjo, E.; Ogasawara-Shimizu, M.; Makita, Y.; Imaichi, S.; Murata, S.; Otake, K.; Kikuchi, K.; Teratani, M.; Masuda, Y.; Kamei, T.; Takagahara, S.; Ikeda, S.; Ohtaki, T.; Matsumoto, H. Development of antibody-siRNA conjugate targeted to cardiac and skeletal muscles. *J. Controlled Release* **2016**, *237*, 1–13.
- (138) Junutula, J. R.; Raab, H.; Clark, S.; Bhakta, S.; Leipold, D. D.; Weir, S.; Chen, Y.; Simpson, M.; Tsai, S. P.; Dennis, M. S.; Lu, Y.; Meng, Y. G.; Ng, C.; Yang, J.; Lee, C. C.; Duenas, E.; Gorrell, J.; Katta, V.; Kim, A.; McDorman, K.; Flagella, K.; Venook, R.; Ross, S.; Spencer, S. D.; Lee Wong, W.; Lowman, H. B.; Vandlen, R.; Sliwowski, M. X.; Scheller, R. H.; Polakis, P.; Mallet, W. Site-specific conjugation of a cytotoxic drug to an antibody improves the therapeutic index. *Nat. Biotechnol.* **2008**, *26*, 925–932.
- (139) Cuellar, T. L.; Barnes, D.; Nelson, C.; Tanguay, J.; Yu, S. F.; Wen, X.; Scales, S. J.; Gesch, J.; Davis, D.; van Brabant Smith, A.; Leake, D.; Vandlen, R.; Siebel, C. W. Systematic evaluation of antibody-

mediated siRNA delivery using an industrial platform of THIOMAB-siRNA conjugates. *Nucleic Acids Res.* **2015**, *43*, 1189–1203.

(140) Fu, Z.; Li, S.; Han, S.; Shi, C.; Zhang, Y. Antibody drug conjugate: the “biological missile” for targeted cancer therapy. *Signal Transduct Target Ther.* **2022**, *7*, 93.

(141) Nanna, A. R.; Kel'in, A. V.; Theile, C.; Pierson, J. M.; Voo, Z. X.; Garg, A.; Nair, J. K.; Maier, M. A.; Fitzgerald, K.; Rader, C. Generation and validation of structurally defined antibody-siRNA conjugates. *Nucleic Acids Res.* **2020**, *48*, S281–S293.

(142) Barker, S. J.; Thayer, M. B.; Kim, C.; Tatarakis, D.; Simon, M. J.; Dial, R.; Nilewski, L.; Wells, R. C.; Zhou, Y.; Afetian, M.; Akkapeddi, P.; Chappell, A.; Chew, K. S.; Chow, J.; Clemens, A.; Discenza, C. B.; Dugas, J. C.; Dwyer, C.; Earr, T.; Ha, C.; Ho, Y. S.; Huynh, D.; Lozano, E. I.; Jayaraman, S.; Kwan, W.; Mahon, C.; Pizzo, M.; Robles-Colmenares, Y.; Roche, E.; Sanders, L.; Stergioulis, A.; Tong, R.; Tran, H.; Zuchero, Y.; Estrada, A. A.; Gadkar, K.; Koth, C. M. M.; Sanchez, P. E.; Thorne, R. G.; Watts, R. J.; Sandmann, T.; Kane, L. A.; Rigo, F.; Dennis, M. S.; Lewcock, J. W.; DeVos, S. L. Targeting the transferrin receptor to transport antisense oligonucleotides across the mammalian blood-brain barrier. *Sci. Transl. Med.* **2024**, *16*, No. eadi2245.

(143) Biscans, A.; Coles, A.; Haraszi, R.; Echeverria, D.; Hassler, M.; Osborn, M.; Khvorova, A. Diverse lipid conjugates for functional extrahepatic siRNA delivery in vivo. *Nucleic Acids Res.* **2019**, *47*, 1082–1096.

(144) Osborn, M. F.; Coles, A. H.; Biscans, A.; Haraszi, R. A.; Roux, L.; Davis, S.; Ly, S.; Echeverria, D.; Hassler, M. R.; Godinho, B.; Nikan, M.; Khvorova, A. Hydrophobicity drives the systemic distribution of lipid-conjugated siRNAs via lipid transport pathways. *Nucleic Acids Res.* **2019**, *47*, 1070–1081.

(145) Nikan, M.; Osborn, M. F.; Coles, A. H.; Godinho, B. M.; Hall, L. M.; Haraszi, R. A.; Hassler, M. R.; Echeverria, D.; Aronin, N.; Khvorova, A. Docosahexaenoic acid conjugation enhances distribution and safety of siRNA upon local administration in mouse brain. *Mol. Ther. Nucleic Acids.* **2016**, *5*, No. e344.

(146) Godinho, B. M. D. C.; Gilbert, J. W.; Haraszi, R. A.; Coles, A. H.; Biscans, A.; Roux, L.; Nikan, M.; Echeverria, D.; Hassler, M.; Khvorova, A. Pharmacokinetic profiling of conjugated therapeutic oligonucleotides: A high-throughput method based upon serial blood microsampling coupled to peptide nucleic acid hybridization assay. *Nucleic Acid Ther.* **2017**, *27*, 323–334.

(147) Godinho, B. M. D. C.; Henninger, N.; Bouley, J.; Alterman, J. F.; Haraszi, R. A.; Gilbert, J. W.; Sapp, E.; Coles, A. H.; Biscans, A.; Nikan, M.; Echeverria, D.; DiFiglia, M.; Aronin, N.; Khvorova, A. Transvascular delivery of hydrophobically modified siRNAs: Gene silencing in the rat brain upon disruption of the blood-brain barrier. *Mol. Ther.* **2018**, *26*, 2580–2591.

(148) Prakash, T. P.; Mullick, A. E.; Lee, R. G.; Yu, J.; Yeh, S. T.; Low, A.; Chappell, A. E.; Ostergaard, M. E.; Murray, S.; Gaus, H. J.; Swayze, E. E.; Seth, P. P. Fatty acid conjugation enhances potency of antisense oligonucleotides in muscle. *Nucleic Acids Res.* **2019**, *47*, 6029–6044.

(149) Hariharan, V. N.; Nakamura, T.; Shin, M.; Tang, Q.; Sontakke, V.; Caiazza, J.; Hildebrand, S.; Khvorova, A.; Yamada, K. Phosphatidylcholine head group chemistry alters the extrahepatic accumulation of lipid-conjugated siRNA. *Mol. Ther. Nucleic Acids.* **2024**, *35*, No. 102230.

(150) Wijesinghe, A.; Kumari, S.; Booth, V. Conjugates for use in peptide therapeutics: A systematic review and meta-analysis. *PLoS One.* **2022**, *17*, No. e0255753.

(151) Wang, L.; Wang, N.; Zhang, W.; Cheng, X.; Yan, Z.; Shao, G.; Wang, X.; Wang, R.; Fu, C. Therapeutic peptides: current applications and future directions. *Signal Transduct Target Ther.* **2022**, *7*, 48.

(152) Klabenkova, K.; Fokina, A.; Stetsenko, D. Chemistry of peptide-oligonucleotide conjugates: A review. *Molecules.* **2021**, *26*, 5420.

(153) Gori, A.; Lodigiani, G.; Colombaroli, S. G.; Bergamaschi, G.; Vitali, A. Cell penetrating peptides: Classification, mechanisms, methods of study, and applications. *ChemMedChem.* **2023**, *18*, No. e202300236.

(154) Bottens, R. A.; Yamada, T. Cell-penetrating peptides (CPPs) as therapeutic and diagnostic agents for cancer. *Cancers (Basel).* **2022**, *14*, 5546.

(155) Gan, L.; Wu, L. C. L.; Wood, J. A.; Yao, M.; Treleaven, C. M.; Estrella, N. L.; Wentworth, B. M.; Hanson, G. J.; Passini, M. A. A cell-penetrating peptide enhances delivery and efficacy of phosphorodiamidate morpholino oligomers in mdx mice. *Mol. Ther. Nucleic Acids.* **2022**, *30*, 17–27.

(156) Klein, A. F.; Varela, M. A.; Arandel, L.; Holland, A.; Naouar, N.; Arzumanov, A.; Seoane, D.; Revillod, L.; Bassez, G.; Ferry, A.; Jauvin, D.; Gourdon, G.; Puymirat, J.; Gait, M. J.; Furling, D.; Wood, M. J. Peptide-conjugated oligonucleotides evoke long-lasting myotonic dystrophy correction in patient-derived cells and mice. *J. Clin. Invest.* **2019**, *129*, 4739–4744.

(157) Li, X.; Kheirabadi, M.; Dougherty, P. G.; Kamer, K. J.; Shen, X.; Estrella, N. L.; Peddigari, S.; Pathak, A.; Blake, S. L.; Sizensky, E.; Genio, C. D.; Gaur, A. B.; Dhanabal, M.; Girgenrath, M.; Sethuraman, N.; Qian, Z. The endosomal escape vehicle platform enhances delivery of oligonucleotides in preclinical models of neuromuscular disorders. *Mol. Ther. Nucleic Acids.* **2023**, *33*, 273–285.

(158) Khairkhan, N.; Namvar, A.; Bolhassani, A. Application of cell penetrating peptides as a promising drug carrier to combat viral infections. *Mol. Biotechnol.* **2023**, *65*, 1387–1402.

(159) Gaynor, K. U.; Vaysburd, M.; Harman, M. A. J.; Albecka, A.; Jeffrey, P.; Beswick, P.; Papa, G.; Chen, L.; Mallery, D.; McGuinness, B.; Van Rietschoten, K.; Stanway, S.; Brear, P.; Lulla, A.; Ciazynska, K.; Chang, V. T.; Sharp, J.; Neary, M.; Box, H.; Herriott, J.; Kijak, E.; Tatham, L.; Bentley, E. G.; Sharma, P.; Kirby, A.; Han, X.; Stewart, J. P.; Owen, A.; Briggs, J. A. G.; Hyvonen, M.; Skynner, M. J.; James, L. C. Multivalent bicyclic peptides are an effective antiviral modality that can potentially inhibit SARS-CoV-2. *Nat. Commun.* **2023**, *14*, 3583.

(160) Mudd, G. E.; Scott, H.; Chen, L.; van Rietschoten, K.; Ivanova-Berndt, G.; Dzionek, K.; Brown, A.; Watcham, S.; White, L.; Park, P. U.; Jeffrey, P.; Rigby, M.; Beswick, P. Discovery of BT8009: A nectin-4 targeting bicyclic toxin conjugate for the treatment of cancer. *J. Med. Chem.* **2022**, *65*, 14337–14347.

(161) Arrowhead. 2024 Summer Series of R&D Webinars Part 1 – Muscle Programs. <https://ir.arrowheadpharma.com/static-files/c8100bbd-963d-4275-9233-c0b3f18f6d31> (23 May 2024).

(162) Arrowhead. Arrowhead Pharmaceuticals Initiates Phase 1/2a Study of ARO-DM1 for Treatment of Type 1 Myotonic Dystrophy. <https://ir.arrowheadpharma.com/news-releases/news-release-details/arrowhead-pharmaceuticals-initiates-phase-12a-study-aro-dm1> (8 March 2024).

(163) Pei, D.; Buyanova, M. Overcoming endosomal entrapment in drug delivery. *Bioconjug. Chem.* **2019**, *30*, 273–283.

(164) Finicle, B. T.; Eckenstein, K. H.; Revenko, A. S.; Anderson, B. A.; Wan, W. B.; McCracken, A. N.; Gil, D.; Fruman, D. A.; Hanessian, S.; Seth, P. P.; Edinger, A. L. Simultaneous inhibition of endocytic recycling and lysosomal fusion sensitizes cells and tissues to oligonucleotide therapeutics. *Nucleic Acids Res.* **2023**, *51*, 1583–1599.

(165) Hermans, G.; Postel, R.; Van De Langemheen, H.; Asadian Birjand, M. Conjugate of saponin, oligonucleotide and galnac. WO2022055351A1, 2022.

(166) Malecova, B.; Burke, R. S.; Cochran, M.; Hood, M. D.; Johns, R.; Kovach, P. R.; Doppalapudi, V. R.; Erdogan, G.; Arias, J. D.; Darimont, B.; Miller, C. D.; Huang, H.; Geall, A.; Younis, H. S.; Levin, A. A. Targeted tissue delivery of RNA therapeutics using antibody-oligonucleotide conjugates (AOCs). *Nucleic Acids Res.* **2023**, *51*, 5901–5910.

(167) Mullard, A. Antibody-oligonucleotide conjugates enter the clinic. *Nat. Rev. Drug Discovery* **2022**, *21*, 6–8.

(168) Bird, T. D. Myotonic dystrophy Type 1. In *GeneReviews*(®), Adam, M. P., Feldman, J., Mirzaa, G. M., Pagon, R. A., Wallace, S. E., Bean, L. J. H., Gripp, K. W., Amemiya, A., Eds.; University of Washington, Seattle: Seattle (WA), 1993.

(169) Liao, Q.; Zhang, Y.; He, J.; Huang, K. Global prevalence of myotonic dystrophy: An updated systematic review and meta-analysis. *Neuroepidemiology.* **2022**, *56*, 163–173.

(170) Johnson, N.; Day, J.; Hamel, J.; Thornton, C.; Subramony, S. H.; Soltanzadeh, P.; Statland, J.; Freimer, M.; Quan, D.; Knisely, B.; Davidson, A.; McEvoy, B.; Tai, L.-J.; Ackermann, E. J. Initial results of

the phase 2 open-label extension study of AOC 1001 in adults with myotonic dystrophy Type 1: MARINA-OLE. In *MDA Clinical & Scientific Conference*, Orlando, FL, 2024; Poster number T307.

(171) Zhu, Y.; Kwan, T.; Meng, Q.; Tai, L.; Cho, H.; Lee, M.; Younis, H.; Levin, A.; Flanagan, M. VP48 AOC 1001 demonstrates DMPK reduction and spliceopathy improvement in a phase 1/2 study in myotonic dystrophy type 1 (DM1) (MARINA) **2023**, 33, S70.

(172) Wood, M.; Varela, M.; Holland, A.; Raz, R.; Furling, D.; Klein, A.; Gait, M. Conjugate and uses thereof. WO 2021/028666 A1. 2021.

(173) Holland, A.; Lonkar, P.; Sweeney, C. B. S.; Gilbert, J.; Svenstrup, N.; Goyal, J. Single- and repeat-dose nonclinical data for PGN-EDO51 demonstrated potential for the treatment of Duchenne muscular dystrophy (DMD). In *MDA Clinical & Scientific Conference*, Orlando, FL, 2024; Poster number S17.

(174) Holland, A.; Godfrey, C.; Lonkar, P.; Svenstrup, N.; Larkindale, J.; McArthur, J.; Bracegirdle, S.; Goyal, J. PGN-EDO51: Preclinical data supporting the development of an enhanced delivery oligonucleotide (EDO) for the treatment of Duchenne muscular dystrophy (DMD). *Neurology* **2023**, 100, P3–8.002.

(175) Mellion, M.; Larkindale, J.; Lonkar, P.; Goyal, J.; Holland, A.; Foy, J.; Garg, B.; Yu, S.; Frank, A.; Abbott, C.; Svenstrup, N.; Cormier, J.; Vacca, S.; McArthur, J. Positive results from a first-in-human study support continued development of PGN EDO51 for the treatment of Duchenne muscular dystrophy (DMD). In *MDA Clinical & Scientific Conference*, Orlando, FL, 2024; Poster number 95.

(176) Larkindale, J.; Lonkar, P.; Goyal, J.; Holland, A.; Foy, J.; Garg, B.; Yu, S.; Frank, A.; Abbott, C.; Svenstrup, N.; Cormier, J.; Vacca, S.; Mellion, M. P44 Phase 1 study of PGN-EDO51 demonstrates tolerability, delivery and high levels of exon skipping for treatment of Duchenne muscular dystrophy (DMD). *Neuromuscular Disorders* **2023**, 33, S69.

(177) Desjardins, C. A.; Yao, M.; Hall, J.; O'Donnell, E.; Venkatesan, R.; Spring, S.; Wen, A.; Hsia, N.; Shen, P.; Russo, R.; Lan, B.; Picariello, T.; Tang, K.; Weeden, T.; Zanolli, S.; Subramanian, R.; Ibraghimov-Beskrovnaia, O. Enhanced exon skipping and prolonged dystrophin restoration achieved by TfR1-targeted delivery of antisense oligonucleotide using FORCE conjugation in mdx mice. *Nucleic Acids Res.* **2022**, 50, 11401–11414.

(178) Nicholas, A.; Bush, E. W.; Kasahara, D. I.; Schienebeck, C. M. RNAi agents for inhibiting expression of receptor for advanced glycation end-products, compositions thereof, and methods of use. US20220396791A1, 2022.

(179) Barrientos, T.; Laothamatas, I.; Koves, T. R.; Soderblom, E. J.; Bryan, M.; Moseley, M. A.; Muoio, D. M.; Andrews, N. C. Metabolic catastrophe in mice lacking transferrin receptor in muscle. *EBioMedicine* **2015**, 2, 1705–1717.

(180) Li, Y.; Cheng, J. X.; Yang, H. H.; Chen, L. P.; Liu, F. J.; Wu, Y.; Fan, M.; Wu, H. T. Transferrin receptor 1 plays an important role in muscle development and denervation-induced muscular atrophy. *Neural Regen Res.* **2021**, 16, 1308–1316.

(181) Naylor, M.; Mix, C.; Han, B.; Dugar, A. P45 A phase 1/2 study of DYNE-251 in males with DMD mutations amenable to exon 51 skipping: DELIVER study design. *Neuromuscular Disord.* **2023**, 33, S69–S70.

(182) Sabrina Haque, U.; Kohut, M.; Yokota, T. Comprehensive review of adverse reactions and toxicology in ASO-based therapies for Duchenne Muscular Dystrophy: From FDA-approved drugs to peptide-conjugated ASO. *Curr. Res. Toxicol.* **2024**, 7, No. 100182.

(183) Flanigan, K.; Campbell, C.; Deconinck, N.; Waele, L. D.; Lorentzos, M.; Phan, H.; Shieh, P.; Mix, C.; Ray, S.; Wang, D.; Farwell, W.; Dugar, A.; Naylor, M. L.; Guglieri, M. Initial data from the DELIVER trial of DYNE-251 in males with DMD mutations amenable to exon 51 skipping. In *MDA Clinical & Scientific Conference*, Orlando, FL, 2024; Poster number M156.

(184) Dyne Therapeutics. *DELIVER Phase 1/2 trial - interim results*. <https://www.neurologylive.com/view/dyne-251-increases-dystrophin-level-expression-phase-1-2-deliver-trial-duchenne> (6 January 2024).

(185) Avidity Biosciences. *FORTITUDETM phase 1/2 trial – interim results*. https://www.fshdsociety.org/wp-content/uploads/2024/06/Day-1.6-Avidity-FORTITUDE-Interim-Results_FSHD-Connect-6.15.24.pdf (20 June 2024).

(186) Halseth, A.; Ackermann, E.; Brandt, T.; Chen, C.-Y.; Stahl, M.; Ditrapani, K.; Hughes, S.; Tawil, A.; Statland, J. Phase 1/2 study to evaluate the safety, tolerability, pharmacokinetics, and pharmacodynamic effects of AOC 1020 administered intravenously to adult patients with facioscapulohumeral muscular dystrophy (FORTITUDE) trial design. *Neurology* **2023**, 100, P3–8.007.

(187) Brown, K. M.; Nair, J. K.; Janas, M. M.; Anglero-Rodriguez, Y. I.; Dang, L. T. H.; Peng, H.; Theile, C. S.; Castellanos-Rizaldos, E.; Brown, C.; Foster, D.; Kurz, J.; Allen, J.; Maganti, R.; Li, J.; Matsuda, S.; Stricos, M.; Chickering, T.; Jung, M.; Wasserman, K.; Rollins, J.; Woods, L.; Kelin, A.; Guenther, D. C.; Mobley, M. W.; Petrusis, J.; McDougall, R.; Racie, T.; Bombardier, J.; Cha, D.; Agarwal, S.; Johnson, L.; Jiang, Y.; Lentini, S.; Gilbert, J.; Nguyen, T.; Chigas, S.; LeBlanc, S.; Poreci, U.; Kasper, A.; Rogers, A. B.; Chong, S.; Davis, W.; Sutherland, J. E.; Castoreno, A.; Milstein, S.; Schlegel, M. K.; Zlatev, I.; Charisse, K.; Keating, M.; Manoharan, M.; Fitzgerald, K.; Wu, J. T.; Maier, M. A.; Jadhav, V. Expanding RNAi therapeutics to extrahepatic tissues with lipophilic conjugates. *Nat. Biotechnol.* **2022**, 40, 1500–1508.

(188) Tcw, J.; Goate, A. M. Genetics of beta-amyloid precursor protein in Alzheimer's disease. *Cold Spring Harb Perspect Med.* **2017**, 7, No. a024539.

(189) Cohen, S.; Ducharme, S.; Brosch, J. R.; Vijverberg, E. G. B.; Apostolova, L. G.; Sostelly, A.; Goteti, S.; Makarova, N.; Avbersek, A.; Guo, W.; Bostwick, B.; Mummery, C. J. Interim phase 1 part A results for ALN APP, the first investigational RNAi therapeutic in development for Alzheimer's disease. *Alzheimer's Dement.* **2023**, 19, No. e082650.

(190) Mummery, C.; Ducharme, S.; Brosch, J.; Vijverberg, E.; Sostelly, A.; Goteti, S.; Makarova, N.; Avbersek, A.; Guo, W.; Bostwick, B.; Cohen, S. Phase 1 safety, tolerability, and pharmacological results of ALN-APP, the first investigational RNA interference therapeutic in development for early-onset Alzheimer's disease. *Journal of prevention of alzheimer's disease* **2023**, 10, S42.

(191) Pothin, E.; Lesuisse, D.; Lafaye, P. Brain delivery of single-domain antibodies: A focus on VHH and VNAR. *Pharmaceutics* **2020**, 12, 937.

(192) Cohen, R. D. M.; Guillaume, J.; Tamsamani, J.; Lechorche, P.; Khrestchatsky, M. Transferrin receptor-binding molecules, conjugates thereof and their uses. WO2020/144233A1, 2020.

(193) Alterman, J. F.; Godinho, B.; Hassler, M. R.; Ferguson, C. M.; Echeverria, D.; Sapp, E.; Haraszti, R. A.; Coles, A. H.; Conroy, F.; Miller, R.; Roux, L.; Yan, P.; Knox, E. G.; Turanov, A. A.; King, R. M.; Gernoux, G.; Mueller, C.; Gray-Edwards, H. L.; Moser, R. P.; Bishop, N. C.; Jaber, S. M.; Gounis, M. J.; Sena-Esteves, M.; Pai, A. A.; DiFiglia, M.; Aronin, N.; Khvorova, A. A divalent siRNA chemical scaffold for potent and sustained modulation of gene expression throughout the central nervous system. *Nat. Biotechnol.* **2019**, 37, 884–894.

(194) Ferguson, C. M.; Godinho, B. M.; Alterman, J. F.; Coles, A. H.; Hassler, M.; Echeverria, D.; Gilbert, J. W.; Knox, E. G.; Caiazza, J.; Haraszti, R. A.; King, R. M.; Taghian, T.; Puri, A.; Moser, R. P.; Gounis, M. J.; Aronin, N.; Gray-Edwards, H.; Khvorova, A. Comparative route of administration studies using therapeutic siRNAs show widespread gene modulation in Dorset sheep. *JCI Insight* **2021**, 6, No. e152203.

(195) Wadman, M. Antisense rescues babies from killer disease. *Science* **2016**, 354, 1359–1360.

(196) Finkel, R. S.; Chiriboga, C. A.; Vajsaar, J.; Day, J. W.; Montes, J.; De Vivo, D. C.; Bishop, K. M.; Foster, R.; Liu, Y.; Ramirez-Schrempp, D.; Schneider, E.; Bennett, C. F.; Wong, J.; Farwell, W. Treatment of infantile-onset spinal muscular atrophy with nusinersen: final report of a phase 2, open-label, multicentre, dose-escalation study. *Lancet Child Adolesc Health.* **2021**, 5, 491–500.

(197) Hariharan, V. N.; Shin, M.; Chang, C. W.; O'Reilly, D.; Biscans, A.; Yamada, K.; Guo, Z.; Somasundaran, M.; Tang, Q.; Monopoli, K.; Krishnamurthy, P. M.; Devi, G.; McHugh, N.; Cooper, D. A.; Echeverria, D.; Cruz, J.; Chan, I. L.; Liu, P.; Lim, S. Y.; McConnell, J.; Singh, S. P.; Hildebrand, S.; Sousa, J.; Davis, S. M.; Kennedy, Z.;

- Ferguson, C.; Godinho, B.; Thillier, Y.; Caiazz, J.; Ly, S.; Muhuri, M.; Kelly, K.; Humphries, F.; Cousineau, A.; Parsi, K. M.; Li, Q.; Wang, Y.; Maehr, R.; Gao, G.; Korkin, D.; McDougall, W. M.; Finberg, R. W.; Fitzgerald, K. A.; Wang, J. P.; Watts, J. K.; Khvorova, A. Divalent siRNAs are bioavailable in the lung and efficiently block SARS-CoV-2 infection. *Proc. Natl. Acad. Sci. U. S. A.* **2023**, *120*, No. e2219523120.
- (198) Almeida, A.; Li, Z.; Bush, E. W.; Pei, T.; Glebocka, A.; Nicholas, A.; Carlson, J.; Fowler-Watters, M. Alpha-v beta-6 integrin ligands and uses thereof. WO2018085415A1, 2023.
- (199) Buckley, S. T.; Ehrhardt, C. The receptor for advanced glycation end products (RAGE) and the lung. *J. Biomed Biotechnol.* **2010**, *2010*, No. 917108.
- (200) Perkins, T. N.; Donnell, M. L.; Oury, T. D. The axis of the receptor for advanced glycation endproducts in asthma and allergic airway disease. *Allergy.* **2021**, *76*, 1350–1366.
- (201) Kasahara, D. I.; Nicholas, A.; Reed, T.; Trilling, Z.; Branca-Afrazi, M.; Hamilton, H.; Hegge, J.; Hamilton, J.; Pei, T.; Bush, E. W. Silencing RAGE expression with a lung-targeted RNAi trigger delivery platform suppresses pulmonary allergic inflammation. *Am. J. Respir Crit Care Med.* **2022**, *205*, A5013.
- (202) Kasahara, D.; Turner, B.; Hamilton, H.; Reed, T.; Klossowski, S.; Williams, C.; Cousineau, C.; Chen, J.; Hegge, J.; Branca-Afrazi, M.; Hamilton, J.; Pei, T.; Bush, E. A lung-targeted RNAi therapeutics delivery platform compatible with subcutaneous (s.c.) administration mediates pulmonary RAGE silencing in rodents and nonhuman primates. *Eur. Respir. J.* **2023**, *62*, No. PA1290.
- (203) O'Carroll, M. R.; Huetsch, J. C.; Hamilton, J.; Ta, A.; Moser, L.; Alagarsamy, S.; Salathe, M. A. A first-in-human study of ARO-RAGE, a novel inhaled RNA-interference therapy for asthma. *Am. J. Respir Crit Care Med.* **2024**, *209*, A1376.
- (204) Tang, Q.; Fakih, H. H.; Zain Ui Abideen, M.; Hildebrand, S. R.; Afshari, K.; Gross, K. Y.; Sousa, J.; Maebius, A. S.; Bartholdy, C.; Sogaard, P. P.; Jackerott, M.; Hariharan, V.; Summers, A.; Fan, X.; Okamura, K.; Monopoli, K. R.; Cooper, D. A.; Echeverria, D.; Bramato, B.; McHugh, N.; Furgal, R. C.; Dresser, K.; Winter, S. J.; Biscans, A.; Chuprin, J.; Haddadi, N. S.; Sherman, S.; Yildiz-Altay, U.; Rashighi, M.; Richmond, J. M.; Bouix-Peter, C.; Blanchard, C.; Clauss, A.; Alterman, J. F.; Khvorova, A.; Harris, J. E. Rational design of a JAK1-selective siRNA inhibitor for the modulation of autoimmunity in the skin. *Nat. Commun.* **2023**, *14*, 7099.
- (205) Lamb, Y. N. Inclisiran: First approval. *Drugs.* **2021**, *81*, 389–395.
- (206) Ray, K. K.; Wright, R. S.; Kallend, D.; Koenig, W.; Leiter, L. A.; Raal, F. J.; Bisch, J. A.; Richardson, T.; Jaros, M.; Wijngaard, P. L. J.; Kastelein, J. J. P. Two phase 3 trials of inclisiran in patients with elevated LDL cholesterol. *N Engl J. Med.* **2020**, *382*, 1507–1519.
- (207) Raal, F. J.; Santos, R. D.; Blom, D. J.; Marais, A. D.; Charng, M. J.; Cromwell, W. C.; Lachmann, R. H.; Gaudet, D.; Tan, J. L.; Chasan-Taber, S.; Tribble, D. L.; Flaim, J. D.; Crooke, S. T. Mipomersen, an apolipoprotein B synthesis inhibitor, for lowering of LDL cholesterol concentrations in patients with homozygous familial hypercholesterolaemia: a randomised, double-blind, placebo-controlled trial. *Lancet.* **2010**, *375*, 998–1006.
- (208) Alhamadani, F.; Zhang, K.; Parikh, R.; Wu, H.; Rasmussen, T. P.; Bahal, R.; Zhong, X. B.; Manautou, J. E. Adverse drug reactions and toxicity of the Food and Drug Administration-approved antisense oligonucleotide drugs. *Drug Metab. Dispos.* **2022**, *50*, 879–887.
- (209) PR Newswire. Comanche Biopharma Announces FDA Clearance of Investigational New Drug (IND) Application for CBP-4888, an siRNA Investigational Therapy for the Treatment of Preeclampsia. <https://www.prnewswire.com/news-releases/comanche-biopharma-announces-fda-clearance-of-investigational-new-drug-ind-application-for-cbp-4888-an-sirna-investigational-therapy-for-the-treatment-of-preeclampsia-301784873.html> (30 March, 2023).
- (210) Arrowhead. 2024 Summer Series of R&D Webinars Part IV – Obesity Programs. <https://ir.arrowheadpharma.com/static-files/d8586ad7-50f8-4973-9576-e49542a893e7> (August, 2024).
- (211) Deaton, A. M.; Dubey, A.; Ward, L. D.; Dornbos, P.; Flannick, J.; Yee, E.; Tica, S.; Noetzi, L.; Parker, M. M.; Hoffing, R. A.; Willis, C.; Plekan, M. E.; Holleman, A. M.; Hinkle, G.; Fitzgerald, K.; Vaishnav, A. K.; Nioi, P. Rare loss of function variants in the hepatokine gene INHBE protect from abdominal obesity. *Nat. Commun.* **2022**, *13*, 4319.
- (212) Hashimoto, O.; Funaba, M.; Sekiyama, K.; Doi, S.; Shindo, D.; Satoh, R.; Itoi, H.; Oiwa, H.; Morita, M.; Suzuki, C.; Sugiyama, M.; Yamakawa, N.; Takada, H.; Matsumura, S.; Inoue, K.; Oyadomari, S.; Sugino, H.; Kurisaki, A. Activin E Controls Energy Homeostasis in Both Brown and White Adipose Tissues as a Hepatokine. *Cell Rep.* **2018**, *25*, 1193–1203.
- (213) Yogosawa, S.; Mizutani, S.; Ogawa, Y.; Izumi, T. Activin receptor-like kinase 7 suppresses lipolysis to accumulate fat in obesity through downregulation of peroxisome proliferator-activated receptor gamma and C/EBPalpha. *Diabetes.* **2013**, *62*, 115–123.
- (214) Cheng, S. Y.; Caiazz, J.; Biscans, A.; Alterman, J. F.; Echeverria, D.; McHugh, N.; Hassler, M.; Jolly, S.; Giguere, D.; Cipi, J.; Khvorova, A.; Punzo, C. Single intravitreal administration of a tetravalent siRNA exhibits robust and efficient gene silencing in mouse and pig photoreceptors. *Mol. Ther Nucleic Acids.* **2024**, *35*, No. 102088.
- (215) Scranton, S.; Hou, J.; Choudhary, D.; Green, C.; Feldman, M.; Panigrahi, D. A phase 2, randomized, prospective, double-masked, vehicle-controlled study to assess the safety and efficacy of topical NEXAGON® in subjects with persistent corneal epithelial defects (PCED) resulting from severe ocular chemical and/or thermal injuries. *Invest Ophthalmol Visual Sci.* **2023**, *64*, 3119.
- (216) Yu, M.; Qin, J.; Liu, X.; Ramsden, D.; Williams, B.; Zlatev, I.; Guenther, D.; Matsuda, S.; Tymon, R.; Darcy, J.; Wong, C.; Tsung, J.; Zawaneh, P.; Chong, S.; Theile, C. S.; Taneja, N.; Rogers, A.; Liu, J.; Castellanos-Rizaldos, E.; Bond, S.; So, K.; Denoncourt, J.; Castoreno, A.; Manoharan, M.; Wu, J. T.; Fitzgerald, K.; Maier, M. A.; Jadhav, V.; Nair, J. K. Evaluating the oral delivery of GalNAc-conjugated siRNAs in rodents and non-human primates. *Nucleic Acids Res.* **2024**, *52*, 5423–5437.
- (217) Alam, R.; Alvarado, A.; Andrews, F. H.; Babb, N. A.; Balasubramaniam, D.; Chauvigne-Hines, L.; Cortez, G. S.; Croy, J. E.; Driver, D. A.; Fellows, R. E.; Girard, D.; Gonzalez Valcarcel, I. C.; Kulkarni, N. H.; Lawrence, S. A.; Liu, F.; Tani, H.; York, J. S. Transferrin Receptor Binding Proteins and Conjugates. WO2024036096A2, 2023.
- (218) Nittoli, T.; Degrad, S. J.; Afsari, H. S.; Anand, P.; Bos, J.; Choi, S.; Mallett, E.; Markotan, T.; Spink, J.; Vega, M.; Saha, M.; Schuster, R.; Jiang, J.; Zhou, J.; Zumsteg, A. B.; Coppi, A.; Jorgenson, S.; Federici, S.; Zhang, H.; Moustakas, J.; D'Souza, C.; Kyrtatsos, C.; Olson, W. Site-specific antibody conjugations using bacterial transglutaminase and the Diels-Alder cycloaddition reaction. *Cell Reports Physical Science.* **2023**, *4*, No. 101544.
- (219) Dall'Acqua, W. F.; Kiener, P. A.; Wu, H. Properties of human IgG1s engineered for enhanced binding to the neonatal Fc receptor (FcRn). *J. Biol. Chem.* **2006**, *281*, 23514–23524.
- (220) Wilkinson, I.; Anderson, S.; Fry, J.; Julien, L. A.; Neville, D.; Qureshi, O.; Watts, G.; Hale, G. Fc-engineered antibodies with immune effector functions completely abolished. *PLoS One.* **2021**, *16*, No. e0260954.
- (221) Liu, L. Pharmacokinetics of monoclonal antibodies and Fc-fusion proteins. *Protein Cell.* **2018**, *9*, 15–32.
- (222) Blanco, E.; Shen, H.; Ferrari, M. Principles of nanoparticle design for overcoming biological barriers to drug delivery. *Nat. Biotechnol.* **2015**, *33*, 941–951.
- (223) Glassman, P. M.; Myerson, J. W.; Ferguson, L. T.; Kiseleva, R. Y.; Shuvaev, V. V.; Brenner, J. S.; Muzykantov, V. R. Targeting drug delivery in the vascular system: Focus on endothelium. *Adv. Drug Deliv Rev.* **2020**, *157*, 96–117.
- (224) Cahill, P. A.; Redmond, E. M. Vascular endothelium - Gatekeeper of vessel health. *Atherosclerosis.* **2016**, *248*, 97–109.
- (225) Barua, S.; Mitragotri, S. Challenges associated with penetration of nanoparticles across cell and tissue barriers: A review of current status and future prospects. *Nano Today.* **2014**, *9*, 223–243.
- (226) Wang, Q.; Liang, Q.; Dou, J.; Zhou, H.; Zeng, C.; Pan, H.; Shen, Y.; Li, Q.; Liu, Y.; Leong, D. T.; Jiang, W.; Wang, Y. Breaking through

the basement membrane barrier to improve nanotherapeutic delivery to tumours. *Nat. Nanotechnol.* **2024**, *19*, 95–105.

(227) Simionescu, M.; Popov, D.; Sima, A. Endothelial transcytosis in health and disease. *Cell Tissue Res.* **2009**, *335*, 27–40.

(228) McIntosh, D. P.; Tan, X. Y.; Oh, P.; Schnitzer, J. E. Targeting endothelium and its dynamic caveolae for tissue-specific transcytosis in vivo: a pathway to overcome cell barriers to drug and gene delivery. *Proc. Natl. Acad. Sci. U. S. A.* **2002**, *99*, 1996–2001.

(229) Kong, G.; Braun, R. D.; Dewhirst, M. W. Characterization of the effect of hyperthermia on nanoparticle extravasation from tumor vasculature. *Cancer Res.* **2001**, *61*, 3027–3032.

(230) Akinc, A.; Maier, M. A.; Manoharan, M.; Fitzgerald, K.; Jayaraman, M.; Barros, S.; Ansell, S.; Du, X.; Hope, M. J.; Madden, T. D.; Mui, B. L.; Semple, S. C.; Tam, Y. K.; Ciufolini, M.; Witzigmann, D.; Kulkarni, J. A.; van der Meel, R.; Cullis, P. R. The Onpatro story and the clinical translation of nanomedicines containing nucleic acid-based drugs. *Nat. Nanotechnol.* **2019**, *14*, 1084–1087.

(231) Sago, C. D.; Lokugamage, M. P.; Loughrey, D.; Lindsay, K. E.; Hincapie, R.; Krupczak, B. R.; Kalathoor, S.; Sato, M.; Echeverri, E. S.; Fitzgerald, J. P.; Gan, Z.; Gamboa, L.; Paunovska, K.; Sanhueza, C. A.; Hatit, M. Z. C.; Finn, M. G.; Santangelo, P. J.; Dahlman, J. E. Augmented lipid-nanoparticle-mediated in vivo genome editing in the lungs and spleen by disrupting Cas9 activity in the liver. *Nat. Biomed Eng.* **2022**, *6*, 157–167.

(232) Huayamates, S. G.; Lokugamage, M. P.; Rab, R.; Da Silva Sanchez, A. J.; Kim, H.; Radmand, A.; Loughrey, D.; Lian, L.; Hou, Y.; Achyut, B. R.; Ehrhardt, A.; Hong, J. S.; Sago, C. D.; Paunovska, K.; Echeverri, E. S.; Vanover, D.; Santangelo, P. J.; Sorscher, E. J.; Dahlman, J. E. High-throughput screens identify a lipid nanoparticle that preferentially delivers mRNA to human tumors in vivo. *J. Controlled Release* **2023**, *357*, 394–403.

(233) Kheirrolomoom, A.; Kare, A. J.; Ingham, E. S.; Paulmurugan, R.; Robinson, E. R.; Baikoghli, M.; Inayathullah, M.; Seo, J. W.; Wang, J.; Fite, B. Z.; Wu, B.; Tumbale, S. K.; Raie, M. N.; Cheng, R. H.; Nichols, L.; Borowsky, A. D.; Ferrara, K. W. In situ T-cell transfection by anti-CD3-conjugated lipid nanoparticles leads to T-cell activation, migration, and phenotypic shift. *Biomaterials*. **2022**, *281*, No. 121339.

(234) Mui, B. L.; Tam, Y. K.; Jayaraman, M.; Ansell, S. M.; Du, X.; Tam, Y. Y.; Lin, P. J.; Chen, S.; Narayanannair, J. K.; Rajeev, K. G.; Manoharan, M.; Akinc, A.; Maier, M. A.; Cullis, P.; Madden, T. D.; Hope, M. J. Influence of polyethylene glycol lipid desorption rates on pharmacokinetics and pharmacodynamics of siRNA lipid nanoparticles. *Mol. Ther Nucleic Acids*. **2013**, *2*, No. e139.

(235) Chen, D.; Ganesh, S.; Wang, W.; Amiji, M. The role of surface chemistry in serum protein corona-mediated cellular delivery and gene silencing with lipid nanoparticles. *Nanoscale*. **2019**, *11*, 8760–8775.

(236) Moore, D. H. Species difference in serum protein patterns. *J. Biol. Chem.* **1945**, *161*, 21–32.

(237) Liu, K.; Nilsson, R.; Lazaro-Ibanez, E.; Duan, H.; Miliotis, T.; Strimfors, M.; Lerche, M.; Salgado Ribeiro, A. R.; Ulander, J.; Linden, D.; Salvati, A.; Sabirsh, A. Multiomics analysis of naturally efficacious lipid nanoparticle coronas reveals high-density lipoprotein is necessary for their function. *Nat. Commun.* **2023**, *14*, 4007.

(238) Yamada, K.; Hariharan, V. N.; Caiazz, J.; Miller, R.; Ferguson, C. M.; Sapp, E.; Fakih, H. H.; Tang, Q.; Yamada, N.; Furgal, R. C.; Paquette, J. D.; Biscans, A.; Bramato, B. M.; McHugh, N.; Summers, A.; Lochmann, C.; Godinho, B.; Hildebrand, S.; Jackson, S. O.; Echeverria, D.; Hassler, M. R.; Alterman, J. F.; DiFiglia, M.; Aronin, N.; Khvorova, A. Enhancing siRNA efficacy in vivo with extended nucleic acid backbones. *Nat. Biotechnol.* **2024**. DOI: 10.1038/s41587-024-02336-7.

NOTE ADDED AFTER ASAP PUBLICATION

Due to a production error, the version of this paper that was published ASAP January 8, 2025, contained errors in Figures 2 and 3. The corrected version was reposted January 9, 2025.